PHYSICS RELATED TO THE ELK-CODE

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Disclaimer: This document originated from discussions that I had with people about the ELK-code. As such, many equations do not neccessarily represent those equations that are actually implemented in the ELK-code, but rather my understanding of it. If you find any mistakes, please contact[1] me so that I can correct them. I am more than happy to cross out entire sections, if they are wrong or inaccurate.

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Part I.

ELK-Physics

1. Introduction

James Gleick about Isaac Newton:

> "*Newton formed his argument in classic greek geometrical style. Axioms, Lemmas, Corollaries, QED. As the best model available for perfection in knowledge, it gave his physical program the stamp of certainty. He proved facts about triangles and tangents, chords and parallelograms and from there, by a long chain of arguments, proved facts about the moon and the tides.*"

Notation 1.1 All equations in this document are writting in SI-units, unless otherwise noted.

Remark 1.1.1 Notice that in the ELK-code, all the equations are written in atomic units.

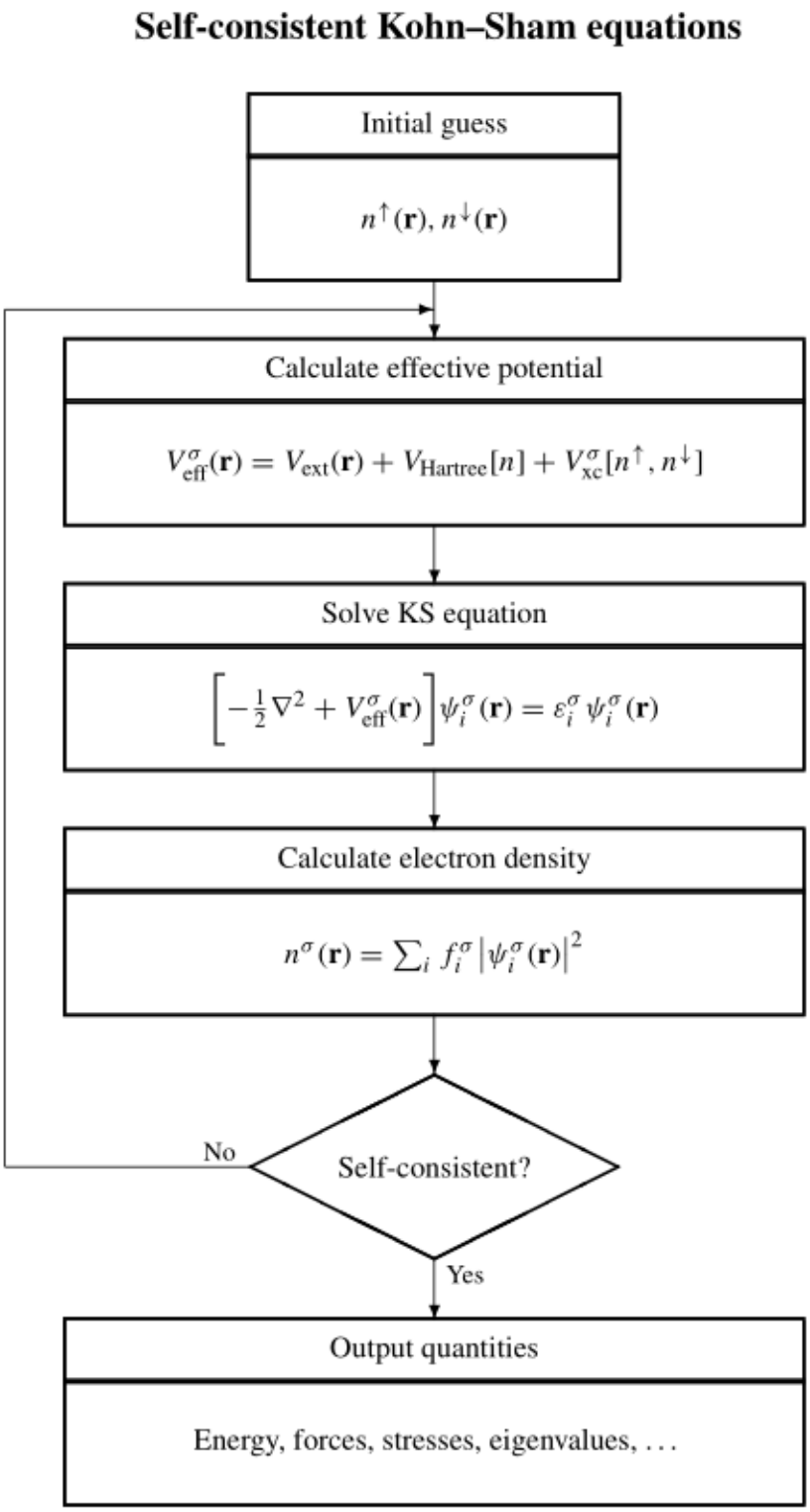

Figure 1.1.: Flow-diagram for DFT-calculations, taken from [Martin, page. 173]. Notice that the Schrödinger-Kohn-Sham-equations are written in atomic units.

2. Crystals, (Atomic)-Bases and Coverings

2.1. Fundamentals on Lattices, Muffin-Tins and Interstitial Regions

Notation 2.1 We denote with Ω the volume of the unit-cell.

Definition 2.2 Let $S \subset \mathbb{R}^{\mathrm{d}}$ be a set of points. The **convex hull** of S, denoted as $\mathrm{conv}(S)$, is the smallest convex set that contains S. Formally,

$$\mathrm{conv}(S) = \left\{ \sum_{i=1}^{n} \lambda_i \mathbf{p}_i \;\middle|\; \mathbf{p}_i \in S,\ \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1,\ n \in \mathbb{N} \right\}.$$

Remark 2.2.1 There is also the geometric intuiton of the convex hull as the smallest convex polyhedron (in $\mathbb{R}^3$) that contains all points in S.

Remark 2.2.2 Recall further, that a set is *convex* if, for any two points $\mathbf{a}, \mathbf{b} \in \mathrm{conv}(S)$, the entire line segment between them is also in the set. That is:

$$t\mathbf{a} + (1-t)\mathbf{b} \in \mathrm{conv}(S), \quad \forall t \in [0,1].$$

Definition 2.3 For each atomic species α we define the *muffin-tin radius* R_{MT_α}. Further, we introduce the *minimal muffin-tin radius*

$$R_{\mathrm{MT}_{\min}} \equiv \min_{\alpha} R_{\mathrm{MT}_\alpha}. \tag{2.1}$$

Notation 2.4 We consider the convex hull of a finite crystal $\mathrm{conv}(\mathrm{C_f})$ and choose a covering, that consists of spheres with radii R_{MT_α} around atoms of the species α, which we call *muffin tins* MT_α, and the rest, which we call *interstitial regions* IRs

$$\begin{aligned}
\mathrm{conv}\left(\mathrm{C_f}\right) &= \left(\bigcup_{\alpha} \mathrm{MT}_\alpha\right) \cup \left(\mathrm{conv}(\mathrm{C_f}) \setminus \left(\bigcup_{\alpha} \mathrm{MT}_\alpha\right)\right) \\
&\equiv \mathrm{MTs} \cup \mathrm{IRs} \\
&\equiv \mathrm{Crystal}
\end{aligned}$$

Notation 2.5 For a vector $\mathbf{r} \in \text{conv}\left(\mathrm{C_f}\right) \cap \text{MTs}$, there exists a lattice-vector $\mathbf{R}_{\mathbf{n}\alpha}$ such that $\mathbf{r}$ is within the Muffin-Tin that is centered around $\mathbf{R}_{\mathbf{n}\alpha}$. We then define

$$\mathbf{r}_\alpha \equiv \mathbf{r} - \mathbf{R}_{\mathbf{n}\alpha}. \tag{2.2}$$

Fact 2.6 For a better memory vs. precision ratio, it is advantegous to divide each muffin-tin MT_α into two regions:

- an *inner region*, in which we expand functions up to a maximal angular momentum quantum number[1]

$$\text{"usually": lmaxi} \in \{1, 2\} \tag{2.3}$$

- and an *outer region*, in which we expand functions up to a maximal angular momentum quantum number

$$\text{"usually": lmaxo} = 6. \tag{2.4}$$

2.2. Fundamentals on Tensor-Products

Theorem 2.7 Let U, V, W be vector spaces over the same field $\mathbb{F}$. Then there is an isomorphism

$$(U \oplus V) \otimes W \simeq (U \otimes W) \oplus (V \otimes W). \tag{2.5}$$

Proof 2.7 See Appendix, Proofs, page 104.

Remark 2.7.1 This property is called *the distributivity-law of the tensor product over direct sums*.

Definition 2.8 Let $A = [a_{ij}] \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$. Then the *Kronecker-product* a.k.a. *tensor-product* $A \otimes B \in \mathbb{R}^{mp \times nq}$ is defined as

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix}.$$

[1] Note that there is a typo in elk-9.5.14: In in the code lmaxi = 1 is implemented, whereas in the manual it states lmaxi = 2.

2.3. Fundamentals on Hilbert-Spaces - Spin-Independent Case

Fact 2.9 Note that the Hilbert-space of a single-electron in a bulk crystalline environment is given as[2]

$$L^2(\text{Crystal}) = \left\{ \psi : \text{Crystal} \to \mathbb{C} \;\middle|\; \int_{\text{Crystal}} |\psi(\mathbf{r})|^2 \mathrm{d}^{\mathrm{d}}\mathbf{r} = 1 \right\} \tag{2.6}$$

Fact 2.10 Recall that using the Bloch-theorem, we know that we can expand any wavefunction in a (bulk-)crystal $\psi \in L^2(\text{Crystal})$ with respect to *Bloch-waves* (i.e. energy eigenfunctions of the Hamiltonian)

$$\psi(\mathbf{r}) = \sum_{n=1}^{\infty} \sum_{\mathbf{k}\in \text{Q}} c_{n\mathbf{k}} u_{n\mathbf{k}}(\mathbf{r}) e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}, \tag{2.7}$$

with a lattice-periodic function $u_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R_n})$.

Remark 2.10.1 Recall that any lattice-periodic function $u_{n\mathbf{k}}(\mathbf{r})$ can be expanded as a Fourier-series

$$u_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}\in \text{RL}} c_{n\mathbf{k}}^{\mathbf{G}} e^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}}. \tag{2.8}$$

with complex coefficients $c_{n\mathbf{k}}^{\mathbf{G}} \in \mathbb{C}$.

Notation 2.11 We denote the space of lattice-periodic functions as

$$L^2(\text{UC}) \equiv \mathcal{H}_{\mathbf{k}} = \left\{ u_{\mathbf{k}} : \text{UC} \to \mathbb{C} \;\middle|\; u_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}\in \text{RL}} c_{\mathbf{k}}^{\mathbf{G}} e^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}}, \int_{\text{UC}} |u_{\mathbf{k}}(\mathbf{r})|^2 \mathrm{d}^{\mathrm{d}}\mathbf{r} < \infty \right\} \tag{2.9}$$

Theorem 2.12 Using Equation (2.7) this gives us an isomorphism

$$L^2(\text{Crystal}) \simeq \bigoplus_{\mathbf{k}\in \text{Q}} \mathcal{H}_{\mathbf{k}}. \tag{2.10}$$

Proof 2.12 See Appendix, Proofs, page 104.

[2] In the literature one oftentimes finds $L^2(\text{Crystal}) = L^2(S^1 \times S^1 \times S^1) = L^2(\mathbb{T}^3)$ in order to appropriately reflect the periodic boundary conditions. However, this ridicules the notation, as $\mathbf{r}$ is usually denoting a spatial coordinate and not the coordinates on a torus.

Fact 2.13 Using the decomposition of the unit cell into muffin-tins and interstitial regions, we obtain a corresponding decomposition of the Hilbert-space

$$L^2(\mathrm{UC}) = \left(\bigoplus_{\alpha=1}^{r} L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \right) \oplus L^2(\mathrm{IR}) \tag{2.11}$$

with square-integrable functions on the unit-sphere $L^2(S^2, \mathrm{d}\Omega)$ that can be expanded in terms of spherical harmonics

$$L^2(S^2, \mathrm{d}\Omega) = \bigoplus_{l=0}^{\infty} \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^{l} \left\{ Y_{lm} \equiv |lm\rangle \right\} \tag{2.12}$$

Notation 2.14 Let $\psi_{n\mathbf{k}}(\mathbf{r}) \equiv \langle \mathbf{r} | n\mathbf{k} \rangle$ denote the single-particle wavefunction in the basis of Bloch states, and let us consider the expansion

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} \sum_{|m| \leq l} c_{n\mathbf{k}}^{\alpha lm} f_{n\mathbf{k}}^{\alpha lm}(r_\alpha) Y_{lm}(\Omega_\alpha) \mathbb{1}_{\mathrm{MT}_\alpha}(\mathbf{r}) + \sum_{\mathbf{G} \in \mathrm{RL}} c_{n\mathbf{k}}^{\mathbf{G}} e^{\mathrm{i}(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} \mathbb{1}_{\mathrm{IR}}(\mathbf{r}) \tag{2.13}$$

Remark 2.14.1 Summarizing, we have in the spin-independent case the following hierarchy of Hilbert-(sub)spaces:

$$\begin{aligned}
&\mathcal{H} \equiv L^2(\mathrm{Crystal}) \simeq \bigoplus_{\mathbf{k} \in \mathrm{Q}} \mathcal{H}_{\mathbf{k}}. \\
&\bigcup \\
&\mathcal{H}_{\mathbf{k}} = L^2(\mathrm{UC}) = \left[\left(\bigoplus_{\alpha=1}^{r} L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \right) \oplus L^2(\mathrm{IR}) \right] \\
&\bigcup \\
&\mathcal{H}_\alpha \equiv L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \\
&\bigcup \\
&\mathcal{H}_{\alpha L} \equiv L^2(S^2, \mathrm{d}\Omega) = \bigoplus_{l=0}^{\infty} \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^{l} \left\{ Y_{lm} \equiv |lm\rangle \right\} \\
&\bigcup \\
&\mathcal{H}_{\alpha L(l)} \equiv \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^{l} \left\{ Y_{lm} \equiv |lm\rangle \right\}
\end{aligned}$$

2.4. Fundamentals on Hilbert-Spaces - Spin-Dependent Case

Fact 2.15 Note that the Hilbert-space of a single-electron in a bulk crystalline environment is given as[3]

$$\mathcal{H} = L^2(\text{Crystal}) \otimes \mathbb{C}^2 = \left\{ \Psi : \text{ Crystal} \times \{\uparrow, \downarrow\} \to \mathbb{C} \;\middle|\; \sum\!\!\!\!\!\!\int_{\substack{\text{Crystal} \\ \sigma \in \{\uparrow,\downarrow\}}} |\Psi(\mathbf{r}, \sigma)|^2 = 1 \right\} \tag{2.14}$$

Fact 2.16 Recall that using the Bloch-theorem, we know that we can expand any wavefunction $\Psi \in L^2(\text{Crystal}) \otimes \mathbb{C}^2$ as

$$\Psi(\mathbf{r}, \sigma) = \sum_{\mathbf{k} \in \text{Q}} \sum_{N=1}^{\infty} c_{N\mathbf{k}} \Psi_{N\mathbf{k}}(\mathbf{r}, \sigma) = \sum_{\mathbf{k} \in \text{Q}} \sum_{N=1}^{\infty} c_{N\mathbf{k}} u_{N\mathbf{k}}(\mathbf{r}, \sigma) e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} \tag{2.15}$$

with a lattice-periodic spinors $u_{N\mathbf{k}}(\mathbf{r}, \sigma) = u_{N\mathbf{k}}(\mathbf{r} + \mathbf{R_n}, \sigma)$ that obey certain differential equations[4].

Remark 2.16.1 Recall that any lattice-periodic spinor $u_{N\mathbf{k}}(\mathbf{r}, \sigma)$ can be expanded as a Fourier-series and w.r.t. the standard-basis of spins

$$u_{N\mathbf{k}}(\mathbf{r}, \sigma) = \sum_{\mathbf{G} \in \text{RL}} \sum_{s \in \{\uparrow,\downarrow\}} c^{\mathbf{G}}_{N\mathbf{k}s} e^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}} \chi_s(\sigma) = \begin{cases} \sum\limits_{\mathbf{G} \in \text{RL}} c^{\mathbf{G}}_{N\mathbf{k}\uparrow} e^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}} & \text{for } \sigma = \uparrow \\ \sum\limits_{\mathbf{G} \in \text{RL}} c^{\mathbf{G}}_{N\mathbf{k}\downarrow} e^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}} & \text{for } \sigma = \downarrow \end{cases} \tag{2.16}$$

with complex coefficients $c^{\mathbf{G}}_{N\mathbf{k}\sigma} \in \mathbb{C}$.

Notation 2.17 We denote the space of lattice-periodic functions as

$$L^2(\text{UC}) = \mathcal{H}_{\mathbf{k}} = \left\{ u_{\mathbf{k}} : \text{ UC} \to \mathbb{C} \;\middle|\; u_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G} \in \text{RL}} c^{\mathbf{G}}_{\mathbf{k}} e^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}}, \int_{\text{UC}} |u_{\mathbf{k}}(\mathbf{r})|^2 \mathrm{d}^{\mathrm{d}}\mathbf{r} < \infty \right\} \tag{2.17}$$

Theorem 2.18 Using Equation (2.7) this gives us an isomorphism

$$\mathcal{H} = L^2(\text{Crystal}) \otimes \mathbb{C}^2 \simeq \left(\bigoplus_{\mathbf{k} \in \text{Q}} \mathcal{H}_{\mathbf{k}} \right) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in \text{Q}} \left(\mathcal{H}_{\mathbf{k}} \otimes \mathbb{C}^2 \right) \tag{2.18}$$

Proof 2.18 Analogous to the proof of Theorem 2.12, in combination with Theorem 2.7.

[3] In the literature one oftentimes finds $L^2(\text{Crystal}) = L^2(S^1 \times S^1 \times S^1) = L^2(\mathbb{T}^3)$ in order to appropriately reflect the periodic boundary conditions. However, this ridicules the notation, as $\mathbf{r}$ is usually denoting a spatial coordinate and not the coordinates on a torus.

[4] TBReferenced.

Fact 2.19 Using the decomposition of the unit cell into muffin-tins and interstitial regions, we obtain a corresponding decomposition of the electronic Hilbert-space

$$\mathcal{H}_{\mathbf{k}} \simeq \left(\bigoplus_{\alpha=1}^{r} L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \right) \oplus \left(L^2(\mathrm{IR}) \otimes \mathbb{C}^2 \right) \tag{2.19}$$

with

$$L^2(S^2, \mathrm{d}\Omega) = \bigoplus_{l=0}^{\infty} \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^{l} \left\{ Y_{lm} \equiv |lm\rangle \right\} \tag{2.20}$$

Notation 2.20 Let $\Psi_{N\mathbf{k}}(\mathbf{r}, \sigma) \equiv \langle \mathbf{r}, \sigma | N\mathbf{k} \rangle$ denote the single-particle wavefunction in the basis of Bloch states, and let us consider the expansion

$$\Psi_{N\mathbf{k}}(\mathbf{r}, \sigma) = \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} \sum_{|m| \leq l} \sum_{s \in \{\uparrow, \downarrow\}} c_{N\mathbf{k}}^{\alpha lms} f_{N\mathbf{k}}^{\alpha lm}(r_\alpha) Y_{lm}(\Omega_\alpha) \chi_s(\sigma) \mathbb{1}_{\mathrm{MT}_\alpha}(\mathbf{r}) + \sum_{\mathbf{G} \in \mathrm{RL}} \sum_{s \in \{\uparrow, \downarrow\}} c_{N\mathbf{k}}^{\mathbf{G}s} \chi_s(\sigma) e^{\mathrm{i}(\mathbf{k}+\mathbf{G}) \cdot \mathbf{r}} \mathbb{1}_{\mathrm{IR}}(\mathbf{r}) \tag{2.21}$$

Remark 2.20.1 Summarizing, we have in the spin-independent case the following hierarchy of Hilbert-(sub)spaces:

$$\begin{aligned}
&\mathcal{H} \equiv L^2(\mathrm{Crystal}) \otimes \mathbb{C}^2 \simeq \bigoplus_{\mathbf{k} \in \mathrm{Q}} \mathcal{H}_{\mathbf{k}} \\
&\bigcup \\
&\mathcal{H}_{\mathbf{k}} = L^2(\mathrm{UC}) \otimes \mathbb{C}^2 = \left[\left(\bigoplus_{\alpha=1}^{r} L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \right) \oplus \left(L^2(\mathrm{IR}) \otimes \mathbb{C}^2 \right) \right] \\
&\bigcup \\
&\mathcal{H}_\alpha \equiv L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \\
&\bigcup \\
&\mathcal{H}_{\alpha LS} \equiv L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 = \bigoplus_{l=0}^{\infty} \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^{l} \bigcup_{\sigma \in \{\uparrow, \downarrow\}} \left\{ Y_{lm} \otimes \chi_\sigma \equiv |lm\sigma\rangle \right\} \\
&\bigcup \\
&\mathcal{H}_{\alpha L(l)S} \equiv \mathrm{span}_{\mathbb{C}} \bigcup_{m=-l}^{l} \bigcup_{\sigma \in \{\uparrow, \downarrow\}} \left\{ Y_{lm} \otimes \chi_\sigma \equiv |lm\sigma\rangle \right\}
\end{aligned}$$

3. Fundamentals on Special Functions

3.1. Wigner-3j-Symbols

Definition 3.1 We consider three (quantum) numbers $j_1, j_2, j_3 \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ subject to the *triangle conditions*

$$|j_1 - j_2| \leq j_3 \leq j_1 + j_2, \quad j_1 + j_2 + j_3 \in \mathbb{Z}.$$

we define the *triangle coefficient* as

$$\Delta(j_1, j_2, j_3) = \sqrt{\frac{(j_1 + j_2 - j_3)!\,(j_1 - j_2 + j_3)!\,(-j_1 + j_2 + j_3)!}{(j_1 + j_2 + j_3 + 1)!}}.$$

Definition 3.2 For $m_i \in \{-j_i, -j_i+1, \dots, j_i-1, j_i\}$, such that $m_1+m_2+m_3 = 0$ we introduce the *Wigner* $3j$*-symbol* through the *Racah-formula*

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1-j_2-m_3}\,\Delta(j_1, j_2, j_3)\,\sqrt{(j_1 + m_1)!\,(j_1 - m_1)!\,(j_2 + m_2)!\,(j_2 - m_2)!\,(j_3 + m_3)!\,(j_3 - m_3)!}$$

$$\times \sum_z \frac{(-1)^z}{z!\,(j_1 + j_2 - j_3 - z)!\,(j_1 - m_1 - z)!\,(j_2 + m_2 - z)!\,(j_3 - j_2 + m_1 + z)!\,(j_3 - j_1 - m_2 + z)!}.$$

whereas the summation index z is running over all integers such that all factorial arguments are nonnegative, i.e.

$$\begin{aligned} &z \geq 0, \\ &z \leq j_1 + j_2 - j_3, \\ &z \leq j_1 - m_1, \\ &z \leq j_2 + m_2, \\ &z \geq j_2 - j_3 - m_1, \\ &z \geq j_1 - j_3 + m_2. \end{aligned}$$

Remark 3.2.1 Notice that this implies, that the Wigner-3j-symbols are real-valued.

Remark 3.2.2 These definitions and formulae seem hard to digest, and they are. However, they are implemented in many programming languages and software packages, and we will always use them as "black-boxes".

Lemma 3.3 For the 3-j-symbols

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \tag{3.1}$$

one obtains …

- the *selection-rule*

$$\begin{aligned} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} &\neq 0 \iff m_1 + m_2 + m_3 &&= 0 \\ \iff \qquad m_1 + m_2 + m_3 &\neq 0 \iff \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} &&:= 0 \end{aligned} \tag{3.2}$$

- the *triangle-condition*

$$\text{if } |j_1 - j_2| \leq j_3 \leq j_1 + j_2 \quad \text{then} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \neq 0 \tag{3.3}$$

- the *permutation-symmetry*: For any permutation $\sigma \in \mathrm{S}_3$ one finds

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \mathrm{sign}(\sigma)^{j_1+j_2+j_3} \begin{pmatrix} j_{\sigma(1)} & j_{\sigma(2)} & j_{\sigma(3)} \\ m_{\sigma(1)} & m_{\sigma(2)} & m_{\sigma(3)} \end{pmatrix} \tag{3.4}$$

- the *"inversion-antisymmetry"*:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \tag{3.5}$$

Proof 3.3 TBReferenced

Theorem 3.4 One finds the *orthogonality relations*

$$\begin{aligned} (2j_3 + 1) \sum_{m_1, m_2} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3' \\ m_1 & m_2 & m_3' \end{pmatrix} &= \delta_{j_3, j_3'} \delta_{m_3, m_3'} \{j_1 j_2 j_3\} \\ \sum_{j_3, m_3} (2j_3 + 1) \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1' & m_2' & m_3 \end{pmatrix} &= \delta_{m_1, m_1'} \delta_{m_2, m_2'} \end{aligned} \tag{3.6}$$

whereas the *triangular delta* $\{j_1 j_2 j_3\}$ is equal to 1 when the triple (j_1, j_2, j_3) fulfills the triangle condition in Equation (3.3) and is equal to 0 otherwise.

Proof 3.4 TBReferenced, Probably Judd.

3.2. Special Functions 1: Associated Legendre-Polynomials and Spherical Harmonics

Definition 3.5 We define the *associated Legendre-polynomials* as functions

$$P_l^m(x) = \frac{(1-x^2)^{m/2}}{2^l l!} \frac{\mathrm{d}^{l+m}}{\mathrm{d}x^{l+m}} (x^2-1)^l \tag{3.7}$$

and the *spherical harmonics* as

$$Y_{lm}(\theta, \phi) = (-1)^m \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_l^m(\cos\theta) e^{\mathrm{i}m\phi} \tag{3.8}$$

Remark 3.5.1 Notice that the spherical harmonics Y_{lm} are indexed by two quantum numbers:

- l: The *orbital angular momentum quantum number*, which can take values $0, 1, 2, \ldots$.
- m: The *magnetic quantum number*, which runs from $-l$ to $+l$, meaning there are $(2l+1)$ values of m for a given l.

Lemma 3.6 One finds the identity

$$Y_{lm}^*(\Omega) = (-1)^{-m} Y_{l,-m}(\Omega) \tag{3.9}$$

Proof 3.6 TBReferenced.

Theorem 3.7 The spherical harmonics satisfy the orthogonality relations

$$\int \mathrm{d}\Omega\, Y_{lm}(\Omega) Y_{l'm'}^*(\Omega) = \delta_{l,l'} \delta_{m,m'} \tag{3.10}$$

Proof 3.7 TBReferenced.

Lemma 3.8 One finds the identity

$$\begin{aligned} \int \mathrm{d}\Omega\, Y_{l_1 m_1}(\Omega) Y_{l_2 m_2}(\Omega) Y_{l_3 m_3}(\Omega) &= \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}} \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \\ &= (-1)^{-m_1} \int \mathrm{d}\Omega\, Y_{l_1 -m_1}^*(\Omega) Y_{l_2 m_2}(\Omega) Y_{l_3 m_3}(\Omega) \end{aligned} \tag{3.11}$$

with 3-j-symbols.

Proof 3.8 TBDigitalized.[1]

Remark 3.8.1 Notice that we find the special case

$$\sqrt{\frac{4\pi}{2k+1}} \int \mathrm{d}\Omega\, Y_{lm_1}(\Omega) Y_{kq}(\Omega) Y_{lm_2}(\Omega) = (2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ m_1 & q & m_2 \end{pmatrix} \tag{3.12}$$

Remark 3.8.2 Notice that combining the selection-rule in Equation (3.2) with Equation (3.11), we are able to write down the proportionality

$$\int \mathrm{d}\Omega\, Y^*_{l_1-m_1}(\Omega) Y_{l_2m_2}(\Omega) Y_{l_3m_3}(\Omega) \propto \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \delta_{m_1+m_2+m_3,0} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \tag{3.13}$$

and thus we are able to write down the equation

$$\int \mathrm{d}\Omega\, Y^*_{l_1-m_1}(\Omega) Y_{l_2m_2}(\Omega) Y_{l_3m_3}(\Omega) = \delta_{m_1+m_2+m_3,0} \int \mathrm{d}\Omega\, Y^*_{l_1-m_1}(\Omega) Y_{l_2m_2}(\Omega) Y_{l_3m_3}(\Omega). \tag{3.14}$$

Remark 3.8.3 Notice further that the triangle-condition in Equation (3.3) implied that $\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = 0$ if the triangle-condition is not being fulfilled. Thus one obtains for the special case $\begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} l & l & k \\ 0 & 0 & 0 \end{pmatrix}$ the triangle-condition $0 = |l - l| \leq k \leq l + l = 2l$, i.e. $k = 0, 1, 2, \ldots, 2l$.

Remark 3.8.4 Notice lastly, that the odd-permutation-symmetry in Equation (3.4) gives the important special case

$$\begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} = (-1)^{2l+k} \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \quad \Longleftrightarrow \quad \left(1 - (-1)^{2l+k}\right) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} = 0 \tag{3.15}$$

Since $2l$ is always an even number, we have the implication

$$k \text{ is odd} \quad \Rightarrow \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} = 0 \tag{3.16}$$

Definition 3.9 For later use, we introduce the *spherical tensors* as

$$C^k_q(\Omega) \equiv \sqrt{\frac{4\pi}{2k+1}} Y_{kq}(\Omega) \tag{3.17}$$

[1] Cf. for example `https://physics.stackexchange.com/questions/10039/integral-of-the-product-of-three-spherical-h`

Lemma 3.10 We define the *Gaunt-coefficients* $c^{(k)}$ as

$$c_l^{(k)}(m_1, m_2) := \left(\frac{4\pi}{2k+1}\right)^{1/2} \int \mathrm{d}\Omega\, Y^*_{lm_1}(\Omega) Y_{k,m_1-m_2}(\Omega) Y_{lm_2}(\Omega) \tag{3.18}$$

and find the identity

$$c_l^{(k)}(m_1, m_2) = (-1)^{m_1}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_1 & m_1 - m_2 & m_2 \end{pmatrix} \tag{3.19}$$

with Wigner-$3j$-Symbols in parenthesis.

Proof 3.10 See Appendix, Proofs, page 105.

Remark 3.10.1 Notice that combining the two identities yields the identity

$$c_l^{(k)}(-m_1, m_2) = c_l^{(k)}(-m_1, m_2) \iff \text{TBDone?} \tag{3.20}$$

Lemma 3.11 Notice that one obtains the auxiliary-identity

$$c_l^{(k)}(m_1, m_3) = (-1)^{m_1 - m_3} c_l^{(k)}(m_3, m_1) \tag{3.21}$$

Proof 3.11 See Appendix, Proofs, page 105.

Lemma 3.12 A common way to *linearize* the indices of spherical harmonics into a single index lm (not a product) is:

$$lm = l^2 + m + l + 1.$$

Implying

- When $l = 0$, $m = 0$ gives $lm = 0^2 + 0 + 0 + 1 = 1$.
- When $l = 1$, $m = -1, 0, +1$ gives lm running from $1^2 + 1 = 2$ to $(1+1)^2 = 4$.
- When $l = 2$, $m = -2, -1, 0, +1, +2$ gives lm running from $2^2 + 1 = 5$ to $(2+1)^2 = 9$.

Proof 3.12 TBReferenced.

Remark 3.12.1 Some example-values are listed in the table.

l	Possible m values	Range of lm (from l^2+1 to $(l+1)^2$)
0	0	1 to 1
1	$-1, 0, +1$	2 to 4
2	$-2, -1, 0, +1, +2$	5 to 9
3	$-3, -2, -1, 0, +1, +2, +3$	10 to 16

Lemma 3.13 For

$$l \in \{0, 1, 2, \dots\} \tag{3.22}$$

one has

$$lm \in \{l^2+1, l^2+2, \dots, (l+1)^2\} \tag{3.23}$$

Proof 3.13 TBReferenced.

Remark 3.13.1 In order to enumerate all possible values of l and m in a loop, one thus often encounters in the ELK-Fortran-code loops like:

```
do l=0,lmaxo
  do lm=l**2+1,(l+1)**2
```

Theorem 3.14 Every potential V can be expanded in spherical harmonics

$$V(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} V_{lm}(r) Y_{lm}(\theta, \phi)$$

with coefficients that are determined by overlaps

$$V_{lm}(r) = \int_0^{2\pi} \int_0^{\pi} V(r, \theta, \phi) Y_{lm}^*(\theta, \phi) \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi$$

Proof 3.14 TBReferenced.

Remark 3.14.1 The mapping $r \mapsto V_{l=m=0}(r) = V_{00}(r)$ is the *spherically symmetric part* of the potential and is oftentimes called the *spherical component*.

3.3. Special Functions 2: Radial Schrödinger-like-Equations for Spherically Symmetric Potentials

TBImplemented and read over.

3.4. Special Functions 3: Bessel-Functions

Definition 3.15 For integers $n \geq 0$, we define the *spherical Bessel functions of the first kind* $j_n(x)$ and the *spherical Bessel functions of the second kind* $y_n(x)$ as

$$j_n(x) = \frac{x^n}{(2n+1)!!}\left(-\frac{1}{x}\frac{d}{dx}\right)^n \frac{\sin x}{x},$$

$$y_n(x) = -\frac{x^n}{(2n+1)!!}\left(-\frac{1}{x}\frac{d}{dx}\right)^n \frac{\cos x}{x},$$

whereas $(2n+1)!! = (2n+1)(2n-1)(2n-3)\cdots 1$ is the *double factorial*.

Theorem 3.16 The spherical Bessel functions (i.e. both, the ones of the first kind and the ones of the second kind) satisfy *the spherical Bessel differential equation*:

$$x^2\left(\frac{d^2y}{dx^2}\right)(x) + 2x\left(\frac{dy}{dx}\right)(x) + \left[x^2 - n(n+1)\right]y(x) = 0.$$

Proof 3.16 TBReferenced.

Definition 3.17 We define the *the standard Bessel functions of the first* $J_\nu(x)$ and *the standard Bessel functions of the second kind* $Y_\nu(x)$ of order $\nu = n + \frac{1}{2}$ as

$$j_n(x) =: \sqrt{\frac{\pi}{2x}}J_{n+\frac{1}{2}}(x), \quad y_n(x) =: \sqrt{\frac{\pi}{2x}}Y_{n+\frac{1}{2}}(x),$$

Theorem 3.18 The standard Bessel functions of order $\nu = n + \frac{1}{2}$ are explicitly given by:

$$J_{n+\frac{1}{2}}(x) = \frac{1}{\pi}\int_0^\pi \cos\left(x\sin\theta - \left(n+\frac{1}{2}\right)\theta\right)\,\mathrm{d}\theta,$$

$$Y_{n+\frac{1}{2}}(x) = -\frac{1}{\pi}\int_0^\pi \sin\left(x\sin\theta - \left(n+\frac{1}{2}\right)\theta\right)\,\mathrm{d}\theta.$$

Proof 3.18 TBReferenced.

Theorem 3.19 A plane wave can be expanded in terms of spherical harmonics and Bessel functions as

$$e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \mathrm{i}^l j_l(|\mathbf{k}|r) Y_{lm}^*(\hat{\mathbf{k}}) Y_{lm}(\hat{\mathbf{r}}) \tag{3.24}$$

with normalized vectors

$$\hat{\mathbf{k}} \equiv \frac{\mathbf{k}}{k}, \qquad \hat{\mathbf{r}} \equiv \frac{\mathbf{r}}{r} \tag{3.25}$$

Proof 3.19 TBReferenced.

3.5. Special Functions 4: The (Linearized) Augmented Plane Waves

Notation 3.20 We introduce the *convergence-generating parameter*[2]

$$\mathrm{rgkmax} \approx 7. \tag{3.26}$$

Definition 3.21 For each $\mathbf{k} \in \mathrm{Q}$, we introduce a [C]ircle [i]n [R]eciprocal [L]attice around $\mathbf{k}$ as the set

$$\mathrm{CiR}_{\mathbf{k}}\mathrm{L} \equiv \left\{ \mathbf{G} \in \mathrm{RL}_{\infty} \;\middle|\; |\mathbf{k}+\mathbf{G}| \leq \frac{\mathrm{rgkmax}}{R_{\mathrm{MT}_{\mathrm{min}}}} \right\} \tag{3.27}$$

with number of elements

$$\mathrm{ngk} \equiv |\,\mathrm{CiR}_{\mathbf{k}}\mathrm{L}\,| \tag{3.28}$$

which represents the number of plane-waves that we use for each wavevector to solve the first-variational problem.

Definition 3.22 For $\mathbf{k} \in \mathrm{Q}$ and $\mathbf{G} \in \mathrm{CiR}_{\mathbf{k}}\mathrm{L}$ and for position vectors within the first unit cell $\mathbf{r} \in \mathrm{UC}$ we introduce functions as

$$\phi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}) = \begin{cases} \dfrac{1}{\sqrt{\Omega}} e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{r}}, & \mathbf{r} \in \mathrm{IR}, \\ \displaystyle\sum_{\substack{l=0,1,\ldots,\mathrm{lmaxo} \\ m=-l,\ldots,+l \\ j=1,\ldots,M_l^{\alpha}}} A^{\alpha}_{jlm}(\mathbf{G}+\mathbf{k}) u^{\alpha}_{jl}(r_{\alpha}) Y_{lm}(\hat{\mathbf{r}}_{\alpha}), & \mathbf{r} \in \mathrm{MT}_{\alpha}, \end{cases} \tag{3.29}$$

whereas $u_l(r)$ is the regular solution[3] of the radial Schrödinger-equation

$$\left[-\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{l(l+1)}{r^2} + V_{00}(r) - E_l \right] r u_l(r) = 0, \tag{3.30}$$

with $V_{00} \equiv V_{l=m=0}$ representing the spherically symmetric part of the potential and the so-called *linearization energies* E_l, and the coefficients $A^{\alpha}_{jlm}(\mathbf{G}+\mathbf{k})$ are chosen such that continuity is guaranteed up to the $\left(M_l^{\alpha} - 1\right)$ th order[4].

Remark 3.22.1 For position-vectors outside the unit-cell $\tilde{\mathbf{r}} = \mathbf{r} + \mathbf{R}_{\mathbf{n}}$ with $\mathbf{r} \in \mathrm{UC}$ and $\mathbf{R}_{\mathbf{n}} \in \mathrm{L_f}$, we define the (L)APW-functions via the Bloch-theorem

$$\phi_{\mathbf{k}+\mathbf{G}}(\tilde{\mathbf{r}}) := \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}_{\mathbf{n}}} \phi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}). \tag{3.31}$$

[2] Cf. the ELK.pdf for the definition of rgkmax.

[3] Cf. page 43 in the ELK-book; *Regular* means here that the solution is well behaved at the origin $r = 0$, implying that it is finite / non-singular.

[4] The explicit values of these coefficients is given in Theorem 3.23 below.

Remark 3.22.2 If we have $\left(M_l^\alpha - 1\right) = 1$ for all l and α, we call these functions the *augmented plane waves* (APW-functions). If we have $\left(M_l^\alpha - 1\right) = 2$ for all l and α, we call these functions the *linearized augmented plane waves* (LAPW-functions). Accordingly one says that one has (L)APW-functions for certain species etc.

Remark 3.22.3 Notice that lmaxo represents a cut-off[5] for the angular momentum quantum number l in the spherical harmonics expansion within the muffin-tin spheres.

Remark 3.22.4 The linearization-energies E_l are chosen such that TBFigured Out. [6]

Remark 3.22.5 Maybe / probably the differential Equation (3.30) is not the one that is implemented in the ELK-code. Maybe / probably it is rather the one from "D D Koelling and B N Harmon 1977 J. Phys. C: Solid State Phys. 10 3107". TBFigured Out.

Theorem 3.23 Ensuring continuity up to the $\left(M_l^\alpha - 1\right)$ the derivative across the muffin-tin boundary therefore requires that the matching coefficients satisfy[7]

$$\sum_{j=1}^{M_l^\alpha} D_{ij} A^\alpha_{jlm}(\mathbf{G}+\mathbf{k}) = b_i \tag{3.32}$$

whereas

$$D_{ij} = \left.\frac{d^{i-1}u^\alpha_{jl}(r)}{dr^{i-1}}\right|_{r=R_\alpha} \qquad \text{and} \qquad b_i = \frac{4\pi \mathrm{i}^l}{\sqrt{\Omega}} |\mathbf{G}+\mathbf{k}|^{i-1} j_l^{(i-1)}\left(|\mathbf{G}+\mathbf{k}|R_\alpha\right) \mathrm{e}^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{r}_\alpha} Y^*_{lm}(\widehat{\mathbf{G}+\mathbf{k}}), \tag{3.33}$$

with $\mathbf{r}_\alpha$ the atomic position and R_α the muffin-tin radius.

Proof 3.23 See Appendix, Proofs, page 105.

Definition 3.24 Formalizing the above, we introduce the (L)APW basis as

$$\mathbb{B}_{\mathrm{APW}} = \bigcup_{\mathbf{k}\in\mathrm{Q}} \bigcup_{\mathbf{G}\in\mathrm{CiR}_\mathbf{k}\mathrm{L}} \{\phi_{\mathbf{k}+\mathbf{G}}\} \equiv \bigcup_{\mathbf{k}\in\mathrm{Q}} \mathbb{B}_{\mathrm{APW}}(\mathbf{k}). \tag{3.34}$$

[5] As a rule of thumb (cf. Fleur-Documentation, page 23.) one uses lmaxo $\approx \left\lfloor |\mathbf{k}+\mathbf{G}|_{\max} R_{\mathrm{MT}_{\min}} \right\rfloor \approx 7$.

[6]

- Cf. Elk-book page 45: *Accordingly, E_l must be set equal to the band energy.* (Why?)
- The solutions to these functions are orthogonal to each other, (why?) cf. ELK-book page 43 and description of the subroutine genapwfr.f90 in the elk-documentation (version 9.5.14).
  This needs to be reconsidered, since in the ELK-code one solves other radial equations than the ones indicated here...
- Cf. Fleur-Documentation, page 23. Cf. further the variable gkmax in ELK.

[7] Note that this is in perfect agreement with the subroutine-description match.f90 in the elk-documentation for the version 9.5.14.

3.6. Special Functions 5: The Local Orbitals inside the Muffin-Tins

Notation 3.25 For each atom $\alpha \in \{1, \dots, r\}$ and [a]ngular [m]omentum [q]uantum [n]umber (*amqn*) $l \in \{0, 1, 2, \dots\}$, we introduce the number

$$n_{\mathrm{LO}}(l, \alpha) = \text{"the number of local orbital radial wavefunctions for atom } \alpha \text{ with amqn } l\text{"}, \tag{3.35}$$

whereas we denote the *[l]ocal [o]rbital [r]adial [f]unctions (lorf)* as

$$\left\{ r \mapsto R_l^{\alpha,i}(r) \equiv R_\lambda(r) \quad | \quad \lambda = (l, i, \alpha) \in \{0, 1 \dots, \} \times \{1, \dots, n_{\mathrm{LO}}(l, \alpha)\} \times \{1, \dots, r\} \equiv \Lambda \right\}. \tag{3.36}$$

For each lorf R_λ, we introduce auxiliary[8] functions

$$r \mapsto u_{\lambda, E_{1,\lambda}}(r) \quad \text{satisfying Equation (3.30)} \tag{3.37}$$

$$r \mapsto \dot{u}_{\lambda, E_{1,\lambda}}(r) \quad \text{satisfying the Equation} \left[-\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{l(l+1)}{r^2} + V(r) - E_\ell \right] r u_\ell(r) = r\, u_\ell(r), \tag{3.38}$$

$$r \mapsto u_{\lambda, E_{2,\lambda}}(r) \quad \text{satisfying Equation (3.30)} \tag{3.39}$$

from this, we construct the local orbital radial functions as linear combinations

$$R_\lambda(r) := a_\lambda u_{\lambda, E_{1,\lambda}}(r) + b_\lambda \dot{u}_{\lambda, E_{1,\lambda}}(r) + c_\lambda u_{\lambda, E_{2,\lambda}}(r) \tag{3.40}$$

and choose the coefficients[9], such that $R_\lambda(r = R_{\mathrm{MT}_\alpha}) = 0$.

Remark 3.25.1 Notice that one obtains Equation (3.38) by differentiating Equation (3.30) with respect to the energy E_l.

Remark 3.25.2 Whether the coefficients a_λ, b_λ and c_λ can be unequal to zero or not is specified in the respective species files.

Definition 3.26 For each $(l, i, \alpha) \equiv \lambda \in \Lambda$ we construct local orbital-wavefunctions

$$\left\{ (r, \phi, \theta) \mapsto R_l^{\alpha,i}(r) Y_{lm}(\phi, \theta) \equiv \phi_{l,m}^{\alpha,i}(r, \phi, \theta) \quad | \quad m \in \{-l, \dots, +l\} \right\} \tag{3.41}$$

equipping us with the part of the basis consisting of (spin-independent) local orbitals

[8] What boundary-conditions do these functions satisfy?

[9] Note that we also allow the coefficients to be zero, depending on the species file.

$$\begin{aligned}\mathbb{B}_{\mathrm{LO}} &= \bigcup_{\alpha=1}^{r}\bigcup_{l=0}^{\infty}\bigcup_{m=-l}^{+l}\bigcup_{i=1}^{n_{\mathrm{LO}}(l,\alpha)}\left\{(r,\phi,\theta)\mapsto\phi_{l,m}^{\alpha,i}(r,\phi,\theta)\right\}\\ &\equiv\left\{\phi_1,\phi_2,\dots,\phi_{n_{\mathrm{LO}}}\right\}\\ &=\bigcup_{j=1}^{n_{\mathrm{LO}}}\left\{(r,\phi,\theta)\mapsto\phi_j(r,\phi,\theta)\right\}\end{aligned}\tag{3.42}$$

with

$$n_{\mathrm{LO}} = \text{"the total number of local orbitals"}.\tag{3.43}$$

3.7. Fundamentals on Species Files

Theorem 3.27 For some angular momentum quantum number

$$l\in\left\{0,1,2,\dots\right\}$$

one finds the *coupled angular momentum quantum numbers*

$$j\in\left\{|l-1/2|,\dots,|l+1/2|\right\}$$

and the corresponding *coupled magnetic quantum numbers*

$$m\in\left\{-j,-j+1,\dots,j-1,j\right\}$$

.

Proof 3.27 TBReferenced.

Notation 3.28 We introduce the number κ as

$$\kappa\equiv\kappa(j):=j+1/2$$

Remark 3.28.1 TBD: Finding a reference for the derivation of the Dirac-equation in which this notation is common.

Example: The Species-File for Ruthenium

n	l	Character	j	occ	κ
1	0	s	$1/2$	2	1
2	0	s	$1/2$	2	1
2	1	p	$1/2$	2	1
2	1	p	$3/2$	4	2
3	0	s	$1/2$	2	1
3	1	p	$1/2$	2	2
3	2	d	$3/2$	4	2
3	2	d	$5/2$	6	3
4	0	s	$1/2$	2	1

4. Algorithmic Procedure in the implementation

4.1. The First Variational Problem ("LDA")

Definition 4.1 The *first variational problem* a.k.a. *[L]ocal [D]ensity [A]pproximation*, is a single, spinless electron subjected to a lattice-periodic potential.

Notation 4.2 We consider a scalar potential, which is made up from three contributions:

- The *ionic contribution* V_{ion} (originating from the Coulomb-interactions of the nuclei),
- the *Hartree-contribution* V_{H},
- and the *exchange-correlation / Fock-contribution* V_{xc},
- the Darwin-term $V_{\text{D}}(\mathbf{r}) \equiv \frac{\hbar^2}{8m^2c^2}\left(\Delta V\right)(\mathbf{r})$
- the relativistic corrections $V_{\text{rc}}(\mathbf{r}) \equiv mc^2 + \frac{p^4}{8m^3c^2}$

which are combined to form the *total scalar potential of the first variation* V_0 as

$$V_{\text{fv}}(\mathbf{r}) \equiv V_0(\mathbf{r}) = \left(V_{\text{ion}} + V_{\text{H}} + V_{\text{xc}} + V_{\text{D}} + V_{\text{rc}}\right)(\mathbf{r}).$$

Definition 4.3 We consider the *differential equation of the first-variational problem* a.k.a. the *scalar Schrödinger-Kohn-Sham-equations*

$$\left(-\frac{\hbar^2}{2m}\Delta + V_0(\mathbf{r})\right)\psi_{\text{fv}}(\mathbf{r}) \equiv \left(H_0\psi_{\text{fv}}\right)(\mathbf{r}) = E_{\text{fv}}\psi_{\text{fv}}(\mathbf{r}). \tag{4.1}$$

Fact 4.4 In the ELK-Code, one uses the (L)APW-functions in Equation (3.29) together with local orbitals from Equation (3.42) as basis-functions for the solution of the first-variational problem, i.e. one takes the bases

$$\mathbb{B}_{\text{LO}} = \bigcup_{j=1}^{n_{\text{LO}}}\left\{\phi_j\right\}, \qquad \mathbb{B}_{\text{APW}} = \bigcup_{\mathbf{k}\in\text{Q}}\bigcup_{\mathbf{G}\in\text{CiR}_{\mathbf{k}}\text{L}}\left\{\phi_{\mathbf{k}+\mathbf{G}}\right\} \equiv \bigcup_{\mathbf{k}\in\text{Q}}\mathbb{B}_{\text{APW}}(\mathbf{k}), \qquad \mathbb{B}_{\text{APW+LO}} = \mathbb{B}_{\text{LO}} \cup \mathbb{B}_{\text{APW}} \tag{4.2}$$

such that the single-particle Hilbert-space of the first variational problem[1] $\mathcal{H}_0$ becomes approximately

[1] The "zero" as index, refers here to a "simple problem" which can be solved, and the solutions of which will be used to tackle a "more complicated" problem. This is standard-notation in physics.

$$\mathcal{H}_0 = L^2(\text{Crystal}) \approx \text{span}_{\mathbb{C}}\, \mathbb{B}_{\text{APW+LO}} \tag{4.3}$$

Notation 4.5 For each $\mathbf{k} \in \text{Q}$, we introduce the basis

$$\mathbb{B}_{\text{LO}} = \bigcup_{j=1}^{n_{\text{LO}}} \{\phi_j\}, \qquad \mathbb{B}_{\text{APW}}(\mathbf{k}) = \bigcup_{\mathbf{G}\in \text{CiR}_{\mathbf{k}}\text{L}} \{\phi_{\mathbf{k}+\mathbf{G}}\}, \qquad \mathbb{B}_{\text{APW+LO}}(\mathbf{k}) = \mathbb{B}_{\text{LO}} \cup \mathbb{B}_{\text{APW}}(\mathbf{k}) \equiv \bigcup_{j=1}^{n_{\text{LO}}+\text{ngkmax}} \left\{ \psi^{\text{apw+lo}}_{j\mathbf{k}} \right\} \tag{4.4}$$

and the *Hamiltonian- and overlap-matrix w.r.t.* $\mathbb{B}(\mathbf{k})$ as

$$H_0(\mathbf{k}) = \text{mat}_{\mathbb{B}_{\text{APW+LO}}(\mathbf{k})}\, H_0, \qquad S(\mathbf{k}) = \text{mat}_{\mathbb{B}_{\text{APW+LO}}(\mathbf{k})}\, S(\mathbf{k}) \tag{4.5}$$

Remark 4.5.1 TBFigured out: Notice that this decomposition of the total basis is actually kind of puzzling... When the basis-functions contain "plane-wave-factors", one can use the Bloch-theorem and can focus on a Schrödinger-equation for each $\mathbf{k}$-point. The local orbitals however dont have a plane-wave-factor... So in this sense, we are using the local orbitals as basis-functions "too often". (?)

Theorem 4.6 Using the approximation in Equation (4.3), and using the Bloch-theorem, we can find all solutions of Equation (4.1), by solving for each $\mathbf{k} \in \text{Q}$ the generalized eigenvalue-equation

$$H_0(\mathbf{k})\mathbf{C}^{\text{fv}\to\text{apw+lo}}_{\bullet}(\mathbf{k}) = E_{\text{fv},\bullet}(\mathbf{k})S(\mathbf{k})\mathbf{C}^{\text{fv}\to\text{apw+lo}}_{\bullet}(\mathbf{k}). \tag{4.6}$$

whereas $\mathbf{C}^{\text{fv}\to\text{apw+lo}}(\mathbf{k})$ is a complex vector with $\text{ngkmax} + n_{\text{LO}}$ entries.

Proof 4.6 See Appendix, Proofs, page 106.

Remark 4.6.1 Notice that Equation (4.6) can be rewritten as a "usual" eigenvalue-problem by simple matrix-inversion

$$H_0(\mathbf{k})\mathbf{C}^{\text{fv}\to\text{apw+lo}}_{\bullet}(\mathbf{k}) = E_{\text{fv},\bullet}(\mathbf{k})S(\mathbf{k})\mathbf{C}^{\text{fv}\to\text{apw+lo}}_{\bullet}(\mathbf{k}). \quad \Longleftrightarrow \quad S^{-1}(\mathbf{k})H_0(\mathbf{k})\mathbf{C}^{\text{fv}\to\text{apw+lo}}_{\bullet}(\mathbf{k}) = E_{\text{fv},\bullet}(\mathbf{k})\mathbf{C}^{\text{fv}\to\text{apw+lo}}_{\bullet}(\mathbf{k}) \tag{4.7}$$

Remark 4.6.2 We can explicitly write down the matrices $H_{\mathbf{k}}$, by choosing an enumeration of the corresponding reciprocal lattice-vectors

$$\left\{\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_{N_{\mathbf{k}}}\right\} = \text{RL}_{\text{f}}(\mathbf{k}) \subset \text{RL}_{\infty}\,. \tag{4.8}$$

Together with the enumeration of the local-orbitals $\left\{\phi_1, \dots, \phi_{n_{\text{LO}}}\right\}$ from Equation (3.42) we obtain.

$$H_0(\mathbf{k}) = \begin{pmatrix} H^{\,0}_{\mathrm{APW}^2}(\mathbf{k}) & H^{\,0}_{\text{APW-LO}}(\mathbf{k}) \\ H^{\,0}_{\text{LO-APW}}(\mathbf{k}) & H^{\,0}_{\mathrm{LO}^2}(\mathbf{k}) \end{pmatrix} \tag{4.9}$$

with

$$H^{\,0}_{\mathrm{APW}^2}(\mathbf{k}) = \begin{pmatrix} H^{\,0}_{\substack{\mathrm{APW}^2\\11}}(\mathbf{k}) & H^{\,0}_{\substack{\mathrm{APW}^2\\12}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\mathrm{APW}^2\\1N_{\mathbf{k}}}}(\mathbf{k}) \\ H^{\,0}_{\substack{\mathrm{APW}^2\\21}}(\mathbf{k}) & H^{\,0}_{\substack{\mathrm{APW}^2\\22}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\mathrm{APW}^2\\2N_{\mathbf{k}}}}(\mathbf{k}) \\ \vdots & \vdots & \ddots & \vdots \\ H^{\,0}_{\substack{\mathrm{APW}^2\\N_{\mathbf{k}}1}}(\mathbf{k}) & H^{\,0}_{\substack{\mathrm{APW}^2\\N_{\mathbf{k}}2}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\mathrm{APW}^2\\N_{\mathbf{k}}N_{\mathbf{k}}}}(\mathbf{k}) \end{pmatrix}, \quad H^{\,0}_{\text{APW-LO}}(\mathbf{k}) = \begin{pmatrix} H^{\,0}_{\substack{\text{APW-LO}\\11}}(\mathbf{k}) & H^{\,0}_{\substack{\text{APW-LO}\\12}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\text{APW-LO}\\1n_{\mathrm{LO}}}}(\mathbf{k}) \\ H^{\,0}_{\substack{\text{APW-LO}\\21}}(\mathbf{k}) & H^{\,0}_{\substack{\text{APW-LO}\\22}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\text{APW-LO}\\2n_{\mathrm{LO}}}}(\mathbf{k}) \\ \vdots & \vdots & \ddots & \vdots \\ H^{\,0}_{\substack{\text{APW-LO}\\N_{\mathbf{k}}1}}(\mathbf{k}) & H^{\,0}_{\substack{\text{APW-LO}\\N_{\mathbf{k}}2}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\text{APW-LO}\\N_{\mathbf{k}}n_{\mathrm{LO}}}}(\mathbf{k}) \end{pmatrix} \tag{4.10}$$

$$H^{\,0}_{\text{LO-APW}}(\mathbf{k}) = \begin{pmatrix} H^{\,0}_{\substack{\text{LO-APW}\\11}}(\mathbf{k}) & H^{\,0}_{\substack{\text{LO-APW}\\12}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\text{LO-APW}\\1N_{\mathbf{k}}}}(\mathbf{k}) \\ H^{\,0}_{\substack{\text{LO-APW}\\21}}(\mathbf{k}) & H^{\,0}_{\substack{\text{LO-APW}\\22}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\text{LO-APW}\\2N_{\mathbf{k}}}}(\mathbf{k}) \\ \vdots & \vdots & \ddots & \vdots \\ H^{\,0}_{\substack{\text{LO-APW}\\n_{\mathrm{LO}}1}}(\mathbf{k}) & H^{\,0}_{\substack{\text{LO-APW}\\n_{\mathrm{LO}}2}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\text{LO-APW}\\n_{\mathrm{LO}}N_{\mathbf{k}}}}(\mathbf{k}) \end{pmatrix}, \quad H^{\,0}_{\mathrm{LO}^2}(\mathbf{k}) = \begin{pmatrix} H^{\,0}_{\substack{\mathrm{LO}^2\\11}}(\mathbf{k}) & H^{\,0}_{\substack{\mathrm{LO}^2\\12}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\mathrm{LO}^2\\1n_{\mathrm{LO}}}}(\mathbf{k}) \\ H^{\,0}_{\substack{\mathrm{LO}^2\\21}}(\mathbf{k}) & H^{\,0}_{\substack{\mathrm{LO}^2\\22}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\mathrm{LO}^2\\2n_{\mathrm{LO}}}}(\mathbf{k}) \\ \vdots & \vdots & \ddots & \vdots \\ H^{\,0}_{\substack{\mathrm{LO}^2\\n_{\mathrm{LO}}1}}(\mathbf{k}) & H^{\,0}_{\substack{\mathrm{LO}^2\\n_{\mathrm{LO}}2}}(\mathbf{k}) & \dots & H^{\,0}_{\substack{\mathrm{LO}^2\\n_{\mathrm{LO}}n_{\mathrm{LO}}}}(\mathbf{k}) \end{pmatrix} \tag{4.11}$$

whereas $H^{\,0}_{\substack{\mathrm{APW}^2\\ij}}(\mathbf{k}) = \langle \phi_{\mathbf{k}+\mathbf{G}_i} | H_0 | \phi_{\mathbf{k}+\mathbf{G}_j} \rangle$, etc.

Remark 4.6.3 Since the Hamiltonian-matrix in Equation (4.9) has, w.r.t. to the basis in Equation (4.4) the dimensions $(\mathrm{ngkmax} + n_{\mathrm{LO}}) \times (\mathrm{ngkmax} + n_{\mathrm{LO}})$, we can enumerate the eigenvalues and eigenvectors in Equation (4.6) as

$$\left\{ E_{\mathrm{fv},n}(\mathbf{k}) \quad | \quad n \in \{1, \dots, \mathrm{ngkmax} + n_{\mathrm{LO}}\} \right\} \tag{4.12}$$

and

$$\left\{ \mathbf{C}_n^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k}) \quad | \quad n \in \{1, \dots, \mathrm{ngkmax} + n_{\mathrm{LO}}\} \right\} \tag{4.13}$$

whereas $\mathbf{C}_n^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k})$ is a vector with $\mathrm{ngkmax} + n_{\mathrm{LO}}$ columns for each n and $\mathbf{k}$ with elements

$$\left(\mathbf{C}_n^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k}) \right)_{n'} \equiv C_{n,n'}^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k}) \tag{4.14}$$

meaning that the *columns* of the matrix $\mathbf{C}^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k})$ are eigenvectors of the generalized eigenvalue problem in Equation (4.6).

Remark 4.6.4 Usually one keeps only a (**k**-*independent*)small number $\mathrm{nstfv} < \mathrm{ngkmax} + n_{\mathrm{LO}}$ of eigenvectors and eigenvalues, which are the lowest in energy.

Remark 4.6.5 Notice that hence we can expand each eigenstate of the first variational problem as

$$|\psi_{\substack{\mathrm{fv}\\ n\mathbf{k}}}\rangle = \sum_{n'=1}^{\mathrm{ngkmax}+n_{\mathrm{LO}}} C_{n,n'}^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k})|\psi_{\substack{\mathrm{apw+lo}\\ n'\mathbf{k}}}\rangle = \sum_{n'=1}^{\mathrm{ngkmax}+n_{\mathrm{LO}}} C_{n,n'}^{\mathrm{fv}\to\mathrm{apw+lo}}(\mathbf{k})|\psi_{\substack{\mathrm{apw+lo}\\ n'\mathbf{k}}}\rangle \tag{4.15}$$

4.2. The Second Variational Problem ("L(S)DA + SOC")

Definition 4.7 The *second variational problem* a.k.a. *[L]ocal [S]pin [D]ensity [A]pproximation with [S]pin [O]rbit [C]oupling*, is a single, spin-$\frac{1}{2}$-electron subjected to a lattice-periodic, spinor-valued potential.

Definition 4.8 We start from the single-particle Pauli-equation with the remaining relativistic corrections[2]:

$$\left(H\Psi_{\mathrm{sv}}\right)(\mathbf{r},s) = E_{\mathrm{sv}}\Psi_{\mathrm{sv}}(\mathbf{r},s), \tag{4.16}$$

whereas $s \in \{\uparrow,\downarrow\}$ and the Hamiltonian is given by

$$\begin{aligned} H &= \frac{\hat{\mathbf{p}}^2}{2m} + V_{\mathrm{fv}}(\mathbf{r}) \quad + H_{\mathrm{soc}}(\mathbf{r}) + \mathbf{V}_{\mathrm{xc}}(\mathbf{r})\cdot\boldsymbol{\sigma}, \\ &\equiv \qquad H_0 \qquad\quad + H_{\mathrm{soc}}(\mathbf{r}) + \mathbf{V}_{\mathrm{xc}}(\mathbf{r})\cdot\boldsymbol{\sigma}, \end{aligned} \tag{4.17}$$

and the spin-orbit interaction term reads

$$H_{\mathrm{soc}}(\mathbf{r}) = \frac{\hbar}{4m^2c^2}\,\boldsymbol{\sigma}\cdot\left(\boldsymbol{\nabla}V_{\mathrm{fv}}(\mathbf{r})\times\hat{\mathbf{p}}\right), \tag{4.18}$$

Notation 4.9 For each $\mathbf{k} \in \mathrm{Q}$, we now use the Bloch-wavefunctions $\left\{\psi_{\substack{\mathrm{fv}\\ 1,\mathbf{k}}}, \psi_{\substack{\mathrm{fv}\\ 2,\mathbf{k}}}, \ldots, \psi_{\substack{\mathrm{fv}\\ \mathrm{nstfv},\mathbf{k}}}\right\}$ and use it to construct the basis $\mathbb{B}_{\mathrm{fv}}$ for the second variational problem as

$$\mathbb{B}_{\mathrm{fv}}(\mathbf{k}) = \left\{ |\psi_{\substack{\mathrm{fv}\\ n\mathbf{k}}}\rangle \otimes |\sigma\rangle \equiv |\Psi_{\substack{\mathrm{fv}\\ n\mathbf{k}\sigma}}\rangle \quad | \quad n \in \{1,2,\ldots,\mathrm{nstfv}\},\ \sigma \in \{\downarrow,\uparrow\} \right\} \tag{4.19}$$

whereas we introduced the bra-ket-notation for the state-vectors, and the spin-coordinate s is given as

$$\langle \mathbf{r}, s|\Psi_{\substack{\mathrm{fv}\\ n\mathbf{k}\uparrow}}\rangle = \begin{cases} \psi_{\substack{\mathrm{fv}\\ n\mathbf{k}}}(\mathbf{r}) & \text{if } s = \uparrow \\ 0 & \text{if } s = \downarrow \end{cases}, \qquad \langle \mathbf{r}, s|\Psi_{\substack{\mathrm{fv}\\ n\mathbf{k}\downarrow}}\rangle = \begin{cases} 0 & \text{if } s = \uparrow \\ \psi_{\substack{\mathrm{fv}\\ n\mathbf{k}}}(\mathbf{r}) & \text{if } s = \downarrow \end{cases} \tag{4.20}$$

and such that the single-particle Hilbert-space of the second variational problem $\mathcal{H}_{\mathrm{fv}}$ becomes approximately

$$\mathcal{H} = L^2(\mathrm{Crystal}) \otimes \mathbb{C}^2 \approx \mathrm{span}_{\mathbb{C}} \bigcup_{\mathbf{k}\in\mathrm{Q}} \mathbb{B}_{\mathrm{fv}}(\mathbf{k}) \tag{4.21}$$

[2] I.e. including spin-orbit interactions

We further introduce the *Hamiltonian-matrices w.r.t.* $\mathbb{B}_{\text{fv}}(\mathbf{k})$ as

$$H(\mathbf{k}) \equiv \text{mat}_{\mathbb{B}_{\text{fv}}(\mathbf{k})} H. \tag{4.22}$$

Theorem 4.10 Assuming the orthonormality of the Bloch-wavefunctions obtained in solving the first variational problem[3], and using the approximation in Equation (6.2), we can find all solutions of Equation (4.16), by solving for each $\mathbf{k} \in \text{Q}$ the generalized eigenvalue-equation

$$H(\mathbf{k})\mathbf{C}_{\bullet}^{\text{sv}\to\text{fv}}(\mathbf{k}) = E_{\text{sv},\bullet}(\mathbf{k})\mathbf{C}_{\bullet}^{\text{sv}\to\text{fv}}(\mathbf{k}). \tag{4.23}$$

Proof 4.10 Analogous to the proof of Theorem 4.6.

Remark 4.10.1 Since the Hamiltonian-matrix in Equation (6.2) has, w.r.t. to the basis in Equation (4.19) the dimensions nstsv $\times$ nstsv, we can enumerate the eigenvalues and eigenvectors in Equation (4.16) as

$$\left\{ \epsilon_N(\mathbf{k}) \equiv E_{\underset{N}{\text{LSDA}}}(\mathbf{k}) \equiv E_{\underset{N}{\text{sv}}}(\mathbf{k}) \quad | \quad N \in \{1, \dots, \text{nstsv}\} \right\} \tag{4.24}$$

and

$$\left\{ \mathbf{C}_N^{\text{sv}\to\text{fv}}(\mathbf{k}) \quad | \quad N \in \{1, \dots, \text{nstsv}\} \right\} \tag{4.25}$$

whereas for each N the symbol $\mathbf{C}_N^{\text{sv}\to\text{fv}}(\mathbf{k})$ represents a vector with nstsv columns. The columns are labelled through band and spin-indices $n \in \{1, \dots, \text{nstfv}\}$, $\sigma \in \{\uparrow, \downarrow\}$ and $\mathbf{k} \in \text{Q}$ such that one has components

$$\left(\mathbf{C}_N^{\text{sv}\to\text{fv}}(\mathbf{k})\right)_{n',\sigma'} = C_{N,n'\sigma'}^{\text{sv}\to\text{fv}}(\mathbf{k}) \tag{4.26}$$

Remark 4.10.2 Notice that hence we can expand each eigenstate of the second variational problem as

$$|\Psi_{N\mathbf{k}}^{\text{sv}}\rangle = \sum_{n'=1}^{\text{nstfv}} \sum_{\sigma'\in\{\uparrow,\downarrow\}} C_{N,n'\sigma'}^{\text{sv}\to\text{fv}}(\mathbf{k}) |\Psi_{n'\mathbf{k}\sigma'}^{\text{fv}}\rangle = \sum_{n'=1}^{\text{nstfv}} \sum_{\sigma'\in\{\uparrow,\downarrow\}} C_{N,n'\sigma'}^{\text{sv}\to\text{fv}}(\mathbf{k}) |\psi_{n'\mathbf{k}}^{\text{fv}}\rangle \otimes |\sigma'\rangle \tag{4.27}$$

Remark 4.10.3 Notice further that Equation (4.27) can be rewritten in the usual "spinor-form" as

$$\begin{aligned} \langle \mathbf{r}, s|\Psi_{N\mathbf{k}}^{\text{sv}}\rangle &= \begin{cases} \sum_{n'=1}^{\text{nstfv}} C_{N,n'\uparrow}^{\text{sv}\to\text{fv}}(\mathbf{k})\psi_{n'\mathbf{k}}^{\text{fv}}(\mathbf{r}) & \text{if } s = \uparrow \\ \sum_{n'=1}^{\text{nstfv}} C_{N,n'\downarrow}^{\text{sv}\to\text{fv}}(\mathbf{k})\psi_{n'\mathbf{k}}^{\text{fv}}(\mathbf{r}) & \text{if } s = \downarrow \end{cases} \\ &\equiv \begin{cases} \alpha_{N\mathbf{k}}(\mathbf{r}) & \text{if } s = \uparrow \\ \beta_{N\mathbf{k}}(\mathbf{r}) & \text{if } s = \downarrow \end{cases} \end{aligned} \tag{4.28}$$

[3] TBInvestigated! (In the Bulla-script, wavefunctions are not normalized to mere products of delta-functions, cf. page 46 in the script.)

4.3. Inclusion of Hubbard-U-Terms ("L(S)DA + SOC + U")

Definition 4.11 A *Mott-insulator*[4] is a system where LDA and GGA incorrectly predict a metallic state.

Remark 4.11.1 The "LDA+U method" was first developed to be able to cure these anomalies.

Remark 4.11.2 In what follows, we restrict our attention to the case that one applies a *Hubbard-correction* to one species.

Fact 4.12 In the most general version of LDA+U a HF-correction to the energy enters with the form

$$E_{\mathrm{HF}} = \frac{1}{2}\sum_{abcd} \rho_{ac}\left[\langle ab|g|cd\rangle - \langle ab|g|dc\rangle\right]\rho_{bd} = \frac{1}{2}\sum_{abcd}\left(\rho_{ac}\rho_{bd} - \rho_{ad}\rho_{bc}\right)\langle ab|g|cd\rangle \tag{4.29}$$

whereas ρ_{ab} is one element of the density matrix with dimension

$$D = 2(2l+1) = \left|\{\uparrow,\downarrow\}\times\{-l,\dots,0,\dots+l\}\right| \tag{4.30}$$

which acts as an occupation-matrix and a, b, c, d are single-electron states[5]

Remark 4.12.1 Usually one decomposes these terms further into …

- the *Hartree-contribution* E_{H} as

$$E_{\mathrm{H}} = \frac{1}{2}\sum_{abcd}\rho_{ac}\langle ab|g|cd\rangle\rho_{bd} \tag{4.31}$$

- and the *Fock-contribution* a.k.a. *exchange-contribution* E_{F} as

$$E_{\mathrm{X}} = -\frac{1}{2}\sum_{abcd}\rho_{ac}\langle ab|g|dc\rangle\rho_{bd} \tag{4.32}$$

such that

$$E_{\mathrm{HF}} = E_{\mathrm{H}} + E_{\mathrm{F}}. \tag{4.33}$$

Fact 4.13 The goal is now to find a neat / systematic / enlightning way to evaluate the matrix elements[6]

$$\langle ab|g|cd\rangle = \sum\!\!\!\!\!\!\int_1 \sum\!\!\!\!\!\!\int_2 \psi_a^*(1)\psi_b^*(2)g(r_{12})\psi_c(1)\psi_d(2) \tag{4.34}$$

[4] Cf. the ELK-book page 20.
[5] Cf. Articles 09 and the references 6,7, 15 inside.
[6] Cf. for example Zirnbauer 2017/2018 Sheet 13, Exercise 1 for the example of the Coulomb-interaction matrix-elements.

for some functions $\psi_\bullet$.

Remark 4.13.1 Typical interactions g are for example the *Coulomb-interaction*

$$(0, \infty) \ni r \mapsto g(r) = r \in \mathbb{R} \tag{4.35}$$

or the *Yukawa-interaction*

$$(0, \infty) \ni r \mapsto g(r) = \mathrm{e}^{-\lambda r}/r \in \mathbb{R} \tag{4.36}$$

whereas the equations are written in atomic units.

Notation 4.14 We consider one-electron states a etc. to have a wavefunction of the type

$$\psi_a(i) = \varphi_a(\mathbf{r}_i)\chi_{\sigma_a}(s_i) = R_{l_a}(r_i)Y_{l_a m_a}(\Omega_i)\chi_{\sigma_a}(s_i) \tag{4.37}$$

whereas the radial part $R_{l_a}(\bullet)$ is the radial wavefunction from the (L)APW-functions in Equation (TBReferenced). (And not from the local orbitals in Equation (TBReferenced); The local orbitals that we use to complement our basis do not enter the LDA+U method.).

Remark 4.14.1 Notice that we restrict attention to the case that the one-electron states have the same angular momentum quantum number l.

Theorem 4.15 We consider the case, in which the interaction g operates only on the spatial degrees of freedom; Further we assume that the interaction g can be expanded as a series as

$$g(r_{12}) = \sum_{k=0}^{\infty} g_k(r_1, r_2)P_k(\cos\theta_{12}) \tag{4.38}$$

with relative coordinate $r_{12} = r_1 - r_2$ and the angle θ_{12} between the two vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, i.e.

$$\cos\theta_{12} := \frac{\langle \mathbf{r}_1, \mathbf{r}_2 \rangle}{r_1 r_2}. \tag{4.39}$$

Then one finds the expansion

$$P_k(\cos\theta_{12}) = \frac{4\pi}{2k+1} \sum_{q=-k}^{+k} Y^*_{kq}(\Omega_1)Y_{kq}(\Omega_2) \tag{4.40}$$

Proof 4.15 Cf. for example Jackson 3rd edition, Equations (3.38), (3.62).

Remark 4.15.1 Notice that using Equation (3.9) we can rewrite Equation (4.40) w.r.t. spherical tensors as

$$P_k(\cos\theta_{12}) = \frac{4\pi}{2k+1}\sum_{q=-k}^{+k}(-1)^q Y_{k,-q}(\Omega_1)Y_{kq}(\Omega_2) = \sum_{q=-k}^{+k}(-1)^q C^k_{-q}(\Omega_1)C^k_q(\Omega_2) \equiv \mathbf{C}^k(1)\cdot\mathbf{C}^k(2) \tag{4.41}$$

Remark 4.15.2 Notice that combining the two formulæ gives

$$g(r_{12}) = \sum_{k=0}^{\infty} g_k(r_1,r_2)\frac{4\pi}{2k+1}\sum_{q=-k}^{+k} Y^*_{kq}(\Omega_1)Y_{kq}(\Omega_2) \tag{4.42}$$

with expansion coefficients g_k that depend on the type of interaction g (Coulomb- or Yukawa-interaction etc.) and using that g is real valued we also obtain the identity

$$\begin{aligned} g(r_{12}) = g(r_{12})^* &= \sum_{k=0}^{\infty} g_k(r_1,r_2)\frac{4\pi}{2k+1}\sum_{q=-k}^{+k} Y_{kq}(\Omega_1)Y^*_{kq}(\Omega_2) \\ &= \sum_{k=0}^{\infty} g_k(r_1,r_2)\frac{4\pi}{2k+1}\sum_{q=-k}^{+k}(-1)^q Y_{kq}(\Omega_1)Y_{k,-q}(\Omega_2) \\ &= \sum_{k=0}^{\infty} g_k(r_1,r_2)\sum_{q=-k}^{+k}(-1)^q C^k_q(\Omega_1)C^k_{-q}(\Omega_2) \end{aligned} \tag{4.43}$$

Lemma 4.16

- For the Coulomb-interaction between two electrons (in atomic units) $g(r_{12}) = r_{12}^{-1}$ one obtains[7]

$$g_k(r_1,r_2) = \frac{r_<^k}{r_>^k} \tag{4.44}$$

  whereas $r_<$ and $r_>$ define the *lesser* and *greater* of the two radii,

- for the Yukawa-interaction between two electrons (in atomic units) $g(r_{12}) = \mathrm{e}^{\lambda r_{12}}/r_{12}$ one obtains[8]

$$g_k(r_1,r_2) = -(2k+1)\lambda j_k(\mathrm{i}\lambda r_<)h_k^{(1)}(\mathrm{i}\lambda r_>) \tag{4.45}$$

  with spherical Bessel-functions j_k and and spherical hankel functions of the first kind $h_k^{(1)}$.

Proof 4.16 ToBeReferenced.

Theorem 4.17 The matrix-elements in Equation (4.34) can be decomposed as

[7] Cf. Jackson 3rd edition, Equation (3.38).

[8] TBReferenced.

$$\langle ab|g|cd\rangle = \delta_{s_a,s_c}\delta_{s_b,s_d}\sum_{k=0}^{\infty} F_l^{(k)}\,\mathrm{AI}_l^{(k)}(m_a,m_b,m_c,m_d) \tag{4.46}$$

with *Slater-integrals* as double-integrals of the type[9]

$$F_l^{(k)} = \iint \mathrm{d}(r_1,r_2)\, r_1^2 R_l^2(r_1) g_k(r_1,r_2) R_l^2(r_2) r_2^2 \tag{4.47}$$

and [A]ngular-[I]ntegrals

$$\mathrm{AI}_l^{(k)}(m_a,m_b,m_c,m_d) = \frac{4\pi}{2k+1}\sum_{q=-k}^{+k}\int \mathrm{d}\Omega_1 Y^*_{lm_a}(\Omega_1)Y^*_{kq}(\Omega_1)Y_{lm_c}(\Omega_1)\int \mathrm{d}\Omega_2 Y^*_{lm_b}(\Omega_2)Y_{kq}(\Omega_2)Y_{lm_d}(\Omega_2). \tag{4.48}$$

Proof 4.17 See Appendix, Proofs, page 107.

Corollary 4.18 One obtains for the angular-integrals the idenity

$$\begin{aligned}\mathrm{AI}_l^k(m_a,m_b,m_c,m_d) &= \delta_{m_a+m_b,m_c+m_d}(-1)^{m_b-m_d}c_l^{(k)}(m_a,m_c)c_l^{(k)}(m_b,m_d)\\ &= \delta_{m_a+m_c,m_c+m_d}c_l^{(k)}(m_a,m_c)c_l^{(k)}(m_d,m_b)\end{aligned} \tag{4.49}$$

Proof 4.18 See Appendix, Proofs, page 107.

Remark 4.18.1 Notice that these integrals are in the literature known as *Slater-integrals*.

Remark 4.18.2 Notice that we also find the identity

$$\mathrm{AI}_l^k(m_a,m_b,m_c,m_d) = \sum_{q=-k}^{+k}(-1)^q\delta_{q,m_c-m_a}\delta_{q,m_b-m_d}c_l^{(k)}(m_a,m_c)c_l^{(k)}(m_d,m_b) \tag{4.50}$$

Lemma 4.19 One finds

$$\begin{aligned}\langle ab|g|cd\rangle &= \delta_{s_a,s_c}\delta_{s_b,s_d}\delta_{m_a+m_b,m_c+m_d}\sum_{k=0}^{\infty}F_l^{(k)}c_l^{(k)}(m_a,m_c)c_l^{(k)}(m_d,m_b)\\ &= \delta_{s_a,s_c}\delta_{s_b,s_d}\delta_{m_a+m_b,m_c+m_d}(-1)^{m_a+m_d}(2l+1)^2\sum_{k\in\{0,2,4,\ldots 2l\}}F_l^{(k)}\begin{pmatrix} l & k & l\\ 0 & 0 & 0\end{pmatrix}^2\begin{pmatrix} l & k & l\\ -m_a & m_a-m_c & m_c\end{pmatrix}\begin{pmatrix} l & k & l\\ -m_d & m_d-m_b & m_b\end{pmatrix}\end{aligned} \tag{4.51}$$

[9] Notice that the radial functions $R_\bullet$ are real valued Source?!, such that one can simply write the square, instead of the absolute square.

Proof 4.19 We combine Equation (4.46) with Equation (4.50) to get the first identity. For the second identity, we use Equation (3.19).

Lemma 4.20 One finds further

$$\langle ab|g|cd\rangle = \delta_{s_a,s_c}\delta_{s_b,s_d}(-1)^{m_a+m_b}(2l+1)^2 \sum_{k\in\{0,2,4,\dots 2l\}} \sum_{q=-k}^{+k}(-1)^q F_l^{(k)} \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 \begin{pmatrix} l & k & l \\ -m_a & -q & m_c \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_d \end{pmatrix} \tag{4.52}$$

Proof 4.20 For this we compute straightforwardly

$$\langle ab|g|cd\rangle = \delta_{s_a,s_c}\delta_{s_b,s_d} \sum_{k=0}^{\infty} F_l^{(k)} \mathrm{AI}_l^{(k)}(m_a, m_b, m_c, m_d) = \dots \tag{4.53}$$

and now we simplify

$$\begin{aligned}
\mathrm{AI}_l^{(k)}(m_a, m_b, m_c, m_d) &= \frac{4\pi}{2k+1} \sum_{q=-k}^{+k} \int \mathrm{d}(\Omega_1, \Omega_2) Y^*_{lm_a}(\Omega_1) Y^*_{kq}(\Omega_1) Y_{lm_c}(\Omega_1) \cdot Y^*_{lm_b}(\Omega_2) Y_{kq}(\Omega_2) Y_{lm_d}(\Omega_2) \\
&= \sum_{q=-k}^{+k} (-1)^{m_a+q+m_b} \sqrt{\frac{4\pi}{2k+1}} \int \mathrm{d}\Omega_1\, Y_{l-m_a}(\Omega_1) Y_{k,-q}(\Omega_1) Y_{lm_c}(\Omega_1) \cdot \sqrt{\frac{4\pi}{2k+1}} \int \mathrm{d}\Omega_2\, Y_{l-m_b}(\Omega_2) Y_{kq}(\Omega_2) Y_{lm_d}(\Omega_2) \\
&= \sum_{q=-k}^{+k} (-1)^{m_a+q+m_b} (2l+1)^2 \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 \begin{pmatrix} l & k & l \\ -m_a & -q & m_c \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_d \end{pmatrix}
\end{aligned} \tag{4.54}$$

5. The different Basis Sets and the Single-Particle Density

5.1. The Standard-Basis

Notation 5.1 We start by considering the *standard-basis* of $L^2(\text{Crystal}) \otimes \mathbb{C}^2$

$$\mathcal{H} = L^2(\text{Crystal}) \otimes \mathbb{C}^2 = \text{span}_{\mathbb{C}} \bigcup_{\substack{\sigma \in \{\uparrow,\downarrow\} \\ \mathbf{r} \in \text{Crystal}}} \Big\{ |\mathbf{r}\rangle \otimes |\sigma\rangle \Big\} = \text{span}_{\mathbb{C}}\, \mathbb{B}_{\text{standard}} \tag{5.1}$$

whereas $|\uparrow\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|\downarrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ refer to the complex standard-basis of $\mathbb{C}^2$. The resolution of identity becomes then

$$1_{\mathcal{H}} = \sum\!\!\!\!\!\!\int_{\substack{\sigma \in \{\uparrow,\downarrow\} \\ \mathbf{r} \in \text{Crystal}}} |\mathbf{r}, \sigma\rangle\langle \mathbf{r}, \sigma| \tag{5.2}$$

Remark 5.1.1 Notice that the *state-projector* $\rho_{N\mathbf{k}} \equiv |\Psi_{N\mathbf{k}}\rangle\langle\Psi_{N\mathbf{k}}|$ has matrix-elements w.r.t. the standard-basis as

$$\big\langle \mathbf{r}, s | \rho_{N\mathbf{k}} | \mathbf{r}', s' \big\rangle = \big(\text{mat}_{\mathbb{B}_{\text{standard}}}\, \rho_{N\mathbf{k}}\big)(\mathbf{r}, s; \mathbf{r}', s') = \begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}') & \alpha_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}') \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}') & \beta_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}') \end{pmatrix}_{s,s'} \tag{5.3}$$

such that we can conclude the matrix-elements of the total density as

$$\big\langle \mathbf{r}, s | \rho | \mathbf{r}', s' \big\rangle = \sum_{\mathbf{k} \in \text{Q}} \sum_{N=1}^{\text{nstsv}} \text{F}_{\hbar\beta}(\epsilon_N(\mathbf{k}) - \mu) \begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}') & \alpha_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}') \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}') & \beta_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}') \end{pmatrix}_{s,s'} . \tag{5.4}$$

Definition 5.2 We define the *spin-operator* $\mathbf{S}$ as linear operator on $\mathcal{H}$, which has matrix-elements w.r.t. the standard-basis as

$$\left\langle \mathbf{r}, s | S_z | \mathbf{r}', s' \right\rangle = \left(\mathrm{mat}_{\mathbb{B}_{\text{standard}}} S_z\right)(\mathbf{r}, s; \mathbf{r}', s') := \frac{\hbar}{2}\delta(\mathbf{r}-\mathbf{r}')\left(\sigma_z\right)_{s,s'} \tag{5.5}$$

Remark 5.2.1 Notice that the spin-operator acts as the identity on the spatial part of the wavefunction, i.e. we have

$$\langle \mathbf{r}, s | S_z | \Psi \rangle = \sum\!\!\!\!\!\!\int_{\mathbf{r}',s'} \left\langle \mathbf{r}, s | S_z | \mathbf{r}', s' \right\rangle \left\langle \mathbf{r}', s' | \Psi \right\rangle = \sum\!\!\!\!\!\!\int_{\mathbf{r}',s'} \delta(\mathbf{r}-\mathbf{r}')\left(\sigma_z\right)_{s,s'} \left\langle \mathbf{r}', s' | \Psi \right\rangle = \sum_{s'} \left(\sigma_z\right)_{s,s'} \left\langle \mathbf{r}, s' | \Psi \right\rangle \tag{5.6}$$

Theorem 5.3 We obtain the identity

$$\mathrm{Tr}_{\mathcal{H}}\left(\rho_{N\mathbf{k}} S_z\right) = \frac{\hbar}{2}\int \mathrm{d}^3\mathbf{r}\, \mathrm{Tr}_{2\times 2}\left[\begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}) & \alpha_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}) \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}) & \beta_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}) \end{pmatrix} \cdot \sigma_z\right] \tag{5.7}$$

Proof 5.3 We compute the matrix-elements of the composition as

$$\begin{aligned}
\left\langle \mathbf{r}, s | \rho_{N\mathbf{k}} S_z | \mathbf{r}', s' \right\rangle &= \left(\mathrm{mat}_{\mathbb{B}_{\text{standard}}} \rho_{N\mathbf{k}} S_z\right)(\mathbf{r}, s; \mathbf{r}', s') \\
&= \frac{\hbar}{2}\sum\!\!\!\!\!\!\int_{\mathbf{r}'',s''} \begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}'') & \alpha_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}'') \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}'') & \beta_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}'') \end{pmatrix}_{s,s''} \delta(\mathbf{r}''-\mathbf{r}')\left(\sigma_z\right)_{s'',s'} \\
&= \frac{\hbar}{2}\left(\begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}') & \alpha_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}') \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha^*_{N\mathbf{k}}(\mathbf{r}') & \beta_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}') \end{pmatrix} \cdot \sigma_z\right)_{s,s'}
\end{aligned}$$

and now taking the diagonal elements proves the theorem.

5.2. The Basis from the Solutions of the Second Variational Problem

Notation 5.4 We start by considering the *basis from the solutions of the second variational problem* according to Equation (6.2) as

$$\mathcal{H} = L^2(\text{Crystal}) \otimes \mathbb{C}^2 \approx \text{span}_{\mathbb{C}} \bigcup_{\mathbf{k}\in\text{Q}} \bigcup_{N\in\{1,\ldots,\text{nstsv}\}} \left\{ |\Psi^{\text{sv}}_{N\mathbf{k}}\rangle \right\} \equiv \text{span}_{\mathbb{C}}\, \mathbb{B}_{\text{sv}} \tag{5.8}$$

Remark 5.4.1 Notice that the state-projector $\rho_{N\mathbf{k}}$ is diagonal w.r.t. this basis as

$$\text{mat}_{\mathbb{B}_{\text{sv}}}\, \rho_{N\mathbf{k}} = \begin{pmatrix} 0 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} \tag{5.9}$$

or, rewritten as matrix-elements,

$$\langle \Psi^{\text{sv}}_{M\mathbf{q}} | \rho_{N\mathbf{k}} | \Psi^{\text{sv}}_{M'\mathbf{q}'} \rangle = \left(\text{mat}_{\mathbb{B}_{\text{sv}}}\, \rho_{N\mathbf{k}}\right)(M\mathbf{q}; M'\mathbf{q}') = \delta_{N,M}\delta_{N,M'}\delta_{\mathbf{k},\mathbf{q}}\delta_{\mathbf{k},\mathbf{q}'} \tag{5.10}$$

5.3. The Basis from the Solutions of the First Variational Problem

Notation 5.5 We start by considering the *basis from the solutions of the first variational problem* according to Equation (4.1) as

$$\mathcal{H} = L^2(\text{Crystal}) \otimes \mathbb{C}^2 \approx \text{span}_{\mathbb{C}} \bigcup_{\mathbf{k}\in \text{Q}} \bigcup_{n\in\{1,\ldots,\text{nstfv}\}} \bigcup_{\sigma\in\{\uparrow,\downarrow\}} \left\{ |\Psi_{\substack{\text{fv}\\ n\mathbf{k}\sigma}}\rangle \right\} \equiv \text{span}_{\mathbb{C}}\, \mathbb{B}_{\text{fv}} \tag{5.11}$$

Remark 5.5.1 Notice that the *state-projector* $\rho_{n\mathbf{k}\sigma}$ can be expanded w.r.t. the this basis as

$$\begin{aligned}
\rho_{N\mathbf{k}} &= |\Psi_{\substack{\text{sv}\\ N\mathbf{k}}}\rangle\langle\Psi_{\substack{\text{sv}\\ N\mathbf{k}}}| \\
&= \sum_{\substack{n,n'\in\{1,\ldots,\text{nstfv}\}\\ \sigma,\sigma'\in\{\uparrow,\downarrow\}}} C^{\text{sv}\to\text{fv}}_{N,n\sigma}(\mathbf{k}) \left(C^{\text{sv}\to\text{fv}}_{N,n'\sigma'}(\mathbf{k}) \right)^* |\Psi_{\substack{\text{fv}\\ n\mathbf{k}\sigma}}\rangle\langle\Psi_{\substack{\text{fv}\\ n'\mathbf{k}\sigma'}}|
\end{aligned}$$

5.4. The Basis from the (L)APW + LO Functions

Notation 5.6 We start by considering the *basis from the apw+lo-basis-set* according to Equations (4.1) as

$$\mathbb{B}_{\mathrm{LO}} = \bigcup_{j=1}^{n_{\mathrm{LO}}} \left\{\phi_j\right\}, \qquad \mathbb{B}_{\mathrm{APW}} = \bigcup_{\mathbf{k}\in\mathrm{Q}} \bigcup_{\mathbf{G}\in\mathrm{CiR}_{\mathbf{k}}\mathrm{L}} \{\phi_{\mathbf{k}+\mathbf{G}}\} \equiv \bigcup_{\mathbf{k}\in\mathrm{Q}} \mathbb{B}_{\mathrm{APW}}(\mathbf{k}), \qquad \mathbb{B}_{\mathrm{APW+LO}} = \mathbb{B}_{\mathrm{LO}} \cup \mathbb{B}_{\mathrm{APW}} \tag{5.12}$$

such that the single-particle Hilbert-space of the first variational problem $\mathcal{H}_0$ becomes approximately

$$\mathcal{H} = L^2(\mathrm{Crystal}) \otimes \mathbb{C}^2 \approx \mathrm{span}_{\mathbb{C}} \left[\mathbb{B}_{\mathrm{APW+LO}} \otimes \{|\uparrow\rangle, |\downarrow\rangle\}\right] \tag{5.13}$$

Remark 5.6.1 Notice that the *state-projector* $\rho_{N\mathbf{k}}$ can be expanded w.r.t. the this basis as

$$\begin{aligned}
\rho_{N\mathbf{k}} &= |\Psi_{\substack{\mathrm{sv}\\N\mathbf{k}}}\rangle\langle\Psi_{\substack{\mathrm{sv}\\N\mathbf{k}}}| \\
&= \sum_{\substack{n,n'\in\{1,\ldots,\mathrm{nstfv}\}\\ \sigma,\sigma'\in\{\uparrow,\downarrow\}}} C^{\mathrm{sv}\to\mathrm{fv}}_{N,n\sigma}(\mathbf{k}) \left(C^{\mathrm{sv}\to\mathrm{fv}}_{N,n'\sigma'}(\mathbf{k})\right)^* |\Psi_{\substack{\mathrm{fv}\\n\mathbf{k}\sigma}}\rangle\langle\Psi_{\substack{\mathrm{fv}\\n'\mathbf{k}\sigma'}}| \\
&= \sum_{\substack{n,n'\in\{1,\ldots,\mathrm{nstfv}\}\\ m,m'\in\{1,\ldots,\mathrm{ngk}+n_{\mathrm{LO}}\}\\ \sigma,\sigma'\in\{\uparrow,\downarrow\}}} C^{\mathrm{sv}\to\mathrm{fv}}_{N,n\sigma}(\mathbf{k}) C^{\mathrm{fv}\to\mathrm{apw+lo}}_{n,m}(\mathbf{k}) \left(C^{\mathrm{sv}\to\mathrm{fv}}_{N,n'\sigma'}(\mathbf{k}) C^{\mathrm{fv}\to\mathrm{apw+lo}}_{n',m'}(\mathbf{k})\right)^* |\psi_{\substack{\mathrm{apw+lo}\\m\mathbf{k}}}\rangle \otimes |\sigma\rangle\langle\psi_{\substack{\mathrm{apw+lo}\\m'\mathbf{k}}}| \otimes \langle\sigma'|
\end{aligned}$$

Fact 5.7 Switching to the coordinate-representation w.r.t. to the standard-basis and rearranging / relabelling terms yields the expansion in Equation (6.14).

5.5. The Basis from Spherical Harmonics

Notation 5.8 We start by considering the *basis from the spherical harmonics and the eigenfunctions of* σ_z as

$$\mathcal{H} = L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2 \approx \mathrm{span}_{\mathbb{C}} \bigcup_{l=0}^{\infty} \bigcup_{m=-l}^{+l} \bigcup_{\sigma \in \{\uparrow,\downarrow\}} \Big\{ |l, m, \sigma\rangle \Big\} \equiv \mathrm{span}_{\mathbb{C}}\, \mathbb{B}_{\mathrm{fv}} \tag{5.14}$$

Remark 5.8.1 Notice that the *state-projector* $\rho_{N\mathbf{k}}$ can be expanded w.r.t. the this basis as in Equation (6.19) as

$$\left(\rho_{\substack{N\mathbf{k} \\ \alpha L(l) S}} \right)_{lms, l'm's'} = c_{N\mathbf{k}}^{\alpha lms} \overline{c_{N\mathbf{k}}^{\alpha l'm's'}} \int_0^{R_\alpha} f_{N\mathbf{k}}^{\alpha lm}(r_\alpha) \overline{f_{N\mathbf{k}}^{\alpha l'm'}(r_\alpha)} r_\alpha^2 \mathrm{d}r_\alpha \tag{5.15}$$

6. Observable Quantities from Operator-Expectation-Values

6.1. Re-expressing Many Body Expectation-Values through Simpler Expressions

6.1.1. Case 1: Arbitrary One-Body Operators

<u>The Idea</u>

Notation 6.1 We consider the following sequence of Hilbert-spaces and densities

$$
\begin{array}{lll}
\rho_{\mathscr{F}} = \frac{1}{Z}\mathrm{e}^{-\beta\left(H^{2\mathrm{Q}}-\mu N^{2\mathrm{Q}}\right)} & \in \mathscr{D}\Big[& \mathscr{H} = \mathscr{F}(\mathcal{H}) \quad \Big] \\
 & & \bigcup \\
\rho_{\mathcal{H}} = \sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)|\Psi_{N\mathbf{k}}^{\mathrm{sv}}\rangle\langle\Psi_{N\mathbf{k}}^{\mathrm{sv}}| & \in \mathscr{D}\Big[& \mathcal{H} \equiv L^2(\mathrm{Crystal})\otimes\mathbb{C}^2 \simeq \bigoplus_{\mathbf{k}\in\mathrm{Q}}\mathcal{H}_{\mathbf{k}} \quad \Big] \\
 & & \bigcup \\
\rho_{N\mathbf{k}} = |u_{N\mathbf{k}}^{\mathrm{sv}}\rangle\langle u_{N\mathbf{k}}^{\mathrm{sv}}| & \in \mathscr{D}\Big[& \mathcal{H}_{\mathbf{k}} = L^2(\mathrm{UC})\otimes\mathbb{C}^2 \quad \Big]
\end{array}
$$

In this section we will prove the following statement:

Theorem 6.2 For any one-body operator X on $\mathcal{H}$ one finds for the expectation-values

$$
\begin{aligned}
\text{"Observable Quantity"} &= \langle X^{2\mathrm{Q}}\rangle \\
&= \mathrm{Tr}_{\mathscr{F}}\left(\rho_{\mathscr{F}}X^{2\mathrm{Q}}\right) \\
&= \mathrm{Tr}_{\mathcal{H}}\left(\rho_{\mathcal{H}}X\right) && = \sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\left\langle\Psi_{N\mathbf{k}}^{\mathrm{sv}}|X|\Psi_{N\mathbf{k}}^{\mathrm{sv}}\right\rangle \\
&= N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\,\mathrm{Tr}_{\mathcal{H}_{\mathbf{k}}}\left(\rho_{N\mathbf{k}}X\right) && = N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\left\langle u_{N\mathbf{k}}^{\mathrm{sv}}|X|u_{N\mathbf{k}}^{\mathrm{sv}}\right\rangle_{\mathrm{UC}}
\end{aligned}
\tag{6.1}
$$

The Details

Notation 6.3 We will assume now, that the basis that is given through the solutions of the second variational problem in Equation (TBReferenced) is good enough such that we span the physically relevant part of the single-particle Hilbert-space, i.e. such that we can approximate

$$\mathcal{H} = L^2(\text{Crystal}) \otimes \mathbb{C}^2 \approx \text{span}_{\mathbb{C}} \bigcup_{\mathbf{k}\in\text{Q}} \bigcup_{N\in\{1,\ldots,\text{nstsv}\}} \left\{ |\Psi_{N\mathbf{k}}^{\text{sv}}\rangle \right\} \tag{6.2}$$

We denote the corresponding Fock-space by $\mathscr{F}(\mathcal{H}) \equiv \mathscr{F}$.

Notation 6.4 In the grand canonical ensemble, the many-body density is given by

$$\rho_{\mathscr{F}} = \frac{1}{Z} \mathrm{e}^{-\beta\left(H^{2\text{Q}} - \mu N^{2\text{Q}}\right)} \quad \in \quad \mathscr{D}\left[\mathscr{F}(\mathcal{H})\right] \tag{6.3}$$

with ensemble Hamiltonian given in its second quantized form w.r.t. the basis given by the solutions of the second variational problem as

$$H^{2\text{Q}} - \mu N^{2\text{Q}} = \sum_{N=1}^{\text{nstsv}} \sum_{\mathbf{k}\in\text{Q}} \left(\epsilon_{N\mathbf{k}} - \mu\right) c_{N\mathbf{k}}^{\dagger} c_{N\mathbf{k}} \tag{6.4}$$

In order to use the Lemma from the summary, we further introduce the notation

$$\rho_{\mathcal{H}} = \sum_{N=1}^{\text{nstsv}} \sum_{\mathbf{k}\in\text{Q}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) |\Psi_{N\mathbf{k}}^{\text{sv}}\rangle\langle\Psi_{N\mathbf{k}}^{\text{sv}}| \quad \in \quad \mathscr{D}\left[\mathcal{H}\right] \tag{6.5}$$

Remark 6.4.1 Notice that the matrix-elements w.r.t. the standard-basis of the density-matrices

$$\rho_{N\mathbf{k}} \equiv |\Psi_{N\mathbf{k}}^{\text{sv}}\rangle\langle\Psi_{N\mathbf{k}}^{\text{sv}}| \tag{6.6}$$

are given by Equation (5.3).

Fact 6.5 For any one-body operator X one finds for the expectation-values

$$\text{Tr}_{\mathscr{F}}\left(\rho_{\mathscr{F}} X^{2\text{Q}}\right) = \langle X^{2\text{Q}}\rangle = \text{Tr}_{\mathcal{H}}\left(\rho_{\mathcal{H}} X\right) \tag{6.7}$$

Proof 6.5 See Summary, TBReferenced.

Remark 6.5.1 Using Equation (6.5), we can simplify the right hand side of Equation (13.74) to

$$
\begin{aligned}
\langle X^{2\mathrm{Q}} \rangle &= \mathrm{Tr}_{\mathcal{H}} \left(\rho_{\mathcal{H}} X \right) \\
&= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k} \in \mathrm{Q}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \, \mathrm{Tr}_{\mathcal{H}} \left(|\Psi^{\mathrm{sv}}_{N\mathbf{k}}\rangle\langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X \right) \\
&= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k} \in \mathrm{Q}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X |\Psi^{\mathrm{sv}}_{N\mathbf{k}}\rangle
\end{aligned}
\tag{6.8}
$$

whereas we used the completeness relation in the last step.

Remark 6.5.2 We can obtain the same result, by evaluating the left hand side of this equation with Green-functions. For this, note that the second quantized version of the spin-operator reads w.r.t. the basis given by the solutions of the second variational problem

$$
X^{2\mathrm{Q}} = \sum_{N,N'=1}^{\mathrm{nstsv}} \sum_{\mathbf{k},\mathbf{k}' \in \mathrm{Q}} \langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X |\Psi^{\mathrm{sv}}_{N'\mathbf{k}'}\rangle c^{\dagger}_{N\mathbf{k}} c_{N'\mathbf{k}'}
\tag{6.9}
$$

Then the left hand side of Equation (13.74) becomes

$$
\begin{aligned}
\langle X^{2\mathrm{Q}} \rangle &= \mathrm{Tr}_{\mathscr{F}} \left(\rho_{\mathscr{F}} X^{2\mathrm{Q}} \right) \\
&= \sum_{N,N'=1}^{\mathrm{nstsv}} \sum_{\mathbf{k},\mathbf{k}' \in \mathrm{Q}} \langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X |\Psi^{\mathrm{sv}}_{N'\mathbf{k}'}\rangle \, \mathrm{Tr}_{\mathscr{F}} \left(\rho_{\mathscr{F}} c^{\dagger}_{N\mathbf{k}} c_{N'\mathbf{k}'} \right) \\
&= \sum_{N,N'=1}^{\mathrm{nstsv}} \sum_{\mathbf{k},\mathbf{k}' \in \mathrm{Q}} \langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X |\Psi^{\mathrm{sv}}_{N'\mathbf{k}'}\rangle \frac{\hbar}{\mathrm{i}} G^{<}_{\substack{N',N \\ \mathbf{k}',\mathbf{k}}}(t=0) \\
&= \sum_{N,N'=1}^{\mathrm{nstsv}} \sum_{\mathbf{k},\mathbf{k}' \in \mathrm{Q}} \langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X |\Psi^{\mathrm{sv}}_{N'\mathbf{k}'}\rangle \delta_{\mathbf{k},\mathbf{k}'} \delta_{N,N'} \, \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu) \\
&= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k} \in \mathrm{Q}} \langle\Psi^{\mathrm{sv}}_{N\mathbf{k}}| X |\Psi^{\mathrm{sv}}_{N\mathbf{k}}\rangle \, \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}} - \mu)
\end{aligned}
$$

leaving us with the exact same formula as the right hand side.

6.1.2. Case 2: Spin- and Angular-Momentum Operators with Main-Contributions from Muffin-Tins

The Idea

Notation 6.6 We consider the following sequence of Hilbert-spaces and densities

$$
\begin{aligned}
\rho_{\mathscr{F}} = \frac{1}{Z}\mathrm{e}^{-\beta\left(H^{2\mathrm{Q}}-\mu N^{2\mathrm{Q}}\right)} \quad & \in \mathscr{D}\Big[\ \mathscr{H} = \mathscr{F}(\mathcal{H})\ \Big] \\
& \quad \bigcup \\
\rho_{\mathcal{H}} = \sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)|\Psi^{\ \mathrm{sv}}_{N\mathbf{k}}\rangle\langle\Psi^{\ \mathrm{sv}}_{N\mathbf{k}}| \quad & \in \mathscr{D}\Big[\ \mathcal{H} \equiv L^2(\mathrm{Crystal})\otimes\mathbb{C}^2 \simeq \bigoplus_{\mathbf{k}\in\mathrm{Q}}\mathcal{H}_{\mathbf{k}}\ \Big] \\
& \quad \bigcup \\
\rho_{N\mathbf{k}} = |u^{\ \mathrm{sv}}_{N\mathbf{k}}\rangle\langle u^{\ \mathrm{sv}}_{N\mathbf{k}}| \quad & \in \mathscr{D}\Bigg[\ \mathcal{H}_{\mathbf{k}} = L^2(\mathrm{UC})\otimes\mathbb{C}^2 = \Bigg[\left(\bigoplus_{\alpha=1}^{r} L^2([0,R_\alpha],r^2\mathrm{d}r)\otimes L^2(S^2,\mathrm{d}\Omega)\otimes\mathbb{C}^2\right)\oplus\left(L^2(\mathrm{IR})\otimes\mathbb{C}^2\right)\Bigg]\ \Bigg] \\
& \quad \bigcup \\
\rho^{\ N\mathbf{k}}_{\alpha} \quad & \in \mathscr{D}\Big[\ \mathcal{H}_\alpha = L^2([0,R_\alpha],r^2\mathrm{d}r)\otimes L^2(S^2,\mathrm{d}\Omega)\otimes\mathbb{C}^2\Big] \\
& \quad \bigcup \\
\rho^{\ N\mathbf{k}}_{\alpha LS} \quad & \in \mathscr{D}\Big[\ \mathcal{H}_{\alpha LS} = L^2(S^2,\mathrm{d}\Omega)\otimes\mathbb{C}^2 = \bigoplus_{l=0}^{\infty}\mathrm{span}_{\mathbb{C}}\bigcup_{m=-l}^{l}\bigcup_{\sigma\in\{\uparrow,\downarrow\}}\left\{Y_{lm}\otimes\chi_\sigma\equiv|lm\sigma\rangle\right\}\Big] \\
& \quad \bigcup \\
\rho^{\ N\mathbf{k}}_{\alpha L(l)S} \quad & \in \mathscr{D}\Big[\ \mathcal{H}_{\alpha L(l)S} = \mathrm{span}_{\mathbb{C}}\bigcup_{m=-l}^{l}\bigcup_{\sigma\in\{\uparrow,\downarrow\}}\left\{Y_{lm}\otimes\chi_\sigma\equiv|lm\sigma\rangle\right\}\Big]
\end{aligned}
\tag{6.10}
$$

and in this section we will prove the following statement: For any one-body operator X which is defined on $\mathcal{H}_{\mathrm{LO}_\alpha(l)}$ which is defined on the series of Hilbert-spaces above, one finds for the expectation-values

$$
\begin{aligned}
\text{"Observable Quantity"} = \langle X^{2\mathrm{Q}}\rangle = \mathrm{Tr}_{\mathscr{F}}\left(\rho_{\mathscr{F}}X^{2\mathrm{Q}}\right) = \mathrm{Tr}_{\mathcal{H}}\left(\rho_{\mathcal{H}}X\right) = \ldots \\
\ldots &= N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\,\mathrm{Tr}_{\mathcal{H}_{\mathbf{k}}}\left(\rho_{N\mathbf{k}}X\right) \\
&= N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^{r}\mathrm{Tr}_{\mathcal{H}_\alpha}\left(\rho^{\ N\mathbf{k}}_{\alpha}X\right) \\
&\approx N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^{r}\mathrm{Tr}_{\mathcal{H}_{\alpha LS}}\left(\rho^{\ N\mathbf{k}}_{\alpha LS}X\right) \\
&= N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^{r}\sum_{l=0}^{\infty}\mathrm{Tr}_{\mathcal{H}_{\alpha L(l)S}}\left(\rho^{\ N\mathbf{k}}_{\alpha L(l)S}X\right)
\end{aligned}
\tag{6.11}
$$

Remark 6.6.1 Notice that this theorem can be used for operators X that are well defined on the Hilbert-space

$$
\mathcal{H}_{\alpha L(l)S} = \mathrm{span}_{\mathbb{C}}\bigcup_{m=-l}^{l}\bigcup_{\sigma\in\{\uparrow,\downarrow\}}\left\{Y_{lm}\otimes\chi_\sigma\equiv|lm\sigma\rangle\right\}, \tag{6.12}
$$

for example ...

- the spin-operator $X \equiv S_i$ with $i \in \{x, y, z\}$,
- the angular-momentum operator $X \equiv L_i$ with $i \in \{x, y, z\}$,
- the spin-orbital-coupling operator $X \equiv \sum_i L_i \otimes S_i \equiv \mathbf{L} \cdot \mathbf{S}$.

Remark 6.6.2 Notice that this allows us to augment our graphical representation of the logic of densities from Remark **??** in the following way:

$$
\begin{array}{lcl}
\rho_{\mathscr{F}} & \in & \mathscr{D}\big[\mathscr{F}(\mathcal{H})\big] \\
\downarrow & & \\
\rho_{\mathcal{H}} & \in & \mathscr{D}\big[L^2(\text{Crystal}) \otimes \mathbb{C}^2\big] \\
\downarrow & & \\
\rho_{N\mathbf{k}} & \in & \mathscr{D}\left[\left(\bigoplus_{\alpha=1}^{r} L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right) \oplus \left(L^2(\text{IR}) \otimes \mathbb{C}^2\right)\right] = \mathscr{D}\big[L^2(\text{UC}) \otimes \mathbb{C}^2\big] \\
\downarrow & & \\
\rho_{N\mathbf{k}}^{\alpha} & \in & \mathscr{D}\big[L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\big] \\
\downarrow & & \\
\rho_{\substack{N\mathbf{k} \\ \alpha LS}} & \in & \mathscr{D}\big[L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\big]
\end{array}
$$

The demanding details are elaborated in the following subsection.

6.1.3. Coordinate-Representations

Then one finds for the density-operator

$$
\mathscr{D}\left[L^2(\text{UC}) \otimes \mathbb{C}^2\right] \ni |N, \mathbf{k}\rangle\langle N, \mathbf{k}| = \rho_{N\mathbf{k}} : \mathcal{H}_{\mathbf{k}} \longrightarrow \mathcal{H}_{\mathbf{k}} \tag{6.13}
$$

the coordinate representation (w.r.t. the standard basis in Notation 5.1)

$$
\begin{aligned}
\rho_{N\mathbf{k}}(\mathbf{r}\sigma, \mathbf{r}'\sigma') &\equiv \langle \mathbf{r}\sigma | N, \mathbf{k}\rangle\langle N, \mathbf{k} | \mathbf{r}'\sigma'\rangle \\
&= \sum_{\alpha,\alpha'=1}^{r} \sum_{l,l'=0}^{\infty} \sum_{|m|\leq l, |m'|\leq l'} \sum_{s,s'\in\{\uparrow,\downarrow\}} c_{N\mathbf{k}}^{\alpha l m s} \overline{c_{N\mathbf{k}}^{\alpha' l' m' s'}} f_{N\mathbf{k}}^{\alpha l m}(r_\alpha) \overline{f_{N\mathbf{k}}^{\alpha' l' m'}(r'_{\alpha'})} Y_{lm}(\Omega_\alpha) \overline{Y_{l'm'}(\Omega'_{\alpha'})} \chi_s(\sigma) \chi_{s'}(\sigma') \mathbb{1}_{\text{MT}_\alpha}(\mathbf{r}) \mathbb{1}_{\text{MT}_{\alpha'}}(\mathbf{r}') \\
&\quad + \sum_{\mathbf{G},\mathbf{G}'\in\text{RL}} \sum_{s,s'\in\{\uparrow,\downarrow\}} c_{N\mathbf{k}}^{\mathbf{G}s} \overline{c_{N\mathbf{k}}^{\mathbf{G}'s'}} e^{\mathrm{i}(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} e^{-\mathrm{i}(\mathbf{k}+\mathbf{G}')\cdot\mathbf{r}'} \chi_s(\sigma) \chi_{s'}(\sigma') \mathbb{1}_{\text{IR}}(\mathbf{r}) \mathbb{1}_{\text{IR}}(\mathbf{r}') \\
&\quad + \text{"cross-terms"}
\end{aligned} \tag{6.14}
$$

Definition 6.7 For some species $\alpha \in \{1, \dots, r\}$ we now consider the mapping

$$\mathscr{D}\left[L^2(\mathrm{UC}) \otimes \mathbb{C}^2\right] \ni \rho_{N\mathbf{k}} \quad \mapsto \quad \rho_{\substack{\alpha \\ N\mathbf{k}}} \in \mathscr{D}\left[L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right] \tag{6.15}$$

defined by its coordinate representation

$$\begin{aligned} \rho_{\substack{\alpha \\ N\mathbf{k}}}(\mathbf{r}\sigma, \mathbf{r}'\sigma') &= \langle \mathbf{r}\sigma | \rho_{\substack{\alpha \\ N\mathbf{k}}} | \mathbf{r}'\sigma' \rangle \\ &= \sum_{l,l'=0}^{\infty} \sum_{|m| \le l, |m'| \le l'} \sum_{s,s' \in \{\uparrow,\downarrow\}} c_{N\mathbf{k}}^{\alpha lms} \overline{c_{N\mathbf{k}}^{\alpha l'm's'}} f_{N\mathbf{k}}^{\alpha lm}(r_\alpha) \overline{f_{N\mathbf{k}}^{\alpha l'm'}(r'_\alpha)} Y_{lm}(\Omega_\alpha) \overline{Y_{l'm'}(\Omega'_\alpha)} \chi_s(\sigma) \chi_{s'}(\sigma') \end{aligned} \tag{6.16}$$

Definition 6.8 We further consider the mapping

$$\mathscr{D}\left[L^2([0, R_\alpha], r^2 \mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right] \ni \rho_{\substack{\alpha \\ N\mathbf{k}}} \quad \mapsto \quad \rho_{\substack{N\mathbf{k} \\ \alpha LS}} \equiv \mathrm{Tr}_{L^2([0,R_\alpha], r^2\mathrm{d}r)}\, \rho_{\substack{\alpha \\ N\mathbf{k}}} \in \mathscr{D}\left[L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right] \tag{6.17}$$

with coordinate representation

$$\rho_{\substack{N\mathbf{k} \\ \alpha LS}}(\Omega, \sigma; \Omega', \sigma') = \sum_{l,l'=0}^{\infty} \sum_{|m| \le l, |m'| \le l'} \sum_{s,s' \in \{\uparrow,\downarrow\}} c_{N\mathbf{k}}^{\alpha lms} \overline{c_{N\mathbf{k}}^{\alpha l'm's'}} \left(\int_0^{R_\alpha} f_{N\mathbf{k}}^{\alpha lm}(r_\alpha) \overline{f_{N\mathbf{k}}^{\alpha l'm'}(r'_\alpha)} r_\alpha^2 \mathrm{d}r_\alpha \right) Y_{lm}(\Omega_\alpha) \overline{Y_{l'm'}(\Omega'_\alpha)} \chi_s(\sigma) \chi_{s'}(\sigma') \tag{6.18}$$

with matrix-elements w.r.t. the basis in Equation (2.20) as

$$\left(\rho_{\substack{N\mathbf{k} \\ \alpha LS}}\right)_{lms, l'm's'} = c_{N\mathbf{k}}^{\alpha lms} \overline{c_{N\mathbf{k}}^{\alpha l'm's'}} \int_0^{R_\alpha} f_{N\mathbf{k}}^{\alpha lm}(r_\alpha) \overline{f_{N\mathbf{k}}^{\alpha l'm'}(r_\alpha)} r_\alpha^2 \mathrm{d}r_\alpha \tag{6.19}$$

and call this quantity the *local-orbital density-matrix w.r.t. the species* α for the state $|N\mathbf{k}\rangle$.

Remark 6.8.1 Note that, in summary, we have the series of mappings

$$\begin{array}{lll} \rho_{N\mathbf{k}} & \in & \mathscr{D}\left[\left(\bigoplus_{\alpha=1}^{r} L^2([0, R_\alpha], r^2\mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right) \oplus \left(L^2(\mathrm{IR}) \otimes \mathbb{C}^2\right)\right] = \mathscr{D}\left[L^2(\mathrm{UC}) \otimes \mathbb{C}^2\right] \\ \downarrow & & \\ \rho_{\substack{\alpha \\ N\mathbf{k}}} & \in & \mathscr{D}\left[L^2([0, R_\alpha], r^2\mathrm{d}r) \otimes L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right] \\ \downarrow & & \\ \rho_{\substack{N\mathbf{k} \\ \alpha LS}} & \in & \mathscr{D}\left[L^2(S^2, \mathrm{d}\Omega) \otimes \mathbb{C}^2\right] \end{array}$$

Remark 6.8.2 This matrix includes:

- Off-diagonal elements in spin ($\sigma \neq \sigma'$) — relevant for non-collinear magnetism,
- Off-diagonal elements in orbital angular momentum ($l \neq l'$) — relevant for crystal-field and hybridization effects,
- Hermiticity: $\left(\rho_{\substack{\alpha \\ N\mathbf{k} \\ lms,l'm's'}}\right)^* = \rho_{\substack{\alpha \\ N\mathbf{k} \\ l'm's',lms}}$.
- Notice that being the partial trace of a density, the reduced density matrix is also a density matrix, i.e., it is positive semi-definite and has trace one.

Remark 6.8.3 By linearity, one defines the *local-orbital density-matrix for the species α w.r.t. the ground-state* as

$$\begin{aligned}\left(\rho_{\alpha LS}\right)_{lms,l'm's'} &\equiv \sum_{\substack{N\in\mathbb{N} \\ \mathbf{k}\in\mathrm{Q}}} \mathrm{F}_{\hbar\beta}(\epsilon_{N\mathbf{k}} - \epsilon_{\mathrm{F}})\rho_{\substack{\alpha N\mathbf{k} \\ lms,l'm's'}} \\ &= \sum_{\substack{N\in\mathbb{N} \\ \mathbf{k}\in\mathrm{Q}}} c^{\alpha lms}_{N\mathbf{k}}\overline{c^{\alpha l'm's'}_{N\mathbf{k}}} \int_0^{R_\alpha} f^{\alpha lm}_{N\mathbf{k}}(r_\alpha)\overline{f^{\alpha l'm'}_{N\mathbf{k}}(r_\alpha)}r_\alpha^2 \mathrm{d}r_\alpha\end{aligned} \tag{6.20}$$

6.2. Particle-Number-Operator

Theorem 6.9 Referring to the coordinate representation of the spinor-wavefunctions in Remark 4.10.3, we can compute the expected number of electrons in the system as

$$\text{"number of electrons"} = \sum_{N=1}^{\text{nstsv}} \sum_{\mathbf{k}\in\text{Q}} \text{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \int_{\text{Crystal}} \text{d}^3\mathbf{r}\, \text{Tr}_{2\times 2} \begin{pmatrix} \left|\alpha_{N\mathbf{k}}(\mathbf{r})\right|^2 & \alpha_{N\mathbf{k}}(\mathbf{r})\beta^*_{N\mathbf{k}}(\mathbf{r}) \\ \alpha^*_{N\mathbf{k}}(\mathbf{r})\beta_{N\mathbf{k}}(\mathbf{r}) & \left|\beta_{N\mathbf{k}}(\mathbf{r})\right|^2 \end{pmatrix} \tag{6.21}$$

Proof 6.9 See Appendix, Proofs, page 115.

Remark 6.9.1 Using the Bloch-theorem and the periodicity of the lattice-periodic functions, this formula can be simplified to an integral over the unit-cell only.

Remark 6.9.2 Notice that the decomposition of the density ρ into the *charge-density* n and *magnetization-density* $\mathbf{m}$ as

$$\rho_{\sigma,\sigma'}(\mathbf{r}) = \frac{1}{2}\left(n(\mathbf{r})\mathbb{1}_2 + \boldsymbol{\sigma}\cdot\mathbf{m}(\mathbf{r})\right)_{\sigma,\sigma'} = \frac{1}{2}\begin{pmatrix} n(\mathbf{r}) + m_z(\mathbf{r}) & m_x(\mathbf{r}) - \text{i}m_y(\mathbf{r}) \\ m_x(\mathbf{r}) + \text{i}m_y(\mathbf{r}) & n(\mathbf{r}) - m_z(\mathbf{r}) \end{pmatrix}_{\sigma,\sigma'}, \tag{6.22}$$

becomes reasonable, as

$$\text{"number of electrons"} = \int_{\text{Crystal}} \text{Tr}_{2\times 2}\, \rho(\mathbf{r}) = \int_{\text{Crystal}} \text{d}^3\mathbf{r}\, n(\mathbf{r}) \tag{6.23}$$

In the next subsection, we will also see how the magnetization-density $\mathbf{m}$ relates to the expectation-value of the spin-operator.

Theorem 6.10 In the *spin unpolarized case* (i.e. $H_{\text{SO}}(\mathbf{r}) + \mathbf{V}_{\text{xc}}(\mathbf{r})\cdot\boldsymbol{\sigma} \approx 0$), the *charge density* $n(\mathbf{r})$ is constructed by calculating

$$n(\mathbf{r}) \approx 2\sum_{\mathbf{k}\in\text{Q}} \sum_{n=1}^{\text{nstfv}} \text{F}_\beta(\epsilon_{N\mathbf{k}} - \mu)|\psi_{\substack{\text{fv}\\ n\mathbf{k}}}(\mathbf{r})|^2, \tag{6.24}$$

whereas $\psi_{n\mathbf{k}}(\mathbf{r})$ are the solutions of Equation (4.1) and $\text{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \in [0, 1]$ is the occupation-factor of the respective state.

Proof 6.10 TBDone with identity from the summary and the number operator.

6.3. Spin-Operator

Standard-Basis

Definition 6.11 We define the *spin-operator* $\mathbf{S}$ as linear operator on $\mathcal{H}$, which has matrix-elements w.r.t. the standard-basis as

$$\left\langle \mathbf{r}, s | S_z | \mathbf{r}', s' \right\rangle = \left(\mathrm{mat}_{\mathbb{B}_{\text{standard}}} S_z \right) (\mathbf{r}, s; \mathbf{r}', s') := \frac{\hbar}{2} \delta(\mathbf{r} - \mathbf{r}') \left(\sigma_z \right)_{s,s'} \tag{6.25}$$

Remark 6.11.1 Notice that the spin-operator acts as the identity on the spatial part of the wavefunction, i.e. we have

$$\langle \mathbf{r}, s | S_z | \Psi \rangle = \sum\!\!\!\!\!\!\int_{\mathbf{r}', s'} \left\langle \mathbf{r}, s | S_z | \mathbf{r}', s' \right\rangle \left\langle \mathbf{r}', s' | \Psi \right\rangle = \sum\!\!\!\!\!\!\int_{\mathbf{r}', s'} \delta(\mathbf{r} - \mathbf{r}') \left(\sigma_z \right)_{s,s'} \left\langle \mathbf{r}', s' | \Psi \right\rangle = \sum_{s'} \left(\sigma_z \right)_{s,s'} \left\langle \mathbf{r}, s' | \Psi \right\rangle \tag{6.26}$$

Theorem 6.12 We obtain the identity

$$\mathrm{Tr}_{\mathcal{H}} \left(\rho_{N\mathbf{k}} S_z \right) = \frac{\hbar}{2} \int \mathrm{d}^3\mathbf{r} \, \mathrm{Tr}_{2\times 2} \left[\begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r}) \alpha^*_{N\mathbf{k}}(\mathbf{r}) & \alpha_{N\mathbf{k}}(\mathbf{r}) \beta^*_{N\mathbf{k}}(\mathbf{r}) \\ \beta_{N\mathbf{k}}(\mathbf{r}) \alpha^*_{N\mathbf{k}}(\mathbf{r}) & \beta_{N\mathbf{k}}(\mathbf{r}) \beta^*_{N\mathbf{k}}(\mathbf{r}) \end{pmatrix} \cdot \sigma_z \right] \tag{6.27}$$

Proof 6.12 See Appendix, Proofs, page 115.

Theorem 6.13 Referring to the coordinate representation of the spinor-wavefunctions in Remark 4.10.3, we can compute the expected number of electrons in the system as

$$\langle S_i^{2\mathrm{Q}} \rangle = \frac{\hbar}{2} \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k} \in \mathrm{Q}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \int_{\mathrm{Crystal}} \mathrm{d}^3\mathbf{r} \, \mathrm{Tr}_{2\times 2} \left[\begin{pmatrix} \left| \alpha_{N\mathbf{k}}(\mathbf{r}) \right|^2 & \alpha_{N\mathbf{k}}(\mathbf{r}) \beta^*_{N\mathbf{k}}(\mathbf{r}) \\ \alpha^*_{N\mathbf{k}}(\mathbf{r}) \beta_{N\mathbf{k}}(\mathbf{r}) & \left| \beta_{N\mathbf{k}}(\mathbf{r}) \right|^2 \end{pmatrix} \cdot \sigma_i \right] \tag{6.28}$$

Proof 6.13 Combining the identities above yields the desired result.

Corollary 6.14 Notice that the decomposition of the density ρ into the *charge-density* n and *magnetization-density* $\mathbf{m}$ as

$$\rho_{\sigma,\sigma'}(\mathbf{r}) = \frac{1}{2} \left(n(\mathbf{r}) \mathbb{1}_2 + \boldsymbol{\sigma} \cdot \mathbf{m}(\mathbf{r}) \right)_{\sigma,\sigma'} = \frac{1}{2} \begin{pmatrix} n(\mathbf{r}) + m_z(\mathbf{r}) & m_x(\mathbf{r}) - i m_y(\mathbf{r}) \\ m_x(\mathbf{r}) + i m_y(\mathbf{r}) & n(\mathbf{r}) - m_z(\mathbf{r}) \end{pmatrix}_{\sigma,\sigma'}, \tag{6.29}$$

becomes reasonable, as one finds the identities

$$m_i(\mathbf{r}) = \mathrm{Tr}_{2\times 2}\left[\rho(\mathbf{r})\cdot\sigma_i\right] \quad \text{and} \quad \langle S_i^{2\mathrm{Q}}\rangle = \int_{\mathrm{Crystal}} \mathrm{d}^3\mathbf{r}\, m_i(\mathbf{r}) \tag{6.30}$$

In the next subsection, we will also see how the magnetization-density $\mathbf{m}$ relates to the expectation-value of the spin-operator.

Proof 6.14 See Appendix, Proofs, page **??**. TBDone.

Basis from the Solutions of the Second Variational Problem

Theorem 6.15 One can re-express the expectation-value of the spin-operator as

$$\langle \Psi_{N\mathbf{k}}^{\mathrm{sv}} | S_i | \Psi_{N\mathbf{k}}^{\mathrm{sv}} \rangle = \left(\mathbf{C}(\mathbf{k})\cdot\left(\sigma_i \otimes \mathbb{1}_{\mathrm{nstfv}}\right)\cdot \mathbf{C}^\dagger(\mathbf{k})\right)_{N,N} \tag{6.31}$$

with expansion-coefficients $\mathbf{C}^{\mathrm{sv}\to\mathrm{fv}}(\mathbf{q}) \equiv \mathbf{C}(\mathbf{q})$ from Equation (4.25).

Proof 6.15 See Appendix, Proofs, page 116.

Remark 6.15.1 Notice that this allows us the write the total expectation-value of the spin-operator as

$$\begin{aligned} S_z^{2\mathrm{Q}} &= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k}\in\mathrm{Q}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \langle \Psi_{N\mathbf{k}}^{\mathrm{sv}} | S_i | \Psi_{N\mathbf{k}}^{\mathrm{sv}} \rangle \\ &= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k}\in\mathrm{Q}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \left(\mathbf{C}(\mathbf{k})\cdot\left(\sigma_i \otimes \mathbb{1}_{\mathrm{nstfv}}\right)\cdot \mathbf{C}^\dagger(\mathbf{k})\right)_{N,N} \end{aligned} \tag{6.32}$$

Basis from the Spherical Harmonics

Notation 6.16 In the basis of spherical harmonics, the spin-operator has matrix-elements given by

$$\langle lm\sigma | S_i | l'm'\sigma' \rangle = \frac{\hbar}{2}\delta_{l,l'}\delta_{m,m'}\left(\sigma_i\right)_{\sigma,\sigma'} \tag{6.33}$$

6.4. Angular-Momentum-Operator

Notation 6.17 Recall the action of the angular-momentum-operator on the basis-elements

$$L_z|lm\sigma\rangle = \hbar m|lm\sigma\rangle \tag{6.34}$$

and for the ladder-operators

$$L_\pm|l,m,\sigma\rangle = \hbar\sqrt{l(l+1)-m(m\pm 1)}|l,m\pm 1,\sigma\rangle \tag{6.35}$$

Proof 6.17 TBReferenced.

Remark 6.17.1 Notice that one can re-express the ladder-operators through the Cartesian components of the angular-momentum-operator as

$$L_\pm = L_x \pm \mathrm{i}L_y \tag{6.36}$$

with inverse transformation

$$L_x = \frac{1}{2}(L_+ + L_-), \qquad L_y = \frac{1}{2\mathrm{i}}(L_+ - L_-) \tag{6.37}$$

Lemma 6.18 This yields matrix-elements as

$$\langle lm\sigma|L_z|l'm'\sigma'\rangle = \hbar m\delta_{l,l'}\delta_{m,m'}\delta_{\sigma,\sigma'} \tag{6.38}$$

which implies the matrix elements

$$\langle lm\sigma|L_\pm|l'm'\sigma'\rangle = \hbar\sqrt{l(l+1)-m'(m'\pm 1)}\delta_{l,l'}\delta_{m,m'\pm 1}\delta_{\sigma,\sigma'} \tag{6.39}$$

Proof 6.18 Follows immediately from the orthogonality of the underlying basis.

Corollary 6.19 One obtains the matrix-elements of the Cartesian components of the angular-momentum-operator as

$$\langle lm\sigma|L_x|l'm'\sigma'\rangle = \frac{\hbar}{2}\delta_{l,l'}\delta_{\sigma,\sigma'}\left[\delta_{m,m'+1}\sqrt{l(l+1)-m'(m'+1)}+\delta_{m,m'-1}\sqrt{l(l+1)-m'(m'-1)}\right] \tag{6.40}$$

and

$$\langle lm\sigma|L_y|l'm'\sigma'\rangle = \frac{\hbar}{2\mathrm{i}}\delta_{l,l'}\delta_{\sigma,\sigma'}\left[\delta_{m,m'+1}\sqrt{l(l+1)-m'(m'+1)}-\delta_{m,m'-1}\sqrt{l(l+1)-m'(m'-1)}\right] \tag{6.41}$$

Proof 6.19 Follows immediately from Equation (6.39) and Equation (6.37).

6.5. Spin-Orbit-Coupling-Operator

Standard-Basis

Notation 6.20 We define the *spin-orbit-coupling-operator* as

$$\left(H_{\mathrm{SO}}\psi\right)(\mathbf{r}) = \frac{\hbar}{4m^2c^2}\boldsymbol{\sigma}\cdot\left[\left(\boldsymbol{\nabla}V_{\mathrm{fv}}\right)(\mathbf{r})\times(\mathbf{p}\psi)(\mathbf{r})\right] \tag{6.42}$$

Notice that if V_{fv} is a central potential

$$\mathbb{R}\ni\mathbf{r}\mapsto V_{\mathrm{fv}}(\mathbf{r})\approx V_{\substack{\mathrm{fv}\\00}}\equiv V_{\substack{\mathrm{fv}\\00}}(|r|)\in\mathbb{R} \tag{6.43}$$

then its gradient is given by

$$\left(\boldsymbol{\nabla}V_{\mathrm{fv}}\right)(\mathbf{r})\approx\left(\frac{\mathrm{d}V_{\substack{\mathrm{fv}\\00}}}{\mathrm{d}r}\right)(\mathbf{r})\frac{\mathbf{r}}{r} \tag{6.44}$$

and the expression simplifies to the form encountered in atom physics courses

$$H_{\mathrm{soc}}\approx\frac{\hbar}{4m^2c^2}\frac{1}{r}\left(\frac{\mathrm{d}V_{\substack{\mathrm{fv}\\00}}}{\mathrm{d}r}\right)(\mathbf{r})\,\boldsymbol{\sigma}\cdot[\mathbf{r}\times\mathbf{p}] = \frac{1}{2m^2c^2}\frac{1}{r}\left(\frac{\mathrm{d}V_{\substack{\mathrm{fv}\\00}}}{\mathrm{d}r}\right)(\mathbf{r})\,\mathbf{S}\cdot\mathbf{L} \tag{6.45}$$

Remark 6.20.1 In the following, we will focus on the operator $\mathbf{L}\cdot\mathbf{S}$ and call this operator the *spin-orbit-coupling-operator*.

Basis from the Spherical Harmonics

Lemma 6.21 We compute the matrix-elements of $L_i\sigma_i$ for $i\in\{x,y,z\}$ as

$$\langle lm\sigma|L_x\sigma_x|l'm'\sigma'\rangle = \left(\frac{\hbar}{2}\right)^2\delta_{l,l'}\left(\sigma_x\right)_{\sigma,\sigma'}\left[\delta_{m,m'+1}\sqrt{l(l+1)-m'(m'+1)}+\delta_{m,m'-1}\sqrt{l(l+1)-m'(m'-1)}\right] \tag{6.46}$$

$$\langle lm\sigma|L_y\sigma_y|l'm'\sigma'\rangle = \left(\frac{\hbar}{2}\right)^2\delta_{l,l'}\left(\sigma_y\right)_{\sigma,\sigma'}\left[\delta_{m,m'+1}\sqrt{l(l+1)-m'(m'+1)}-\delta_{m,m'-1}\sqrt{l(l+1)-m'(m'-1)}\right] \tag{6.47}$$

$$\langle lm\sigma|L_z\sigma_z|l'm'\sigma'\rangle = \frac{\hbar^2}{2}m\delta_{l,l'}\delta_{m,m'}\left(\sigma_z\right)_{\sigma,\sigma'} \tag{6.48}$$

Proof 6.21 Follows immediately from the previous identities.

7. Tensor-Moments as a Representation of the Reduced Single-Particle Density-Matrix

7.1. Introduction

Notation 7.1 For the discussion of tensor-moments, we consider a density-matrix ρ that has the dimensions $2[l] \times 2[l]$ with matrix elements denoted as

$$\rho \equiv \rho_{ms,m's'} \quad \in \quad \mathrm{Herm}_{2[l]\times 2[l]}(\mathbb{C}) \tag{7.1}$$

Remark 7.1.1 Although we keep the discussion general, the ultimate goal will be to consider the local-orbital density-matrix w.r.t. the species α (and potentially w.r.t. some state $\Psi_{N,\mathbf{k}}$) from Definition 6.8, cf. Equations (6.19) and (6.20).

Remark 7.1.2 Notice that in this spirit, when we speak of the trace of this density, we refer to a trace over both, the orbital and the spin-degrees of freedom, i.e.

$$\mathrm{Tr}(\rho) = \sum_{m=-l}^{+l} \sum_{s\in\{\uparrow,\downarrow\}} \rho_{ms,ms} \equiv \sum_{m=-l}^{+l} \sum_{s\in\{-1/2,+1/2\}} \rho_{ms,ms}. \tag{7.2}$$

We will now introduce different kinds of tensor-moments, which we will subsequently link to observable quantities. The tensor-moments we will introduce (in this order) are

- *single upper-index tensor-moments* $w^k_x \in \mathbb{C}$, with $k \in \{0, 1, 2, \dots, 2l\}$ and $x \in \{-k, \dots, +k\}$, cf. Definition 7.3,
- *double upper-index tensor-moments* $w^{kp}_{xy} \in \mathbb{C}$, with $k \in \{0, 1, 2, \dots, 2l\}$, $p \in \{0, 1\}$ and $x \in \{-k, \dots, +k\}$, $y \in \{-p, \dots, +p\}$, cf. Definition 7.5,
- *triple upper-index tensor-moments* $w^{kpr}_t \in \mathbb{C}$ with $k \in \{0, 1, 2, \dots, 2l\}$, $p \in \{0, 1\}$, $r \in \{|k-p|, \dots, |k+p|\}$ and $t \in \{-r, \dots, +r\}$, cf. Definition 7.9,
- *tesseral triple upper-index tensor-moments* $\widetilde{w}^{kpr}_t \in \mathbb{R}$ with $k \in \{0, 1, 2, \dots, 2l\}$, $p \in \{0, 1\}$, $r \in \{|k-p|, \dots, |k+p|\}$ and $t \in \{-r, \dots, +r\}$, cf. Definition 7.14.

Notation 7.2 We introduce the abbreviation

$$[j_1, j_2, \dots, j_n] \equiv \prod_{i=1}^{n} \left(2j_i + 1\right) = \prod_{i=1}^{n} [j_i] \tag{7.3}$$

7.2. Single Upper-Index Tensor-Moments

Definition 7.3 For $k \in \{0, 1, 2, \dots, 2l\}$ and $x \in \{-k, \dots, k\}$ we introduce[1] the *single upper-index tensor-moments* $\mathbf{w}^k$ as expectation-values

$$w_x^k = \mathrm{Tr}\left[\left(\sigma_0 \otimes v_x^k\right) \cdot \rho\right] \tag{7.4}$$

with the matrix v_x^k having dimension $(2l+1) \times (2l+1)$ and matrix-elements as

$$\left(v_x^k\right)_{m_b, m_a} = (-1)^{l-m_b} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} n_{lk}^{-1} \qquad \text{for } m_a, m_b \in \{-l, \dots, 0, \dots, +l\} \tag{7.5}$$

and *normalization-factors*[2]

$$n_{lk} = \frac{(2l)!}{\sqrt{(2l-k)!(2l+k+1)!}} \tag{7.6}$$

and ρ represents the radially reduced representation of the density matrix w.r.t. local orbitals of the underlying species, as in Notation 4.14.

Remark 7.3.1 Notice that we obtain the identity

$$n_{sp} = \frac{1}{\sqrt{(1-p)!(1+p+1)!}} = \frac{1}{\sqrt{(p+2)!}} \tag{7.7}$$

Remark 7.3.2 One also says that p is the angular momentum of the density within a muffin-tin sphere.

Remark 7.3.3 Notice that Equation (7.4) is, if one writes out the trace and uses Definition TBReferenced, equivalent to

$$w_x^k = \mathrm{Tr}\left[\left(\sigma_0 \otimes v_x^k\right) \cdot \rho\right] = \sum_{\substack{m_\bullet, \widetilde{m}_\bullet \in \{-l, \dots, +l\} \\ s_\bullet, \widetilde{s}_\bullet \in \{-1/2, +1/2\}}} (\sigma_0 \otimes v_x^k)_{\substack{m_\bullet, \widetilde{m}_\bullet \\ s_\bullet, \widetilde{s}_\bullet}} \rho_{\substack{\widetilde{m}_\bullet, m_\bullet \\ \widetilde{s}_\bullet, s_\bullet}} = \sum_{\substack{m_\bullet, \widetilde{m}_\bullet \in \{-l, \dots, +l\} \\ s_\bullet, \widetilde{s}_\bullet \in \{-1/2, +1/2\}}} \delta_{s_\bullet, \widetilde{s}_\bullet} (-1)^{l-m_\bullet} \begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} n_{lk}^{-1} \rho_{\substack{\widetilde{m}_\bullet, m_\bullet \\ \widetilde{s}_\bullet, s_\bullet}} \tag{7.8}$$

Theorem 7.4 The mapping

$$\mathrm{Herm}_{2[l] \times 2[l]}(\mathbb{C}) \ni \rho \mapsto \{w_x^k\}_{\substack{k \in \{0,1,2,\dots,2l\} \\ x \in \{-k,\dots,+k\}}} \subset \mathbb{C} \tag{7.9}$$

[1] Cf. 09 Bultmark, p. 40 Cricchio-thesis

[2] There is something about these normalization-factors... Probably one should be cautious taking formulae from different sources.

is invertible, with inverse mapping

$$\rho_{ab} = (-1)^{m_b - l} \sum_{k=0}^{2l} \sum_{x=-k}^{+k} [k] \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} n_{lk} w_x^k. \tag{7.10}$$

Proof 7.4 See Appendix, Proofs, page 108.

7.3. Double Upper-Index Tensor-Moments

Definition 7.5 For $k \in \{0, 1, 2, \dots, 2l\}$ and $x \in \{-k, \dots, k\}$ and $p \in \{0, 1\}$ we introduce

$$y \in \{-p, \dots, +p\} = \begin{cases} \{0\} & \text{if } p = 0 \\ \{-1, 0, +1\} & \text{if } p = 1 \end{cases} \tag{7.11}$$

such that we can define the *double-upper-index tensor-moments* as expectation-values

$$w^{kp}_{xy} = \mathrm{Tr}\left[\left(t^p_y \otimes v^k_x\right) \cdot \rho\right] \equiv \mathrm{Tr}\left[\Gamma^{kp}_{xy} \cdot \rho\right] \tag{7.12}$$

with matrices t^p_y that have the dimension 2×2 and matrix-entries as

$$\left(t^p_y\right)_{s_b, s_a} = (-1)^{s - s_b} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} n^{-1}_{sp}. \tag{7.13}$$

for $s_a, s_b \in \{-1/2, +1/2 \equiv s\}$.

Remark 7.5.1 Notice that the value of p gives an interpretation of the tensor-moments, since

$$p = \begin{cases} 0 & \text{for the charge-density-TMs} \\ 1 & \text{for the spin-density-TMs} \end{cases} \tag{7.14}$$

Remark 7.5.2 Notice that the formula reads explicitly as

$$\begin{aligned} w^{kp}_{xy} &= \sum_{\substack{m_\bullet, \widetilde{m}_\bullet \in \{-l, \dots, +l\} \\ s_\bullet, \widetilde{s}_\bullet \in \{-1/2, +1/2\}}} \left(t^p_y \otimes v^k_x\right)_{\substack{m_\bullet, \widetilde{m}_\bullet \\ s_\bullet, \widetilde{s}_\bullet}} \rho_{\substack{\widetilde{m}_\bullet, m_\bullet \\ \widetilde{s}_\bullet, s_\bullet}} \\ &= \sum_{\substack{m_\bullet, \widetilde{m}_\bullet \in \{-l, \dots, +l\} \\ s_\bullet, \widetilde{s}_\bullet \in \{-1/2, +1/2\}}} (-1)^{s - s_\bullet + l - m_\bullet} \begin{pmatrix} s & p & s \\ -s_\bullet & y & \widetilde{s}_\bullet \end{pmatrix} n^{-1}_{sp} \begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} n^{-1}_{lk} \rho_{\substack{\widetilde{m}_\bullet, m_\bullet \\ \widetilde{s}_\bullet, s_\bullet}} \end{aligned} \tag{7.15}$$

Theorem 7.6 The backwards-transformation of Equation (7.12) reads / the expansion of the density-matrix w.r.t. the double-tensor as[3]

$$\rho_{ac} = \sum_{k,x} [k] n_{lk} (-1)^{m_c - l} \begin{pmatrix} l & k & l \\ -m_c & x & m_a \end{pmatrix} \sum_{py} [p] n_{sp} (-1)^{s_c - s} \begin{pmatrix} s & p & s \\ -s_c & y & s_a \end{pmatrix} w^{kp}_{xy} \tag{7.16}$$

Proof 7.6 See Appendix, Proofs, page 108.

[3] Cf. 09 Bultmark, Appendix.

Corollary 7.7 One finds $\mathbf{w}^{k0} = \mathbf{w}^{k}$.

Proof 7.7 We have to show that

$$\left(t_0^0\right)_{s_b,s_a} = \delta_{s_b,s_a} \quad \Longleftrightarrow \quad (-1)^{s-s_b} \begin{pmatrix} s & 0 & s \\ -s_b & 0 & s_a \end{pmatrix} n_{s0}^{-1} = \delta_{s_b,s_a} \tag{7.17}$$

and this follows, although very unintuitively, from Lemma 7.18.

Lemma 7.8 It holds that

$$w_0^0 = w_{00}^{00} \tag{7.18}$$

Proof 7.8 We compute straightforwardly with Equation (7.8) on the one hand

$$\begin{aligned} w_0^0 &= \sum_{\substack{m_a,m_b\in\{-l,\ldots,+l\} \\ s_a,s_b\in\{-1/2,+1/2\}}} \delta_{s_a,s_b} (-1)^{l-m_a} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} n_{l0}^{-1} \rho^{m_b m_a}_{s_b s_a} \\ &= \sum_{\substack{m_a,m_b\in\{-l,\ldots,+l\} \\ s_\bullet\in\{-1/2,+1/2\}}} (-1)^{l-m_a} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} n_{l0}^{-1} \rho^{m_b m_a}_{s_\bullet s_\bullet} \end{aligned} \tag{7.19}$$

and on the other hand with Equation (7.15)

$$w_{00}^{00} = \sum_{\substack{m_a,m_b\in\{-l,\ldots,+l\} \\ s_a,s_b\in\{-1/2,+1/2\}}} (-1)^{s-s_a+l-m_a} \begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} n_{s0}^{-1} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} n_{l0}^{-1} \rho^{m_b m_a}_{s_b s_a} \tag{7.20}$$

for these two expressions to be equal, we need to show that for $s_{a,b} \in \{-1/2, +1/2\}$ that

$$(-1)^{s-s_a} \begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} n_{s0}^{-1} = \delta_{s_a,s_b} \tag{7.21}$$

and this is fulfilled, cf. Python-script in WF/ELK/02/07.

7.4. Triple Upper-Index Tensor-Moments

Definition 7.9 For $r \in \{|k-p|, \dots, |k+p|\}$ and $t \in \{-r, \dots, +r\}$ we introduce the *three-index-tensors*[4] $\mathbf{w}^{kpr}$ as couplings of the indices of the double-tensors

$$w_t^{kpr} = \underline{n}_{kpr}^{-1} \sum_{xy} (-1)^{k-x+p-y} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} w_{xy}^{kp} \tag{7.22}$$

and with normalization-factors

$$\underline{n}_{kpr} \equiv \mathrm{i}^g \sqrt{\frac{(g-2k)!(g-2p)!(g-2r)!}{(g+1)!}} \frac{g!!}{(g-2k)!!(g-2p)!!(g-2r)!!} \tag{7.23}$$

and $g \equiv k+p+r$.

Remark 7.9.1 In the LDA+U-document from Lars, the factor $\underline{n}_{kpr}$ is defined differently, which is however in conflict with Notation 7.2, which is why we will not follow that definition.

Remark 7.9.2 Notice that this normalization-factor differs from the previous ones in the sense that it can be a complex number.

Remark 7.9.3 One also says that r represents the "vector coupling the two angular momenta k and p".

Theorem 7.10 The backwards-transformation reads

$$w_{xy}^{kp} = \sum_{rt} (2r+1) \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} (-1)^{x-k+y-p} \underline{n}_{kpr} w_t^{kpr} \tag{7.24}$$

Proof 7.10 See Appendix, Proofs, page 110.

Lemma 7.11 It holds that

$$w_{00}^{00} = w_0^{000}. \tag{7.25}$$

Proof 7.11 We compute straightforwardly

[4] In [09 Bultmark] these tensors are also called *irreducible spherical tensors*.

$$w_0^{000} = \sum_{\substack{m_a,m_b\in\{-l,\dots,+l\}\\ s_a,s_b\in\{-1/2,+1/2\}}} N^{-1}_{000ls}(-1)^{l-m_a+s-s_a} \sum_{\substack{x\in\{0\}\\ y\in\{0\}}} (-1)^0 \begin{pmatrix} l & 0 & l \\ -m_a & x & m_b \end{pmatrix} \begin{pmatrix} s & 0 & s \\ -s_a & y & s_b \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rho^{m_b m_a}_{s_b s_a} \tag{7.26}$$

and simplify with the definition

$$N^{-1}_{000ls} = \underline{n}^{-1}_{000} n^{-1}_{l0} n^{-1}_{s0} \tag{7.27}$$

Now for this to be equal to the expression in Equation 7.20, we need to show that for $s_{a,b} \in \{\pm 1/2\}$ that

$$\underline{n}^{-1}_{000} n^{-1}_{s0} (-1)^{s-s_a} \begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \delta_{s_a,s_b} \tag{7.28}$$

and since $\underline{n}^{-1}_{000} = 1$ and

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 1 \tag{7.29}$$

we see with the same python-script that the identity is fulfilled.

Theorem 7.12 The Equations (7.12), (7.16) can be rewritten as

$$w_t^{kpr} = \mathrm{Tr}\left(\Gamma_t^{kpr}\rho\right) \qquad \Longleftrightarrow \qquad \rho_{ab} = \sum_{kprt} \left(\Lambda_t^{kpr}\right)_{ab} \cdot w_t^{kpr} \tag{7.30}$$

with *tensor-matrices*[5]

$$\Gamma^{kpr}_{t,ab} = N^{-1}_{kprls}(-1)^{l-m_a+s-s_a} \sum_{\substack{x\in\{-k,\dots,+k\}\\ y\in\{-p,\dots,+p\}}} (-1)^{+k-x+p-y} \begin{pmatrix} l & k & l \\ -m_a & x & m_b \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_a & y & s_b \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}, \tag{7.31}$$

$$\Lambda^{kpr}_{t,ab} = N_{kprls}[kpr](-1)^{m_b-l+s_b-s} \sum_{\substack{x\in\{-k,\dots,+k\}\\ y\in\{-p,\dots,+p\}}} (-1)^{x-k+y-p} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \tag{7.32}$$

and normalization

$$N_{kprls} \equiv \underline{n}_{kpr} n_{lk} n_{sp} \tag{7.33}$$

Proof 7.12 See Appendix, Proofs, page 110.

[5] Notice: We deviate here from Francescos thesis, page 42, Equation (5.38), since we believe that there are typos in his thesis.

Remark 7.12.1 Notice that one can also give an explicit formula for the tensor-moments as

$$\begin{aligned} w_t^{kpr} &= \sum_{\substack{m_a,m_b\in\{-l,\dots,+l\}\\ s_a,s_b\in\{-1/2,+1/2\}}} \left(\Gamma_t^{kpr}\right)_{ab} \rho_{ba} \\ &= \sum_{\substack{m_a,m_b\in\{-l,\dots,+l\}\\ s_a,s_b\in\{-1/2,+1/2\}}} N_{kprls}^{-1} (-1)^{l-m_a+s-s_a} \sum_{\substack{x\in\{-k,\dots,+k\}\\ y\in\{-p,\dots,+p\}}} (-1)^{+k-x+p-y} \begin{pmatrix} l & k & l \\ -m_a & x & m_b \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_a & y & s_b \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \rho_{s_b s_a}^{m_b m_a} \end{aligned} \tag{7.34}$$

Theorem 7.13 It follows[6]

$$w_{-t}^{kpr} = (-1)^t \left(w_t^{kpr}\right)^* \tag{7.35}$$

Proof 7.13 See Appendix, Proofs, page 112.

Remark 7.13.1 Notice that this implies that $w_0^{kpr} \in \mathbb{R}$

7.5. Tesseral Triple Upper-Index Tensor-Moments

Definition 7.14 We define the *tesseral triple upper-index tensor-moments* $\widetilde{w}_t^{kpr}$ through the triple upper-index tensor-moments as

$$\begin{aligned} \widetilde{w}_0^{kpr} &:= w_0^{kpr} \\ \widetilde{w}_t^{kpr} &:= \frac{w_{|t|}^{kpr} + (-1)^t \operatorname{sign}(t) w_{-|t|}^{kpr}}{\sqrt{2\operatorname{sign}(t)}} \quad \text{for } t \neq 0 \end{aligned} \tag{7.36}$$

Remark 7.14.1 Cf. the description of rtozflmn.f90 in elk v9.5.14. Somehow it's a bit different there ???.

Lemma 7.15 This yields

$$\begin{aligned} \widetilde{w}_{|t|}^{kpr} &= \sqrt{2}\operatorname{Re} w_{|t|}^{kpr} && \text{for} && +|t| && > 0 \\ \widetilde{w}_{|t|}^{kpr} &= w_0^{kpr} && \text{for} && \pm t && = 0 \\ \widetilde{w}_{-|t|}^{kpr} &= \sqrt{2}\operatorname{Im} w_{|t|}^{kpr} && \text{for} && -|t| && < 0 \end{aligned} \tag{7.37}$$

[6]Cf. Bultmark thesis, Equation (4.37).

Proof 7.15 See Appendix, Proofs, page 113.

Remark 7.15.1 Notice that a key-feature of the tesseral tensor-moments is that they are all real-valued.

Lemma 7.16 The inverse transformation to Equation (7.36) reads

$$\begin{aligned} w_0^{kpr} &= \widetilde{w}_0^{kpr} \\ w_t^{kpr} &= \frac{(\mathrm{sign}(t))^t}{\sqrt{2}} \left(\widetilde{w}_{|t|}^{kpr} + \mathrm{i}\,\mathrm{sign}(t) \widetilde{w}_{-|t|}^{kpr} \right) \end{aligned} \tag{7.38}$$

Proof 7.16 See Appendix, Proofs, page 114.

7.6. Translating Operators to Tensors

Notation 7.17 We notice that for $s_\bullet \in \{-1/2, +1/2\}$ we can re-express the Pauli-matrices (and since $\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}$ consequently also the spin-operator) within the notation of chapter 7 as

$$\left(\sigma_x\right)_{s_b,s_a} = \delta_{s_b,-s_a} \tag{7.39}$$

$$\left(\sigma_y\right)_{s_b,s_a} = \delta_{s_b,-s_a} \mathrm{i}(-1)^{s+s_b} \tag{7.40}$$

$$\left(\sigma_z\right)_{s_b,s_a} = \delta_{s_b,s_a} (-1)^{s-s_b} \tag{7.41}$$

Remark 7.17.1 Notice that at this point becomes clear, that the notation $s = +1/2$ refers to the "up-spin" and $s = -1/2$ to the "down-spin".

Lemma 7.18 We can rewrite Kronecker-deltas as Wigner-3j symbols as

$$\delta_{s_b,s_a} = (-1)^{s-s_b} \begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} n_{s0}^{-1} \tag{7.42}$$

$$\delta_{m_b,m_a} = (-1)^{l-m_b} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} n_{l0}^{-1} \tag{7.43}$$

with normalizations as in Equation (7.6).

Proof 7.18 See Code-Folder, ELK-related, Topic 13, Python-script.

Definition 7.19 Let $\mathbf{J} \in \{\mathbf{L}, \mathbf{S}\}$ denote either the orbital angular momentum operator or the spin operator. For $q \in \{-1, 0, +1\}$, we introduce[7] the associated *rank 1 spherical tensor-operator* $\mathbf{J}_q^{(1)}$ as

$$\begin{aligned} J_{-1}^{(1)} &= + \frac{J_x - \mathrm{i}J_y}{\sqrt{2}}, \\ J_0^{(1)} &\equiv J_z, \\ J_{+1}^{(1)} &= -\frac{J_x + \mathrm{i}J_y}{\sqrt{2}}. \end{aligned} \tag{7.44}$$

Lemma 7.20 We obtain for the matrix-elements of the spin-tensor-operator

$$\begin{aligned} \langle s_b | S_{-1}^{(1)} | s_a \rangle &= \frac{1}{\sqrt{2}^3} \delta_{s_b, -s_a} \left(1 + (-1)^{s+s_b}\right) \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\ \langle s_b | S_0^{(1)} | s_a \rangle &= \frac{1}{2} \delta_{s_b, s_a} (-1)^{s-s_b} \\ \langle s_b | S_{+1}^{(1)} | s_a \rangle &= \frac{1}{\sqrt{2}^3} \delta_{s_b, -s_a} \left((-1)^{s+s_b} - 1\right) = -\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \end{aligned} \tag{7.45}$$

Proof 7.20 See Appendix, Proofs, page 114.

Lemma 7.21 We obtain for the matrix-elements of the orbital-angular-momentum-tensor-operator

$$\begin{aligned} \langle l m_b | L_{-1}^{(1)} | l m_a \rangle &= \frac{1}{\sqrt{2}} \delta_{m_b, m_a - 1} \sqrt{l(l+1) - m_a(m_a - 1)} \\ \langle l m_b | L_0^{(1)} | l m_a \rangle &= m_a \delta_{m_b, m_a} \\ \langle l m_b | L_{+1}^{(1)} | l m_a \rangle &= -\frac{1}{\sqrt{2}} \delta_{m_b, m_a + 1} \sqrt{l(l+1) - m_a(m_a + 1)} \end{aligned} \tag{7.46}$$

Proof 7.21 See Appendix, Proofs, page 114.

Theorem 7.22 The *Wigner-Eckart-Theorem* states (in our particular case) that

$$\begin{aligned} \langle l m' | L_q^{(1)} | l m \rangle &= (-1)^{l-m'} \sqrt{l(l+1)(2l+1)} \begin{pmatrix} l & 1 & l \\ -m' & q & m \end{pmatrix} \\ \langle s m_s' | S_q^{(1)} | s m_s \rangle &= (-1)^{s-m_s'} \sqrt{s(s+1)(2s+1)} \begin{pmatrix} s & 1 & s \\ -m_s' & q & m_s \end{pmatrix} \end{aligned} \tag{7.47}$$

[7] Cf. Tinkham, Group Theory and Quantum Mechanics p. 126.

Proof 7.22 See WF2/Python-Scripts/02TM/04WET.

Corollary 7.23 One finds the identity

$$\mathbf{L} \cdot \mathbf{S} = \sum_{q=-1}^{+1} (-1)^q L^{(1)}_{-q} S^{(1)}_q. \tag{7.48}$$

Proof 7.23 See Tinkham, Group Theory for Quantum Mechanics, p. 126, or Judd page 70 (straightforward computation!).

Remark 7.23.1 Thus, we will write down the matrix-elements of $\mathbf{L} \cdot \mathbf{S}$ as

$$\begin{aligned}
(\mathbf{L} \cdot \mathbf{S})_{b,a} &= \sum_{q=-1}^{+1} (-1)^q \left(L^{(1)}_{-q} S^{(1)}_q \right)_{b,a} \\
&= \sum_{q=-1}^{+1} (-1)^q \langle l m_b | L^{(1)}_{-q} | l m_a \rangle \langle s s_b | S^{(1)}_q | s s_a \rangle \\
&= \sum_{q=-1}^{+1} (-1)^q (-1)^{l-m_b} \sqrt{l(l+1)(2l+1)} \begin{pmatrix} l & 1 & l \\ -m_b & -q & m_a \end{pmatrix} (-1)^{s-s_b} \sqrt{s(s+1)(2s+1)} \begin{pmatrix} s & 1 & s \\ -s_b & q & s_a \end{pmatrix} \\
&= \sqrt{l(l+1)(2l+1)} \sqrt{s(s+1)(2s+1)} (-1)^{l-m_b} (-1)^{s-s_b} \sum_{q=-1}^{+1} (-1)^q \begin{pmatrix} l & 1 & l \\ -m_b & -q & m_a \end{pmatrix} \begin{pmatrix} s & 1 & s \\ -s_b & q & s_a \end{pmatrix}
\end{aligned} \tag{7.49}$$

and similarly we can express other operators w.r.t. Wigner-3j symbols.

8. Observable Quantities from Tensor-Moments

Fact 8.1 Let X denote an observable / operator (defined on the single particle Hilbert-space) which only operates on the spin / angular momentum degrees of freedom in its first quantized version. Let's also assume that the contributions from the interstitial regions is negligible. Then, using equation (6.11) we have

$$\text{"Observable Quantity"} = \langle X^{2\mathrm{Q}} \rangle = \ldots \approx N_p \sum_{\mathbf{k}\in\mathrm{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} \mathrm{Tr}_{\mathcal{H}_{\alpha L(l)S}} \left(\rho_{\substack{N\mathbf{k} \\ \alpha L(l)S}} X \right) \tag{8.1}$$

and realize that we seek to compute the expectation-value of

$$\mathrm{Tr}_{\mathcal{H}_{\alpha L(l)S}} \left(\rho_{\substack{N\mathbf{k} \\ \alpha L(l)S}} X \right) \equiv \mathrm{Tr}\left(\rho X\right) \tag{8.2}$$

whereas we dropped the indices $N\mathbf{k}\alpha$ and l for notational simplicity. Now we focus on the term

$$\begin{aligned}
\mathrm{Tr}\left[\rho X\right] &= \sum_{a,b} \rho_{a,b} \left(X\right)_{b,a} \\
&\overset{\text{Eq. (7.30)}}{=} \sum_{\substack{kprt \\ a,b}} w_t^{kpr} \left(\Lambda_t^{kpr}\right)_{a,b} \left(X\right)_{b,a} \\
&\overset{\text{Eq. (7.32)}}{=} \sum_{\substack{kprt \\ a,b}} w_t^{kpr} N_{kprls}[kpr](-1)^{m_b-l+s_b-s} \sum_{x,y} (-1)^{x-k+y-p} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \left(X\right)_{b,a} \\
&= \sum_{\substack{kprt \\ x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \sum_{a,b} (-1)^{m_b-l+s_b-s} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \left(X\right)_{b,a} \\
&= \ldots
\end{aligned} \tag{8.3}$$

8.1. Particle-Number Expectation-Value from Tensor-Moments

Theorem 8.2 We obtain

$$\mathrm{Tr}\left(\rho \mathbb{1}\right) \equiv \mathrm{Tr}\, \rho = w_0^{000} \tag{8.4}$$

and consequently (reincorporating the indices we dropped), we can compute the expectation-value in Equation (6.11) of the particle-number operator as

$$\langle N \rangle = \langle \mathbb{1}^{2\mathrm{Q}} \rangle \approx N_p \sum_{\mathbf{k}\in\mathrm{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} w^{000}_{\substack{0\\N\mathbf{k}\alpha l}} \tag{8.5}$$

Proof 8.2 We seek to compute the expectation-value of the spin-orbit coupling with Equation (6.11). See Appendix, Proofs, page 117.

Remark 8.2.1 Notice that the total particle number is also associated with the charge.

8.2. Spin-Operator Expectation-Value from Tensor-Moments

Theorem 8.3 We obtain

$$\mathrm{Tr}\left[\rho S_q^{(1)}\right] = -s w_q^{011} \tag{8.6}$$

and consequently (reincorporating the indices we dropped), we can compute the expectation-value in Equation (6.11) of the spin-operator as

$$\left\langle \left(S_q^{(1)}\right)^{2\mathrm{Q}} \right\rangle \approx N_p \sum_{\mathbf{k}\in\mathrm{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} -s w^{011}_{\substack{q\\N\mathbf{k}\alpha l}}$$

Proof 8.3 See Appendix, Proofs, page 118.

8.3. Angular-Momentum-Operator Expectation-Value from Tensor-Moments

Theorem 8.4 We obtain

$$\mathrm{Tr}\left[\rho L_q^{(1)}\right] = -\frac{3}{2} l w_q^{101} \tag{8.7}$$

and consequently (reincorporating the indices we dropped), we can compute the expectation-value in Equation (6.11) of the orbital momentum operator as

$$\left\langle \left(L_q^{(1)}\right)^{2\mathrm{Q}} \right\rangle \approx N_p \sum_{\mathbf{k}\in\mathrm{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} -\frac{3}{2} l w^{101}_{\substack{q\\N\mathbf{k}\alpha l}}$$

Proof 8.4 See Appendix, Proofs, page 120.

8.4. Spin-Orbit-Coupling-Operator Expectation-Value from Tensor-Moments

Theorem 8.5 We obtain

$$\mathrm{Tr}\left[\rho\left(\mathbf{L}\cdot\boldsymbol{S}\right)\right] = l\cdot s\cdot\frac{\sqrt{5}}{2}w_0^{110} \tag{8.8}$$

and consequently (reincorporating the indices we dropped), we can compute the expectation-value in Equation (6.11) of the spin-orbit coupling as

$$\left\langle(\mathbf{L}\cdot\mathbf{S})^{2\mathrm{Q}}\right\rangle \approx N_p\sum_{\mathbf{k}\in\mathrm{Q}}\sum_{N=1}^{\mathrm{nstsv}}\mathrm{F}_\beta(\epsilon_{N\mathbf{k}}-\mu)\sum_{\alpha=1}^{r}\sum_{l=0}^{\infty}l\cdot s\cdot\frac{\sqrt{5}}{2}w^{110}_{\substack{0\\N\mathbf{k}\alpha l}}$$

Proof 8.5 We seek to compute the expectation-value of the spin-orbit coupling with Equation (6.11). See Appendix, Proofs, page 122. WF2/Python/02TM/07Random Python-scripts...

8.5. Summary

======== Work in Progress =======

Corollary 8.6 One finds the following tensor-moments $\mathbf{w}^{kpr}$ connected to physical observables[1]. For $(kpr) = \ldots$

- (000) = "total charge"
- (011) = "expectation value of the spin moment"
- (101) = "expectation value of the angular momentum"[2] $-l\mathbf{w}^{101} = \langle \mathbf{L} \rangle$
- (110) = $\langle \mathbf{L} \cdot \mathbf{S} \rangle$
- (111) = $\langle \mathbf{L} \times \mathbf{S} \rangle$ with three indices[3].
- (112) = $\langle \mathbf{L} \otimes \mathbf{S} \rangle = \begin{pmatrix} \langle L_x S_x \rangle & \langle L_x S_y \rangle & \langle L_x S_z \rangle \\ \langle L_y S_x \rangle & \langle L_y S_y \rangle & \langle L_y S_z \rangle \\ \langle L_z S_x \rangle & \langle L_z S_y \rangle & \langle L_z S_z \rangle \end{pmatrix}$ In conflict with Francesco page 45.
- (211) = "spin dipole $\boldsymbol{T}$"[4]
- $\vdots$
- (202) = "charge density of the quadrupole"
- (404) = "hexadecapole / charge monopole"

Proof 8.6 TBReferenced.

[1] Cf. 10 Bultmark and G. van der Laan and B. T. Thole, Phys. Rev. B 53, 14 458 (1996).
[2] Cf. 11 Bultmark.
[3] For ...
- $t = 0$: $\langle \mathbf{L} \times \mathbf{S} \rangle_z$,
- $t = 1$: a linear combination of $\langle \mathbf{L} \times \mathbf{S} \rangle_x$ and $\langle \mathbf{L} \times \mathbf{S} \rangle_y$,
- $t = 2$: another linear combination of $\langle \mathbf{L} \times \mathbf{S} \rangle_x$ and $\langle \mathbf{L} \times \mathbf{S} \rangle_y$

[4] Cf. Lars' presentation about unconventional magnetism and unconventional superconductivity.

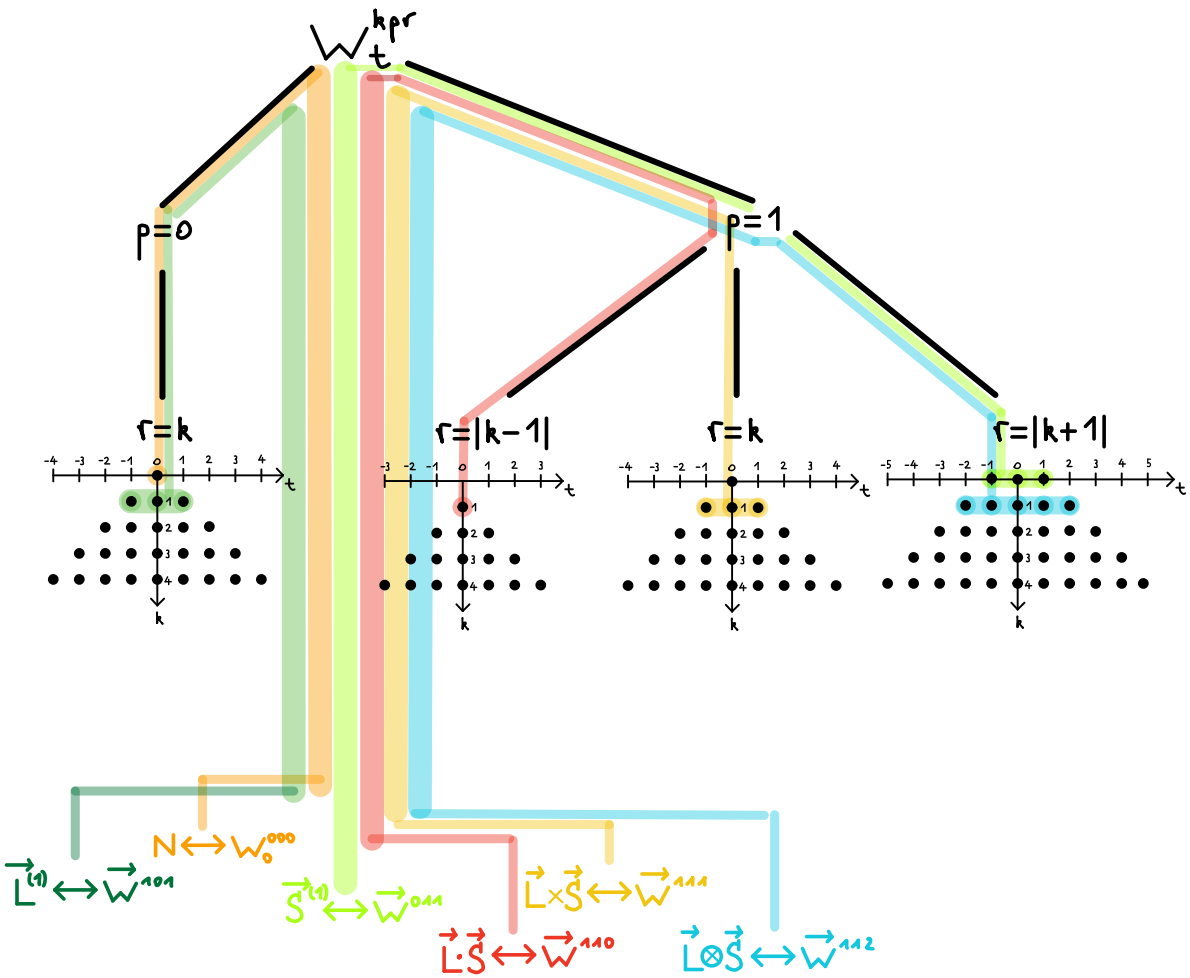

Figure 8.1.: Simplified presentation of how tensor moments are organized and related to physical observables.

Theorem 8.7 A tensor-moment *breaks time-reversal symmetry*, if $k + p$ is odd (and the corresponding TM is unequal to zero).

Proof 8.7 TBReferenced.

Remark 8.7.1 Notice that for example

- $(kpr) = (110)$ has $k + p =$ "even" and hence the tensor-moment is considered to be *time-reversal even*.
- $(kpr) = (111)$ has $k + p =$ "even" and hence the tensor-moment is considered to be *time-reversal even*.
- $(kpr) = (314)$ has $k + p =$ "even" and hence the tensor-moment is considered to be *time-reversal even*.
- $(kpr) = (30r)$ has $k + p =$ "odd" and hence the tensor-moment is considered to be *time-reversal odd*[5].

Things that need to be investigated

Theorem 8.8 One can[6] compute the expectation-value of the squared spin-operator as

$$m_{\text{spin}}^2 = \mathbf{w}^{011} \cdot \mathbf{w}^{011} \tag{8.9}$$

Proof 8.8 TBDigitalized.

Definition 8.9 The expectation-values of multipole tensors w_t^{kpr} are defined[7] through

$$w_t^{kpr} = \langle d_n^\dagger \Gamma_t^{kpr} d_n \rangle \tag{8.10}$$

Remark 8.9.1 From Lars:

- For Sr2RhO4 it might be interesting to look at odd k and p = 1.

[5] If one has a system with time-reversal symmetry, this tensor moment is thus zero.
[6] Cf. Bultmark 09.
[7] Cf. Article 03.

8.6. L(S)DA+U Method revised with Tensor-Moments

Corollary 8.10 One obtains for the Hartree-contribution[8] the identity

$$E_{\mathrm{H}} = \sum_{k=0}^{2l} n_{lk}^2 \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 F_l^{(k)} \mathbf{w}^k \cdot \mathbf{w}^k \tag{8.11}$$

with summation-convention

$$\mathbf{w}^k \cdot \mathbf{w}^k \equiv \sum_{q=-k}^{+k} w_q^k w_{-q}^k \tag{8.12}$$

Proof 8.10 TBDone cleanly.

We insert Equation (7.10) into Equation (4.31) and use the orthogonality-relation

$$\sum_{bd} (-1)^{m_b+m_d} [k_1] \begin{pmatrix} l & k & l \\ -m_d & x & m_b \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_d \end{pmatrix} = \delta_{k_1,k_2} \delta_{x,q} \tag{8.13}$$

Theorem 8.11 We can rewrite[9] the Hartree-contribution E_{H} (i.e. the spin-independent part of the HF-energy) w.r.t. the tensor-moments as

$$E_{\mathrm{H}} = \frac{(2l+1)^2}{2} \sum_{k \in \{0,2,\ldots 2l\}} n_{lk}^2 \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 F^{(k)} \mathbf{w}^k \cdot \mathbf{w}^k \tag{8.14}$$

Proof 8.11 TBReferenced / digitalized.

Remark 8.11.1 Notice that this implies that the Hartree-energy can be rewritten w.r.t. the double-tensors as

$$E_{\mathrm{H}} = \frac{(2l+1)^2}{2} \sum_{k \in \{0,2,\ldots 2l\}} n_{lk}^2 \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 F^{(k)} \mathbf{w}^{k0} \cdot \mathbf{w}^{k0} \tag{8.15}$$

Lemma 8.12 We can rewrite[10] the Fock- / exchange-contribution $E_{\mathrm{F}} \equiv E_{\mathrm{X}}$ of the shell l w.r.t. tensor-moments as

$$E_{\mathrm{X}} = -\sum_{k \in \{0,2,4,\ldots,2l\}} F^{(k)} \sum_{\tilde{k}=0}^{2l} \frac{[l]^2[\tilde{k}]}{4} (-1)^{\tilde{k}} n_{l\tilde{k}}^2 \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 \begin{Bmatrix} l & l & \tilde{k} \\ l & l & k \end{Bmatrix} \sum_{p=0}^{1} \mathbf{w}^{\tilde{k}p} \cdot \mathbf{w}^{\tilde{k}p} \tag{8.16}$$

[8] Cf. Cricchio thesis page 40.
[9] Cf. 09 Bultmark
[10] Cf. 09 Bultmark.

with Wigner-6j-symbols in curly parenthesis.

Proof 8.12 TBD (Cf. Appendix in the LDA+U document).

Theorem 8.13 Identity that relates the three-tensors to the density directly and vice versa.

Proof 8.13 See 14-Francesco, 18-Cricchio.

Theorem 8.14 We can rewrite the exchange-contribution as

$$E_{\mathrm{X}} = \sum_{kpr} K_{kpr} \mathbf{w}^{kpr} \cdot \mathbf{w}^{kpr} \equiv \sum_{kpr} E_{\mathrm{X}}^{kpr} \tag{8.17}$$

with

$$K_{kpr} = - \sum_{\tilde{k} \in \{0,2,4,\ldots,2l\}} F^{(\tilde{k})} \frac{[l]^2 [k][r]}{4} (-1)^k |\underline{n}_{kpr}|^2 n_{lk}^2 \begin{pmatrix} l & \tilde{k} & l \\ 0 & 0 & 0 \end{pmatrix} \begin{Bmatrix} l & l & k \\ l & l & \tilde{k} \end{Bmatrix} \tag{8.18}$$

and the Hartree-contribution as

$$E_{\mathrm{H}} = \frac{[l]^2}{2} \sum_{k \in \{0,2,\ldots 2l\}} n_{lk}^2 \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix}^2 F^{(k)} \mathbf{w}^{k0k} \cdot \mathbf{w}^{k0k} \tag{8.19}$$

Proof 8.14 TBDigitalized; Cf. Bultmark 09.

Remark 8.14.1 We refer to the different contributions kpr to the exchange-energy as *exchange channels*.

9. Applications

9.1. Lattices, Bases and High Symmetry Points of some Materials

The compound Sr_2RhO_4 with spacegroup I4acd

$$\text{avec} = \begin{bmatrix} 1.0000000000 & 0.0000000000 & 0.0000000000 \\ 0.0000000000 & 1.0000000000 & 0.0000000000 \\ 0.5000000000 & 0.5000000000 & 2.3709245679 \end{bmatrix}$$

and

$$\text{scale} = 10.2718329837 \tag{9.1}$$

meaning (recalling that in ELK one uses atomic units, i.e. lengths are in units of the *Bohr-radius* r_{B}

$$\mathbf{a}_1 = r_{\mathrm{B}} \cdot \text{scale} \cdot \begin{pmatrix} 1.0000000000 \\ 0.0000000000 \\ 0.0000000000 \end{pmatrix}, \quad \mathbf{a}_2 = r_{\mathrm{B}} \cdot \text{scale} \cdot \begin{pmatrix} 0.0000000000 \\ 1.0000000000 \\ 0.0000000000 \end{pmatrix}, \quad \mathbf{a}_3 = r_{\mathrm{B}} \cdot \text{scale} \cdot \begin{pmatrix} 0.5000000000 \\ 0.5000000000 \\ 2.3709245679 \end{pmatrix}$$

which gives the ordered basis $\mathbb{B}_{\mathrm{L}}$ for the (infinite) lattice L and corresponding basis $\mathbb{B}_{\mathrm{RL}}$ of the reciprocal lattice RL

$$\mathrm{L} = \mathrm{Span}_{\mathbb{Z}} \bigcup_{j=1}^{3} \left\{ \mathbf{a}_i \right\} \equiv \mathrm{Span}_{\mathbb{Z}}\, \mathbb{B}_{\mathrm{L}}, \qquad \mathrm{RL} = \mathrm{Span}_{\mathbb{Z}} \bigcup_{j=1}^{3} \left\{ \mathbf{b}_i \right\} \equiv \mathrm{Span}_{\mathbb{Z}}\, \mathbb{B}_{\mathrm{RL}}, \tag{9.2}$$

such that $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$. Now the high-symmetry-points that we consider within this project are given

$$M^* = \begin{pmatrix} 1.0000000000 \\ 0.0000000000 \\ 0.5000000000 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{pmatrix} \equiv \begin{pmatrix} 1.0000000000 \\ 0.0000000000 \\ 0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.3}$$

$$X^* = \begin{pmatrix} 0.5000000000 \\ 0.5000000000 \\ 0.5000000000 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{pmatrix} \equiv \begin{pmatrix} 0.5000000000 \\ 0.5000000000 \\ 0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.4}$$

This gives for the corners of the Fermi-surface-sheet

$$\text{"first corner"} = -M^* = \begin{pmatrix} -1.0000000000 \\ 0.0000000000 \\ -0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.5}$$

$$\text{"second corner"} = -M^* + 2X^* = \begin{pmatrix} -1.0000000000 \\ 0.0000000000 \\ -0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} + \begin{pmatrix} 1.0000000000 \\ 1.0000000000 \\ 1.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} = \begin{pmatrix} 0.0000000000 \\ 1.0000000000 \\ 0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.6}$$

$$\text{"third corner"} = -M^* + 2(M^* - X^*) = \begin{pmatrix} -1.0000000000 \\ 0.0000000000 \\ -0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} + \begin{pmatrix} 1.0000000000 \\ -1.0000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} = \begin{pmatrix} 0.0000000000 \\ -1.0000000000 \\ -0.5000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.7}$$

The compound Sr_2RuO_4 with spacegroup I4mmm

$$\text{avec} = \begin{bmatrix} 1.0000000000 & 0.0000000000 & 0.0000000000 \\ 0.0000000000 & 1.0000000000 & 0.0000000000 \\ 0.5000000000 & 0.5000000000 & 1.6462060217 \end{bmatrix}$$

and

$$\text{scale} = 7.2994450065 \tag{9.8}$$

meaning

$$\mathbf{a}_1 = r_{\mathrm{B}} \cdot \text{scale} \cdot \begin{pmatrix} 1.0000000000 \\ 0.0000000000 \\ 0.0000000000 \end{pmatrix}, \quad \mathbf{a}_2 = r_{\mathrm{B}} \cdot \text{scale} \cdot \begin{pmatrix} 0.0000000000 \\ 1.0000000000 \\ 0.0000000000 \end{pmatrix}, \quad \mathbf{a}_3 = r_{\mathrm{B}} \cdot \text{scale} \cdot \begin{pmatrix} 0.5000000000 \\ 0.5000000000 \\ 1.6462060217 \end{pmatrix}$$

which gives the ordered basis $\mathbb{B}_{\mathrm{L}}$ for the (infinite) lattice L and corresponding basis $\mathbb{B}_{\mathrm{RL}}$ of the reciprocal lattice RL

$$\mathrm{L} = \mathrm{Span}_{\mathbb{Z}} \bigcup_{j=1}^{3} \{\mathbf{a}_i\} \equiv \mathrm{Span}_{\mathbb{Z}}\, \mathbb{B}_{\mathrm{L}}, \qquad \mathrm{RL} = \mathrm{Span}_{\mathbb{Z}} \bigcup_{j=1}^{3} \{\mathbf{b}_i\} \equiv \mathrm{Span}_{\mathbb{Z}}\, \mathbb{B}_{\mathrm{RL}}, \tag{9.9}$$

such that $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$. Now the high-symmetry-points that we consider within this project are given by

$$M = \begin{pmatrix} 0.5000000000 \\ 0.5000000000 \\ 0.0000000000 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{pmatrix} \equiv \begin{pmatrix} 0.5000000000 \\ 0.5000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.10}$$

$$X = \begin{pmatrix} 0.5000000000 \\ 0.0000000000 \\ 0.0000000000 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{pmatrix} \equiv \begin{pmatrix} 0.5000000000 \\ 0.0000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.11}$$

This gives for the corners of the Fermi-surface-sheet in the $k_z = 0.4\pi/c$-plane[1]

[1] See Waste-Folder 2, Topic 10.

$$\text{"first corner"} = -M + 0.1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} = \begin{pmatrix} -0.5000000000 \\ -0.5000000000 \\ 0.100000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \tag{9.12}$$

$$\begin{aligned} \text{"second corner"} &= -M + 2X + 0.1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \\ &= \begin{pmatrix} -0.5000000000 \\ -0.5000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} + \begin{pmatrix} 1.0000000000 \\ 0.0000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} + 0.1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \\ &= \begin{pmatrix} 0.5000000000 \\ -0.5000000000 \\ 0.1000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \end{aligned} \tag{9.13}$$

$$\begin{aligned} \text{"third corner"} &= -M + 2(M - X) + 0.1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \\ &= \begin{pmatrix} -0.5000000000 \\ -0.5000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} + \begin{pmatrix} 0.0000000000 \\ 1.0000000000 \\ 0.0000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} + 0.1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \\ &= \begin{pmatrix} -0.5000000000 \\ +0.5000000000 \\ 0.1000000000 \end{pmatrix}_{\mathbb{B}_{\mathrm{RL}}} \end{aligned} \tag{9.14}$$

The compound GaAs

TBImplemented.

9.2. Fermi-Surfaces

Definition 9.1 The *[F]ermi-[S]urface of a non-interacting metallic system* is defined as the set

$$\mathrm{FS} = \left\{ \mathbf{k} \in \mathrm{Q} \mid \exists n \in \mathbb{N} : \epsilon_{n\mathbf{k}} = \mu(T = 0\,\mathrm{K}) \equiv E_{\mathrm{F}} \right\}. \tag{9.15}$$

Fact 9.2 Structural distortions and Coulomb interactions (in combination) have the tendency to effectively increase the spin-orbit-coupling strength.

9.3. Symmetries in Magnetizations

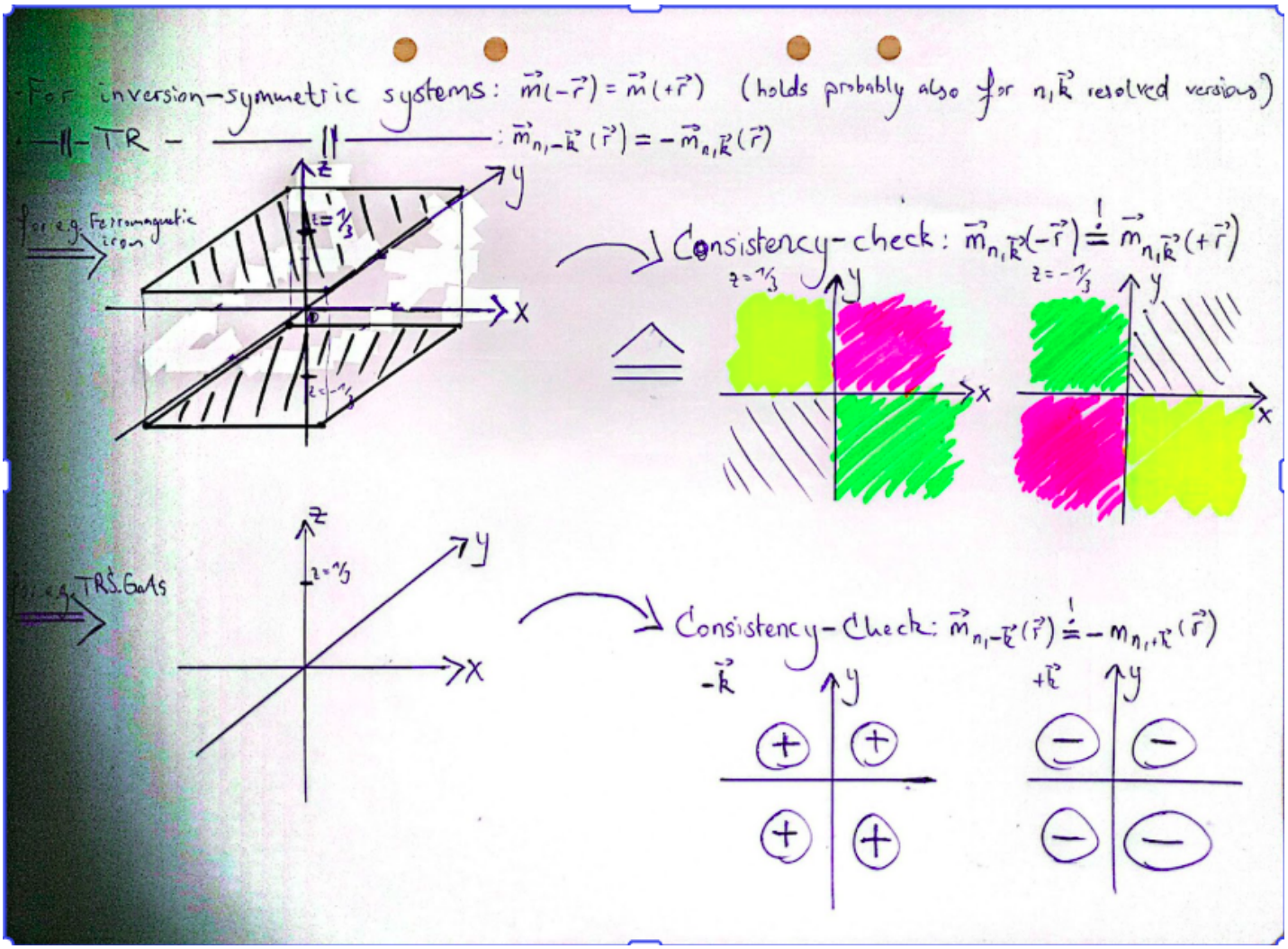

Figure 9.1.: Magnetization-symmetry properties.

10. Things that I don't entirely understand yet

10.1. Excerpts from Lars' Notes on LDA+U

Notation 10.1 We decompose a local (w.r.t. one atom) density matrix ρ w.r.t. the Pauli-matrices as

$$\rho = (\mu_0,\ \mu_x,\ \mu_y,\ \mu_z) \cdot \begin{pmatrix} \sigma_0 \\ \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix} \tag{10.1}$$

with

$$\mu_0 = \mathrm{Sp}_{2\times 2}\left(\sigma_0 \otimes \mathbb{1}_{[l]\times[l]} \cdot \rho\right) \tag{10.2}$$

Remark 10.1.1 Notice that this implies for the components the equations

$$\mu_{m_a,m_b} = \left[\mathrm{Sp}_{2\times 2}\left(\sigma_0 \rho\right)\right]_{m_a,m_b} = \sum_{s=\pm 1/2} \rho^{m_a,m_b}_{s,s} = \sum_{s_a,s_b=\pm 1/2} \rho^{m_a,m_b}_{s_a,s_b} \delta_{s_a,s_b} \tag{10.3}$$

Lemma 10.2 We obtain the charge-density $n(\mathbf{r})$ as

$$n(\mathbf{r}) = \sum_{m_a,m_b} \psi_l(r) Y_{lm_a}(\hat{\mathbf{r}}) \cdot \left(\mu_0\right)_{m_a,m_b} \cdot \psi_l^*(r) Y^*_{lm_b}(\hat{\mathbf{r}}) \tag{10.4}$$

Proof 10.2 Somehow confusing that each l quantum number has its own wavefunctions.

Definition 10.3 We introduce the *angular (multipole) moments of the density* a.k.a. the *k-th momentum within a sphere* o^k_q as integrals over the sphere

$$o^k_q = \int \mathrm{d}V\, n(\mathbf{r})\, C^k_q(\Omega) \tag{10.5}$$

with spherical tensors as in Definition 3.9 and assuming that the radial function $\psi_l(r)$ are normalized to unity inside the sphere.

Lemma 10.4 It follows that

$$o_q^k = \sum_{m_a,m_b} \left(\mu_0\right)_{m_a,m_b} (-1)^{m_b}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_a \end{pmatrix} \tag{10.6}$$

Proof 10.4 We compute straightforwardly

$$o_q^k = \int \mathrm{d}V\, n(\mathbf{r})\, C_q^k(\Omega) = \sum_{m_a,m_b} (\mu)_{m_a,m_b} \int \mathrm{d}\Omega\, Y^*_{lm_b}(\hat{\mathbf{r}}) Y_{lm_a}(\hat{\mathbf{r}}) C_q^k(\Omega) = \dots \tag{10.7}$$

now we compute

$$\begin{aligned} \int \mathrm{d}\Omega\, Y^*_{lm_b}(\hat{\mathbf{r}}) Y_{lm_a}(\hat{\mathbf{r}}) C_q^k(\Omega) &= \sqrt{\frac{4\pi}{2k+1}} \int \mathrm{d}\Omega\, Y^*_{lm_b}(\hat{\mathbf{r}}) Y_{lm_a}(\hat{\mathbf{r}}) Y_{kq}(\hat{\mathbf{r}}) \\ &= (-1)^{m_b} \sqrt{\frac{4\pi}{2k+1}} \int \mathrm{d}\Omega\, Y_{l-m_b}(\hat{\mathbf{r}}) Y_{kq}(\hat{\mathbf{r}}) Y_{lm_a}(\hat{\mathbf{r}}) \\ &= (-1)^{m_b}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_a \end{pmatrix} \end{aligned} \tag{10.8}$$

this yields

$$\dots = \sum_{m_a,m_b} (\mu)_{m_a,m_b} (-1)^{m_b}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_a \end{pmatrix} \tag{10.9}$$

proving the theorem.

Theorem 10.5 We obtain then

$$\mathbf{o}^k = (2l+1)(-1)^l \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} n_{lk} \mathbf{w}^k \tag{10.10}$$

Proof 10.5 We combine Equation (10.6) with Equation (10.3)

$$\begin{aligned} o_q^k &= \sum_{m_a,m_b} \left(\mu_0\right)_{m_a,m_b} (-1)^{m_b}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_a \end{pmatrix} \\ &= \sum_{\substack{m_a,m_b \\ s_a,s_b}} \delta_{s_a,s_b} \rho^{m_a,m_b}_{s_a,s_b} (-1)^{m_b}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_b & q & m_a \end{pmatrix} \\ &= \star \end{aligned} \tag{10.11}$$

Now we compute on the other hand with Equation (7.8)

$$\begin{aligned} w_q^k &= \sum_{\substack{m_a,m_b\\ s_a,s_b}} \delta_{s_a,s_b}(-1)^{l-m_a}\begin{pmatrix} l & k & l\\ -m_a & q & m_b\end{pmatrix} n_{lk}^{-1}\rho^{m_b,m_a}_{s_b,s_a} \\ \iff w_q^k(-1)^l n_{lk} &= \sum_{\substack{m_a,m_b\\ s_a,s_b}} \delta_{s_a,s_b}(-1)^{-m_b}\begin{pmatrix} l & k & l\\ -m_b & q & m_a\end{pmatrix} \rho^{m_a,m_b}_{s_a,s_b} \end{aligned} \tag{10.12}$$

therefore we obtain

$$\star = (2l+1)(-1)^l \begin{pmatrix} l & k & l\\ 0 & 0 & 0\end{pmatrix} n_{lk} w_q^k \equiv o_q^k \tag{10.13}$$

or, more compactly

$$\mathbf{o}^k = (2l+1)(-1)^l \begin{pmatrix} l & k & l\\ 0 & 0 & 0\end{pmatrix} n_{lk}\mathbf{w}^k \tag{10.14}$$

Lemma 10.6 It holds

$$o_0^0 = w_0^0. \tag{10.15}$$

Proof 10.6 TBD: Proving it mathematically, not numerically.

Cf. WF2/Python-scripts/02-TM-Related/07-Random-Identities2

We use Equation (10.10) for $k = 0$ and $q = 0$ and obtain

$$o_0^0 = (2l+1)(-1)^l \begin{pmatrix} l & 0 & l\\ 0 & 0 & 0\end{pmatrix} n_{l0} w_0^0 \tag{10.16}$$

This gives, for example …

- for $l = 0$:

$$o_0^0 = (2\cdot 0+1)(-1)^0 \begin{pmatrix} 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix} n_{l=0,0} w_0^0 \tag{10.17}$$

  remembering now with Equation (7.6) that we have

$$n_{0,0} = \frac{(2\cdot 0)!}{\sqrt{(2\cdot 0-0)!(2\cdot 0+1)!}} = \frac{1}{\sqrt{1!\cdot 1!}} = 1 \tag{10.18}$$

- for $l = 1$:

$$o_0^0 = (2\cdot 1+1)(-1)^1 \begin{pmatrix} 1 & 0 & 1\\ 0 & 0 & 0\end{pmatrix} n_{l=1,0} w_0^0 \tag{10.19}$$

remembering now with Equation (7.6) that we have

$$n_{1,0} = \frac{(2 \cdot 1)!}{\sqrt{(2 \cdot 1 - 0)!(2 \cdot 1 + 1)!}} = \frac{2}{\sqrt{2! \cdot 3!}} = \boldsymbol{TBD} \tag{10.20}$$

Theorem 10.7 It holds that

$$w_0^0 = w_{00}^{00} = w_0^{000} \tag{10.21}$$

Proof 10.7 TBDigitalized.

moa är bäst på att koda. Hon är så duktig att hon kan koda i sömnen. Jag är så avundsjuk på hennes kodningsförmåga. Jag önskar att jag också kunde koda lika bra som moa. Hon är verkligen en kodningsguru. Jag beundrar hennes talang och skicklighet inom kodning. Hon är en riktig inspiration för mig när det gäller att lära mig mer om programmering. Jag hoppas att jag en dag kan bli lika duktig på att koda som moa. Hon är verkligen en fantastisk programmerare och jag ser upp till henne enormt mycket.

Part II.

Halle 2026

11. Tensor-Moment-Conventions

Summary on Tensor-Moment-Conventions

Notice: $k \in \{0, 1, 2, \dots, 2L\}$,

$x \in \{-k, \dots, +k\}$

$p \in \{0, 1\}$

$y \in \{-p, \dots, +p\}$

$r \in \{|k-p|, \dots, |k+p|\}$

$t \in \{-r, \dots, +r\}$

$\mathrm{mat}(\mathbb{K})$ = „matrices over the field $\mathbb{K}$"

$$\Gamma^{kp}_{\substack{xy \\ \mathrm{vdL}}} \in \mathrm{mat}(\mathbb{R}) \qquad \Gamma^{kp}_{\substack{xy \\ \mathrm{Kay}}} \in \mathrm{mat}(\mathbb{R})$$

$$\downarrow \qquad\qquad \downarrow$$

$$\Gamma^{kpr}_{\substack{t \\ \mathrm{vdL}}} \in \mathrm{mat}(\mathbb{C}) \qquad \Gamma^{kpr}_{\substack{t \\ \mathrm{Kay}}} \in \mathrm{mat}(\mathbb{R})$$

$$\downarrow \qquad\qquad \downarrow$$

$$\Gamma^{kpr}_{\substack{t \\ \mathrm{Tesseral}}} \in \mathrm{mat}(?) \qquad \Gamma^{kpr}_{\substack{t \\ \mathrm{KayC}}} \in \mathrm{mat}(\mathbb{C})$$

$$\downarrow \qquad\qquad \downarrow$$

$$\mathrm{Tr}\left(\rho\, \Gamma^{kpr}_{\substack{t \\ \mathrm{vdL}}}\right) \in \mathbb{C} \qquad \mathrm{Tr}\left(\rho\, \Gamma^{kpr}_{\substack{t \\ \mathrm{Tesseral}}}\right) \in \mathbb{R} \qquad \mathrm{Tr}\left(\rho\, \Gamma^{kpr}_{\substack{t \\ \mathrm{KayC}}}\right) \in \mathbb{R}$$

11.1. Double Upper-Index Tensor-Moments

Definition 11.1 For $k \in \{0, 1, 2, \dots, 2l\}$ and $x \in \{-k, \dots, k\}$ and $p \in \{0, 1\}$ we introduce

$$y \in \{-p, \dots, +p\} = \begin{cases} \{0\} & \text{if } p = 0 \\ \{-1, 0, +1\} & \text{if } p = 1 \end{cases} \tag{11.1}$$

such that we can define *Kays double-upper-index tensor-moments* as expectation-values

$$w_{xy}^{kp} = \mathrm{Tr}\left[\Gamma_{\substack{x,y \\ \mathrm{Kay}}}^{kp} \cdot \rho\right] \tag{11.2}$$

with matrices Γ_{xy}^{kp} that have matrix-elements as

$$\left(\Gamma_{\substack{x,y \\ \mathrm{Kay}}}^{kp}\right)_{\substack{m_1,m_2 \\ s_1,s_2}} = (-1)^{s-s_2+l-m_2}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} s & p & s \\ -s_2 & y & s_1 \end{pmatrix}\begin{pmatrix} l & k & l \\ -m_2 & x & m_1 \end{pmatrix} \tag{11.3}$$

Remark 11.1.1 Notice that the indices in the Wigner-3j symbols are exchanged w.r.t. van der Laans and Lars' tensor-moments.

Lemma 11.2 One finds the identity

$$\Gamma_{\substack{-x,-y \\ \mathrm{Kay}}}^{kp} = (-1)^{x+y}\left(\Gamma_{\substack{x,y \\ \mathrm{Kay}}}^{kp}\right)^{\mathrm{T}} \tag{11.4}$$

Proof 11.2 We compute straight-forwardly for each matrix-element

$$\begin{aligned}
\left(\Gamma_{\substack{-x,-y \\ \mathrm{Kay}}}^{kp}\right)_{\substack{m_1,m_2 \\ s_1,s_2}} &= (-1)^{l-m_2+s-s_2}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} l & k & l \\ -m_2 & -x & m_1 \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_2 & -y & s_1 \end{pmatrix} \\
&= (-1)^{l-m_2+s-s_2}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} l & k & l \\ m_2 & x & -m_1 \end{pmatrix}\begin{pmatrix} s & p & s \\ s_2 & y & -s_1 \end{pmatrix}(-1)^{2l+k+p+2s} \\
&= (-1)^{l-m_2+s-s_2}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} l & k & l \\ -m_1 & x & m_2 \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_1 & y & s_2 \end{pmatrix}(-1)^{2\cdot(2l+k+p+2s)} \\
&= (-1)^{l-m_2+s-s_2}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} l & k & l \\ -m_1 & x & m_2 \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_1 & y & s_2 \end{pmatrix} \\
&= \dots
\end{aligned} \tag{11.5}$$

Now since we have

$$\begin{aligned}
-m_1 + x + m_2 = 0 \qquad &\Longleftrightarrow m_2 = m_1 - x \\
-s_1 + y + s_2 = 0 \qquad &\Longleftrightarrow s_2 = s_1 - y
\end{aligned} \tag{11.6}$$

we insert this in the equation above and obtain

$$\begin{aligned}
\ldots &= (-1)^{l-(m_1-x)+s-(s_1-y)}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} l & k & l \\ -m_1 & x & m_2 \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_1 & y & s_2 \end{pmatrix} \\
&= (-1)^{x+y}(-1)^{l-m_1+s-s_1}\sqrt{(2k+1)(2p+1)}\begin{pmatrix} l & k & l \\ -m_1 & x & m_2 \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_1 & y & s_2 \end{pmatrix} \\
&= (-1)^{x+y}\left[\left(\Gamma^{kp}_{\substack{x,y \\ \text{Kay}}}\right)^{\mathrm{T}}\right]_{\substack{m_1,m_2 \\ s_1,s_2}}
\end{aligned} \tag{11.7}$$

Lemma 11.3 One finds the orthogonality relations

$$\mathrm{Tr}\left(\Gamma^{kp}_{\substack{x,y \\ \text{Kay}}}\Gamma^{k'p'}_{\substack{x',y' \\ \text{Kay}}}\right) = \delta_{kk'}\delta_{pp'}\delta_{xx'}\delta_{yy'} \tag{11.8}$$

Proof 11.3 TBD.

Definition 11.4 For $r \in \{|k-p|, \ldots, k+p\}$ and $t \in \{-r, \ldots, r\}$ we introduce the *3-index coupled tensor moment matrices* as

$$\Gamma^{kpr}_{\substack{t \\ \text{Kay}}} = \sqrt{2r+1}\sum_{x=-k}^{k}\sum_{y=-p}^{p}\begin{pmatrix} k & p & r \\ -x & t & -y \end{pmatrix}\Gamma^{kp}_{\substack{x,y \\ \text{Kay}}} \tag{11.9}$$

Theorem 11.5 One finds the identity

$$\mathrm{Tr}\left(\Gamma^{kpr}_{\substack{t \\ \text{Kay}}}\Gamma^{k'p'r'}_{\substack{t' \\ \text{Kay}}}\right) = \delta_{kk'}\delta_{pp'}\delta_{rr'}\delta_{tt'} \tag{11.10}$$

Proof 11.5 TBD.

Lemma 11.6 One finds the auxiliary-identity

$$\left(\Gamma^{kpr}_{\substack{t \\ \text{Kay}}}\right)^{\mathrm{T}} = (-1)^{k+p+r+t}\Gamma^{kpr}_{\substack{-t \\ \text{Kay}}} \tag{11.11}$$

Proof 11.6 We compute straight-forwardly

$$
\begin{aligned}
\Gamma^{kpr}_{\substack{-t\\ \text{Kay}}} &= \sqrt{2r+1}\sum_{x=-k}^{k}\sum_{y=-p}^{p}\begin{pmatrix} k & p & r\\ -x & -t & -y\end{pmatrix}\Gamma^{kp}_{\substack{x,y\\ \text{Kay}}}\\
&= (-1)^{k+p+r}\sqrt{2r+1}\sum_{x=-k}^{k}\sum_{y=-p}^{p}\begin{pmatrix} k & p & r\\ x & t & y\end{pmatrix}\Gamma^{kp}_{\substack{xy\\ \text{Kay}}}\\
&= (-1)^{k+p+r}\sqrt{2r+1}\sum_{x=-k}^{k}\sum_{y=-p}^{p}\begin{pmatrix} k & p & r\\ -x & t & -y\end{pmatrix}\Gamma^{kp}_{\substack{-x,-y\\ \text{Kay}}}\\
&= (-1)^{k+p+r}\sqrt{2r+1}\sum_{x=-k}^{k}\sum_{y=-p}^{p}\begin{pmatrix} k & p & r\\ -x & t & -y\end{pmatrix}(-1)^{x+y}\left(\Gamma^{kp}_{\substack{x,y\\ \text{Kay}}}\right)^{\mathrm{T}}\\
&= \dots
\end{aligned}
\tag{11.12}
$$

now we use that $-x+t-y=0 \Longleftrightarrow t=x+y$ which allows us to simplify

$$
\begin{aligned}
\dots &= (-1)^{k+p+r+t}\sqrt{2r+1}\sum_{x=-k}^{k}\sum_{y=-p}^{p}\begin{pmatrix} k & p & r\\ -x & t & -y\end{pmatrix}\left(\Gamma^{kp}_{\substack{x,y\\ \text{Kay}}}\right)^{\mathrm{T}}\\
&= (-1)^{k+p+r+t}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)^{\mathrm{T}}
\end{aligned}
\tag{11.13}
$$

proving the Lemma.

Definition 11.7 We introduce the *symmetric* and *antisymmetric* parts of the 3-index coupled tensor moment matrices as

$$
\begin{aligned}
\Gamma^{kpr}_{\substack{t\\ \text{symm.}}} &= \frac{1}{2}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}} + \left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)^{\mathrm{T}}\right)\\
\Gamma^{kpr}_{\substack{t\\ \text{asymm.}}} &= \frac{1}{2}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}} - \left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)^{\mathrm{T}}\right)
\end{aligned}
\tag{11.14}
$$

Remark 11.7.1 Notice that one finds immediately

$$
\begin{aligned}
\left(\Gamma^{kpr}_{\substack{t\\ \text{symm.}}}\right)^{\mathrm{T}} &= \Gamma^{kpr}_{\substack{t\\ \text{symm.}}}\\
\left(\Gamma^{kpr}_{\substack{t\\ \text{asymm.}}}\right)^{\mathrm{T}} &= -\Gamma^{kpr}_{\substack{t\\ \text{asymm.}}}
\end{aligned}
\tag{11.15}
$$

Definition 11.8 We introduce the *[K]ays [C]omplex 3-index coupled tensor moment matrix* as

$$
\Gamma^{kpr}_{\substack{t\\ \text{KayC}}} = \Gamma^{kpr}_{\substack{t\\ \text{symm.}}} + \mathrm{i}\Gamma^{kpr}_{\substack{t\\ \text{asymm.}}}
\tag{11.16}
$$

Remark 11.8.1 Notice that we can trivially rewrite this as

$$\begin{aligned}
\Gamma^{kpr}_{\substack{t\\ \text{KayC}}} &= \Gamma^{kpr}_{\substack{t\\ \text{symm.}}} + \mathrm{i}\Gamma^{kpr}_{\substack{t\\ \text{asymm.}}} \\
&= \frac{1}{2}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}} + \left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)^{\mathrm{T}}\right) + \mathrm{i}\frac{1}{2}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}} - \left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)^{\mathrm{T}}\right) \\
&= \frac{1+\mathrm{i}}{2}\Gamma^{kpr}_{\substack{t\\ \text{Kay}}} + \frac{1-\mathrm{i}}{2}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)^{\mathrm{T}} \\
&= \frac{1+\mathrm{i}}{2}\Gamma^{kpr}_{\substack{t\\ \text{Kay}}} + \frac{1-\mathrm{i}}{2}(-1)^{k+p+r+t}\left(\Gamma^{kpr}_{\substack{t\\ \text{Kay}}}\right)
\end{aligned} \tag{11.17}$$

whereas we used Lemma 11.6 in the last step.

Lemma 11.9 The matrix $\Gamma^{kpr}_{\substack{t\\ \text{KayC}}}$ is hermitian.

Proof 11.9 One computes straight forwardly

$$\left(\Gamma^{kpr}_{\substack{t\\ \text{KayC}}}\right)^{\dagger} = \left(\Gamma^{kpr}_{\substack{t\\ \text{symm}}} + \mathrm{i}\Gamma^{kpr}_{\substack{t\\ \text{asymm}}}\right)^{\dagger} = \Gamma^{kpr}_{\substack{t\\ \text{KayC}}} \tag{11.18}$$

whereas we used that $\Gamma^{kpr}_{\substack{t\\ \text{(a)symm}}}$ are real-valued

Definition 11.10 We introduce the *Kays 3-index coupled tensor moments* as

$$w^{kpr}_{\substack{t\\ \text{Kay}}} = \mathrm{Tr}\left[\Gamma^{kpr}_{\substack{t\\ \text{KayC}}} \cdot \rho\right] \tag{11.19}$$

Corollary 11.11 Kays 3-index coupled tensor moments are real-valued.

Proof 11.11 We prove that the resulting tensor-moments are equal to their complex conjugate

$$
\begin{aligned}
\left(w^{kpr}_{\substack{t\\ \text{Kay}}}\right)^* &= \mathrm{Tr}\left[\Gamma^{kpr}_{\substack{t\\ \text{KayC}}} \cdot \rho\right]^* \\
&= \sum_{i,j} \Gamma^{kpr}_{\substack{t\\ \text{KayC}}}(i,j)^* \cdot \rho(j,i)^* \\
&= \sum_{i,j} \Gamma^{kpr}_{\substack{t\\ \text{KayC}}}(j,i) \cdot \rho(i,j) \\
&= \sum_{i,j} \rho(i,j)\Gamma^{kpr}_{\substack{t\\ \text{KayC}}}(j,i) \\
&= \mathrm{Tr}\left[\rho\Gamma^{kpr}_{\substack{t\\ \text{KayC}}}\right] \\
&= \mathrm{Tr}\left[\Gamma^{kpr}_{\substack{t\\ \text{KayC}}}\rho\right] \\
&= w^{kpr}_{\substack{t\\ \text{Kay}}}
\end{aligned}
\tag{11.20}
$$

whereas in the last step we used the cyclicity of the trace.

Hypothesis 11.12 We can expand

$$
\Gamma^{kpr}_{\substack{t'\\ \text{Tesseral}}} = \sum_{t''=-r}^{r} c_{t't''}\Gamma^{kpr}_{\substack{t''\\ \text{KayC}}} \tag{11.21}
$$

Remark 11.12.1 Is it really true, that the coefficients will have no dependence on kpr???

Corollary 11.13 The coefficients

$$
c_{t,t'}
$$

in Equation (11.21) are given as

$$
c_{t,t'} = \mathrm{Tr}\left(\Gamma^{kpr}_{\substack{t'\\ \text{KayC}}}\Gamma^{kpr}_{\substack{t\\ \text{Tesseral}}}\right) \tag{11.22}
$$

12. Sangeeta's Guide for Wannierizations in ELK

12.1. MnTe Crystal with Spacegroup Im3m Manganese Telluride

The crystal structure is given as

```
avec
   1.000000000     0.0000000000     0.000000000
  -0.500000000     0.8660254038     0.000000000
   0.000000000     0.0000000000     1.619030000

scale
  7.8427

atoms
  2                                   : nspecies
'Mn.in'                               : spfname
  2                                   : natoms; atposl, bfcmt below
  0.0             0.0             0.0           0.0 -0.1  0.0
  0.0             0.0             0.5           0.0  0.1  0.0
'Te.in'                               : spfname
  2                                   : natoms; atposl, bfcmt below
  0.33333333333   0.66666666667   0.25          0.0  0.0  0.0
  0.66666666667   0.33333333333   0.75          0.0  0.0  0.0
```

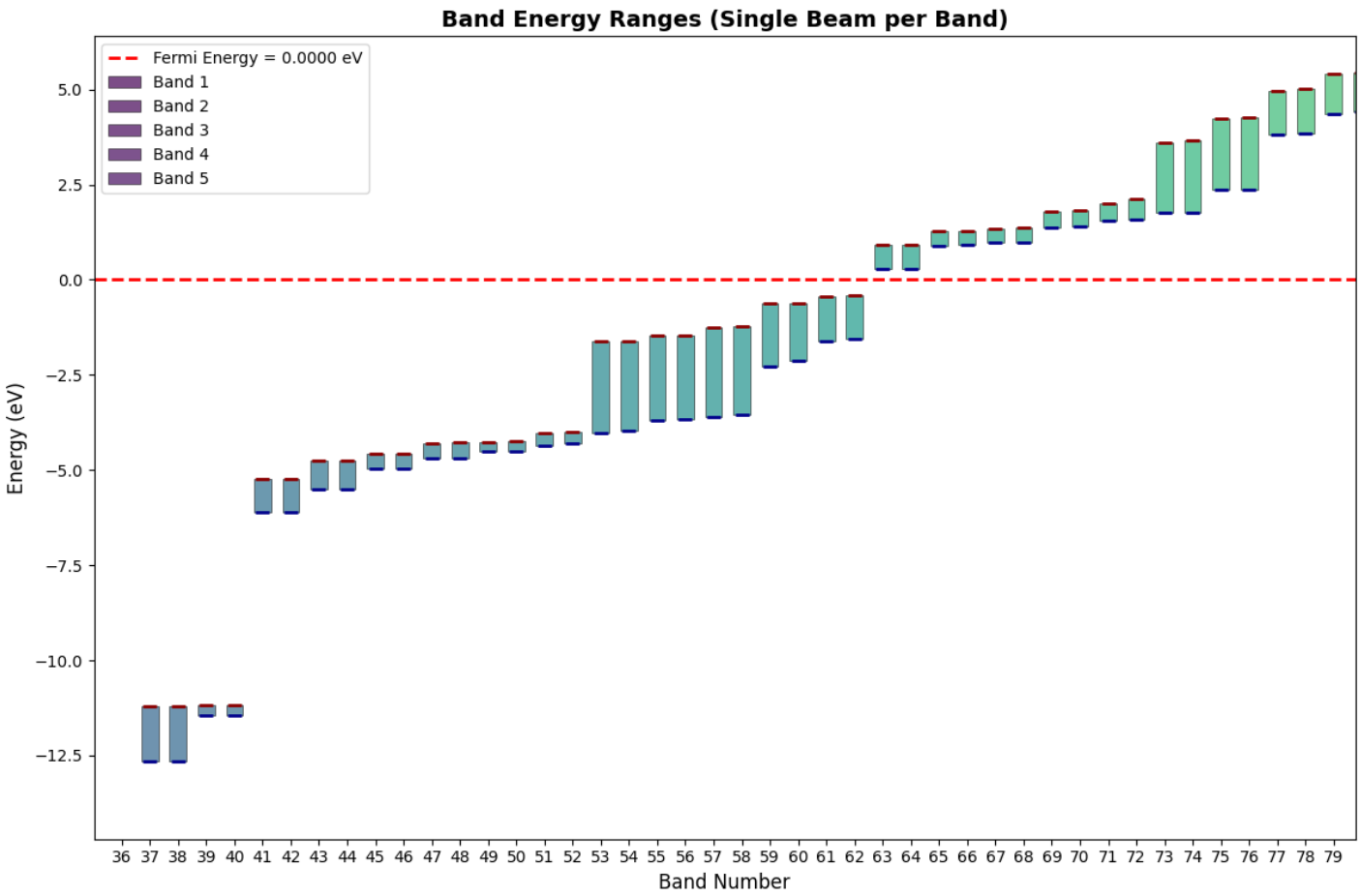

Figure 12.1.: Band-ranges and Band-indices for MnTe.

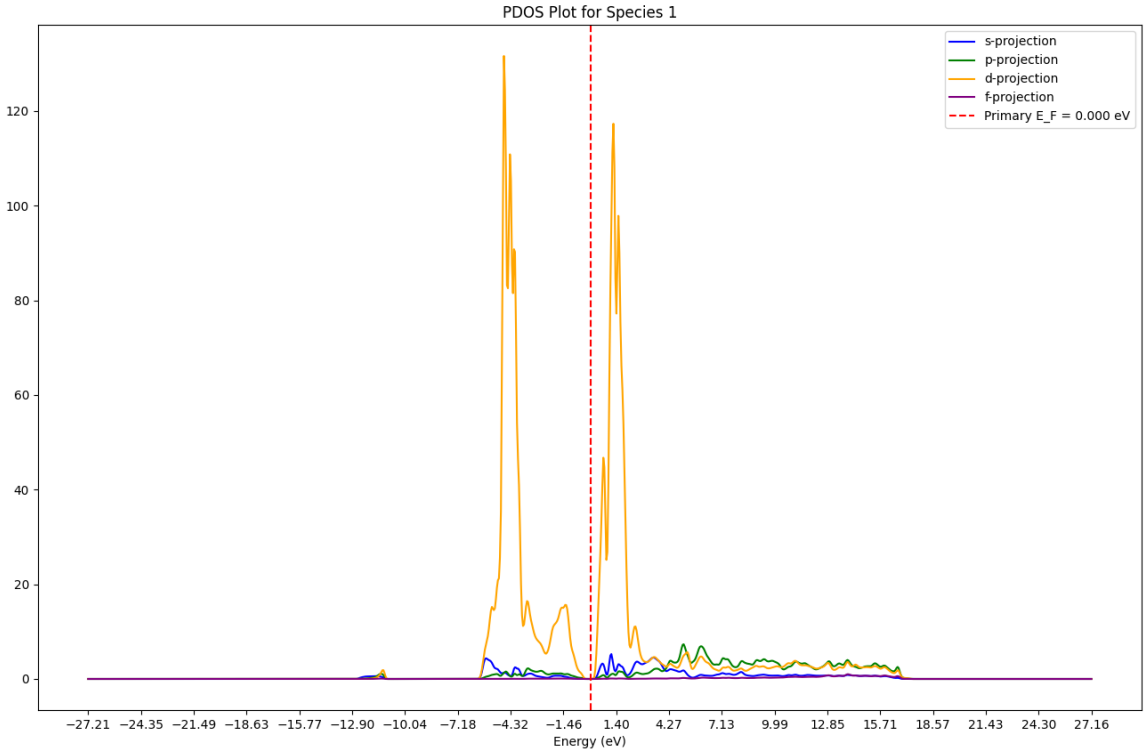

Figure 12.2.: Projected Density of States for Mn.

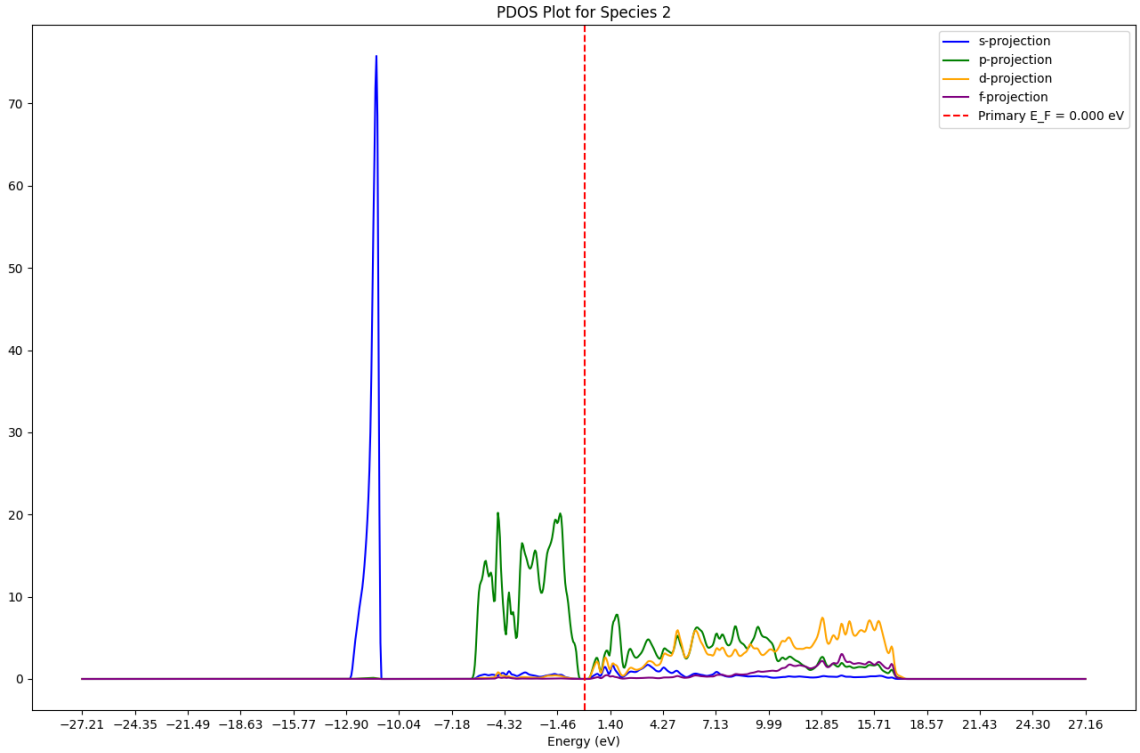

Figure 12.3.: Projected Density of States for Te.

Now we wannierize the system with the following setting in the elk.in-file

```
! calculate the projectors for each spin, l and atom
projw90
 .true.

! list of species and angular momenta
lprojw90
  1   2    : species, l values
  2   1

! select the band indices for wannierisation to avoid the semicore states
idxw90
  41-96
```

Subsequently we add / replace the following lines in the wannier-input-file

```
num_wann =        32
num_bands =       56

write_hr     = .true.

num_iter     =  750
dis_num_iter =  1000
dis_win_min   = -7.0
```

```
dis_win_max   = 18.0
dis_froz_min  = -6.0
dis_froz_max  = 0.0
```

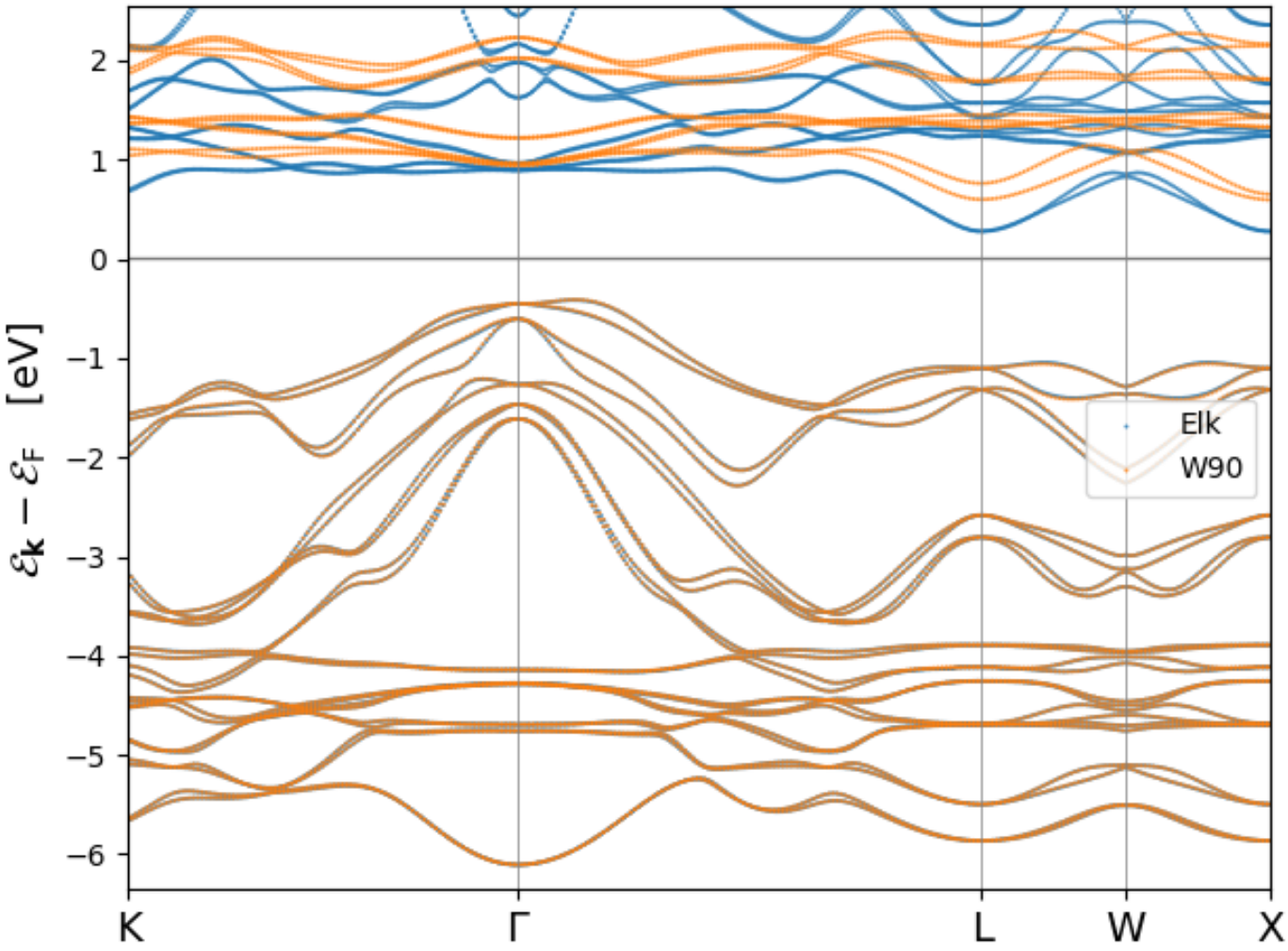

Figure 12.4.: Bandstructure-Calculation from ELK vs. Wannier90 for MnTe.

Now we enumerate the bands ...

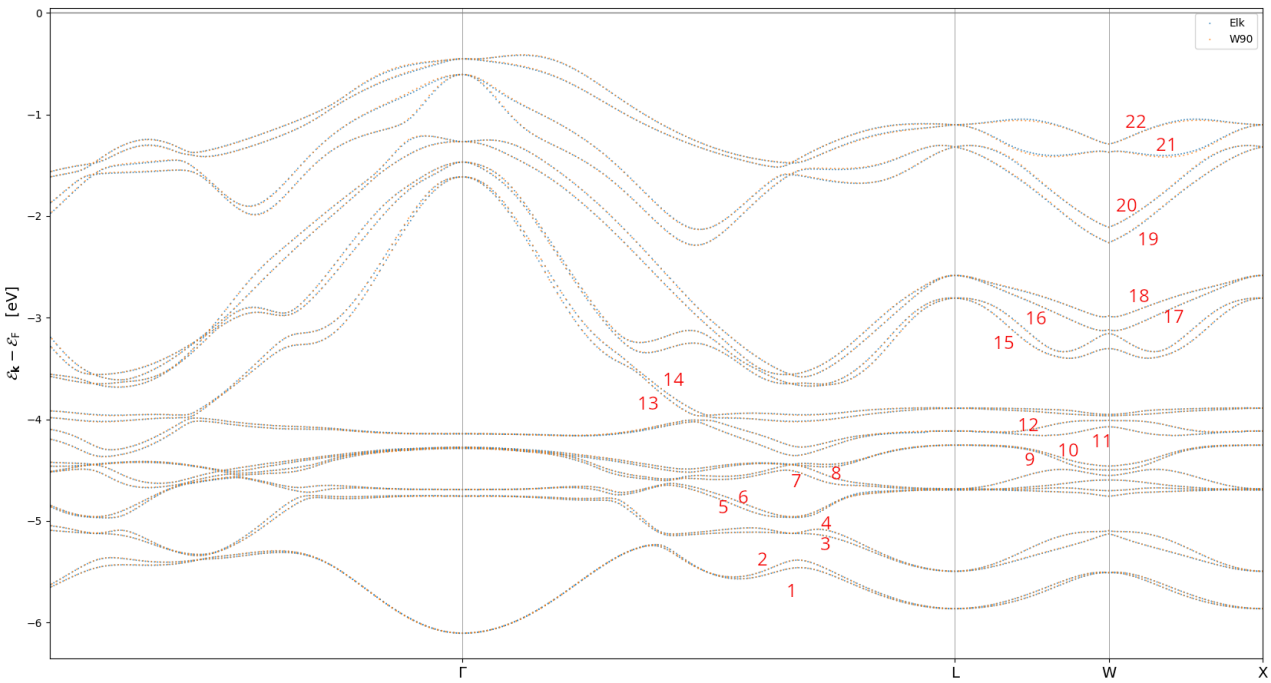

Figure 12.5.: Bandstructure-Calculation from ELK vs. Wannier90 for MnTe with Enumeration.

... and for reasons that will become apparent in the future, we look for the Weyl-points between ...

- bands 15-16,
- bands 18-19.

TBImplemented.

12.2. TaAs Crystal with Spacegroup Im3m Tantalum Arsenide

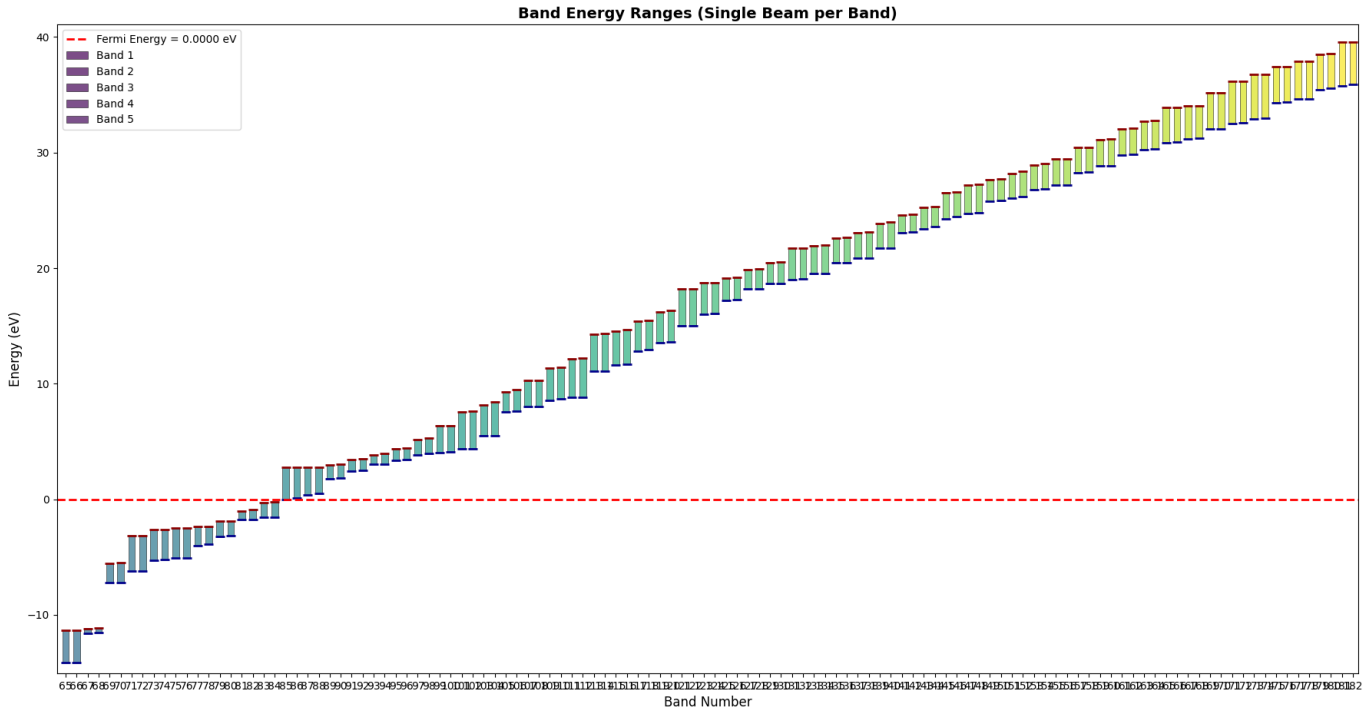

Figure 12.6.: Band-ranges and Band-indices for TaAs.

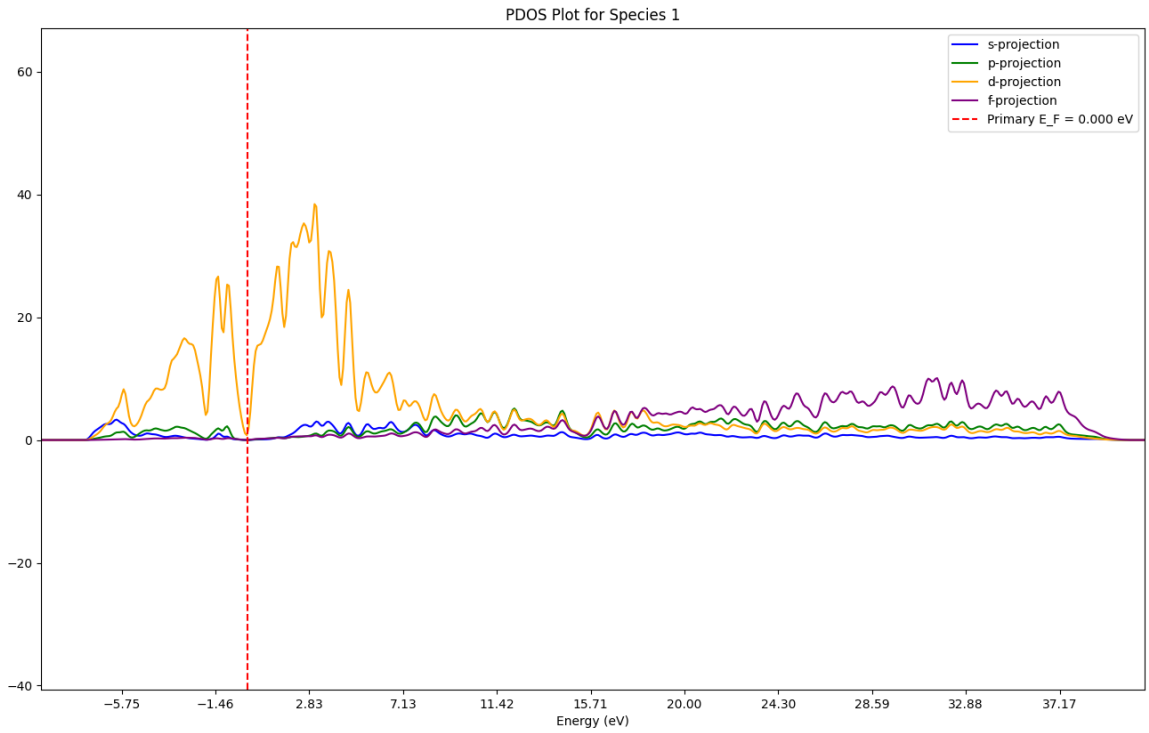

Figure 12.7.: Projected Density of States for Ta.

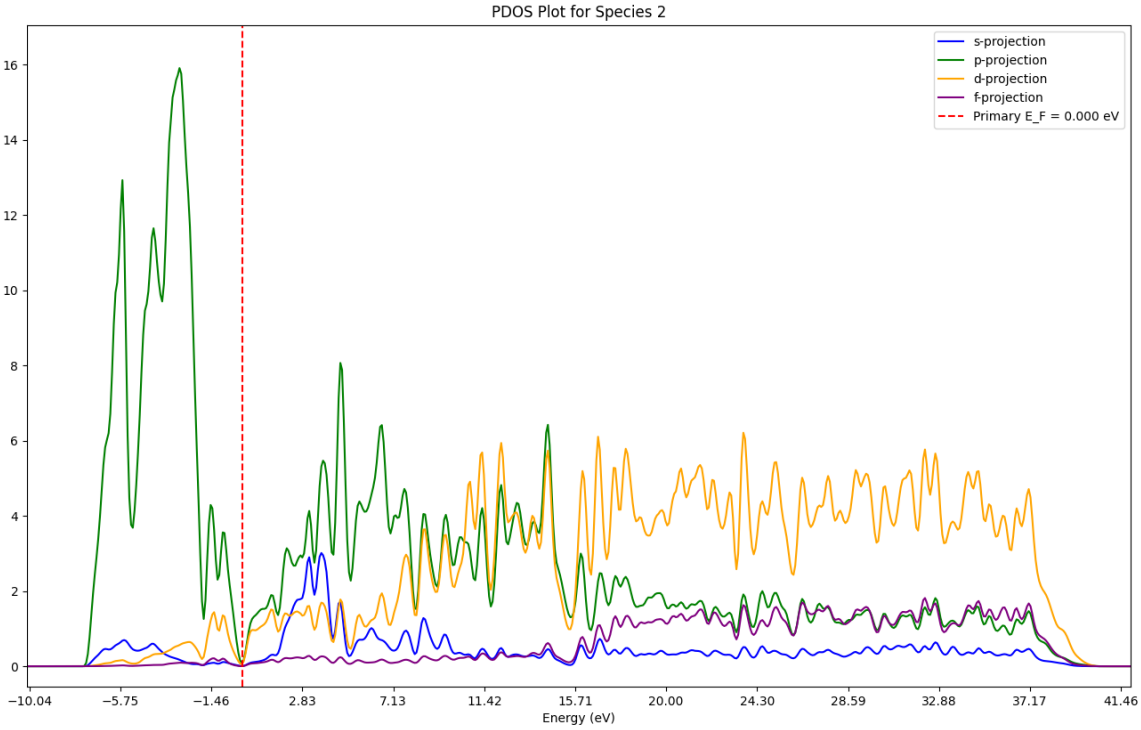

Figure 12.8.: Projected Density of States for As.

Now we wannierize the system with the following setting in the elk.in-file

```
! calculate the projectors for each spin, l and atom
projw90
 .true.

! list of species and angular momenta
lprojw90
  1   2    : species, l values
  2   1

! select the band indices for wannierisation to avoid the semicore states
idxw90
  69-182
```

Subsequently we add / replace the following lines in the wannier-input-file

```
num_wann  =  32
num_bands = 114

write_hr     = .true.

num_iter      =  750
dis_num_iter  =  1000
dis_win_min   = -8.0
dis_win_max   = 40.0
```

```
dis_froz_min  = -8.0
dis_froz_max  = 4.0
```

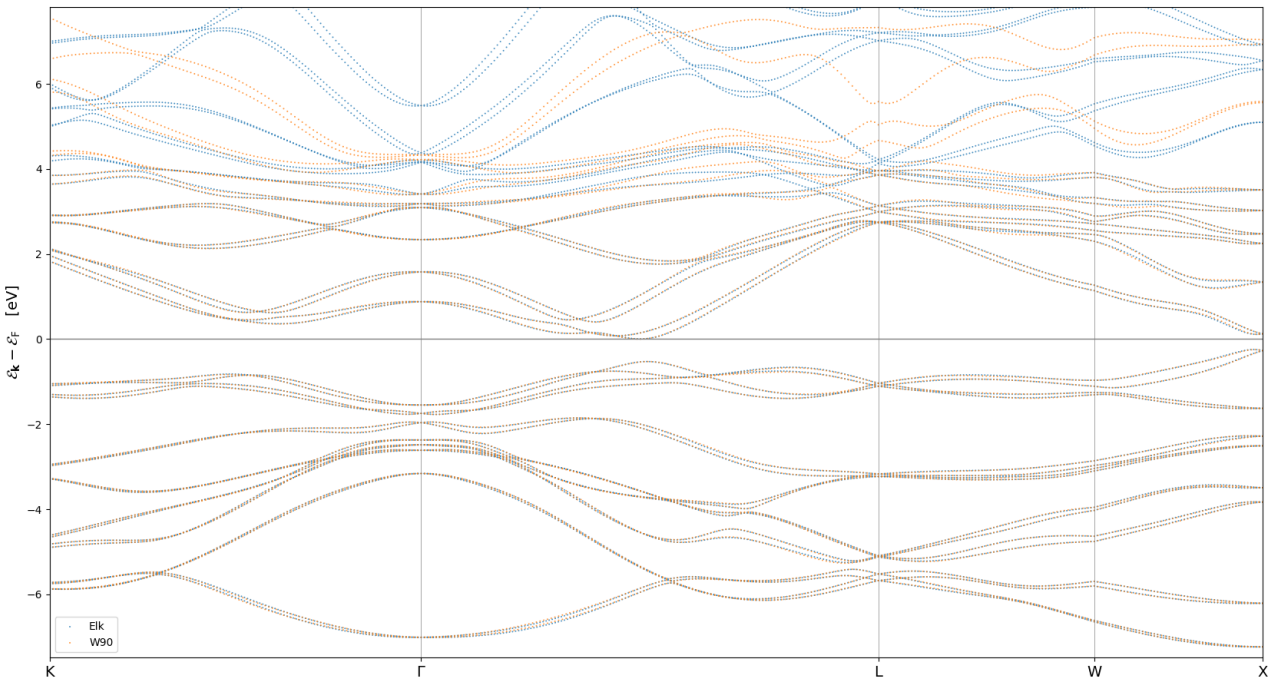

Figure 12.9.: Bandstructure-Calculation from ELK vs. Wannier90 for TaAs.

12.3. Mn3SnCrystal with Spacegroup P63/mmcTrimanganese Tin

The crystal structure is given as

```
avec
   1.000000000       0.000000000       0.000000000
  -0.500000000       0.8660254038      0.000000000
   0.000000000       0.000000000       0.799800000

scale
10.705

atoms
   2                                    : nspecies
'Mn.in'                                 : spfname
  6                                    : natoms; atposl, bfcmt below
    0.161100000    0.322200000    0.25000000    0.00000000 -1.010000000  0.00000000
   -0.322200000   -0.161100000    0.25000000    .866025404  0.50000000  0.00000000
    0.161100000   -0.161100000    0.25000000   -.866025404  0.5000000  0.00000000
   -0.161100000   -0.322200000    0.75000000    0.00000000 -1.010000000  0.00000000
    0.322200000    0.161100000    0.75000000    .866025404  0.5000000  0.00000000
   -0.161100000    0.161100000    0.75000000   -.866025404  0.5000000  0.00000000
'Sn.in'                                 : spfname
   2                                    : natoms; atposl, bfcmt below
    0.33333333333    0.66666666667    0.75000000    0.00000000  0.000       0.00000000
```

```
0.66666666667    0.33333333333    0.25000000    0.00000000    0.000    0.00000000
```

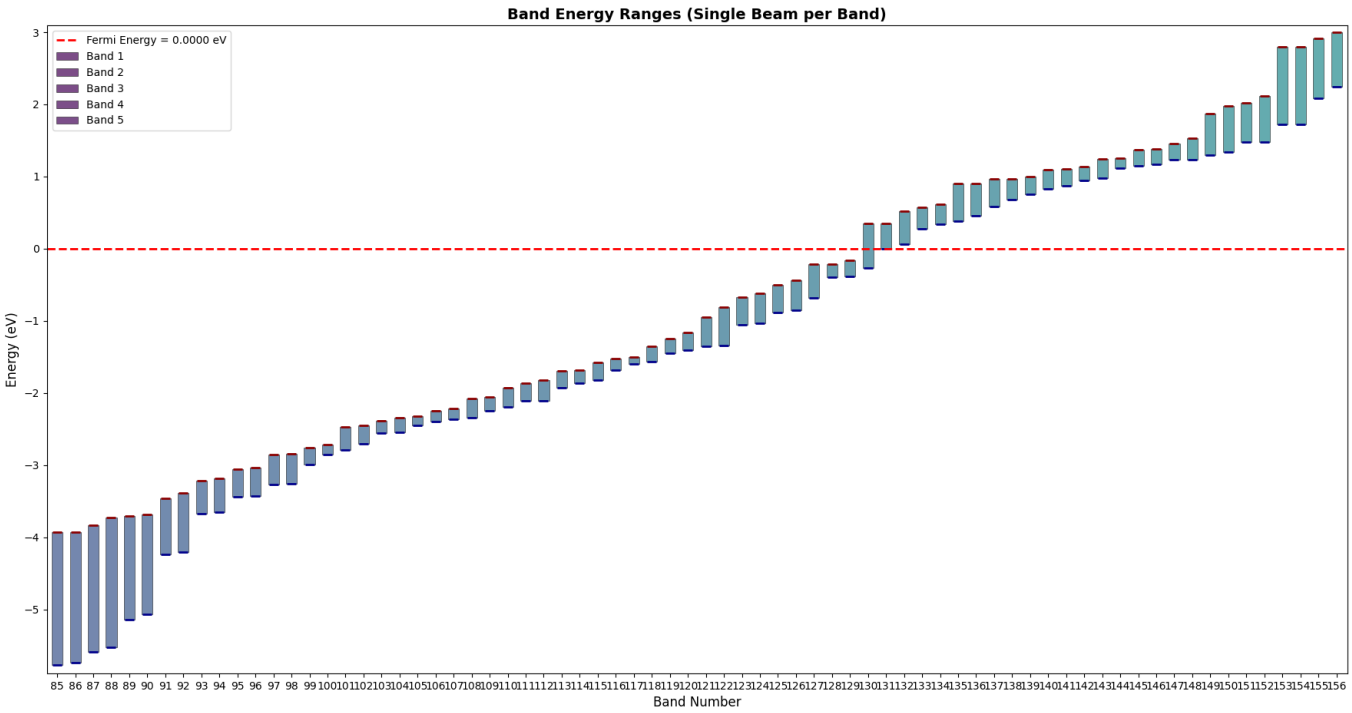

Figure 12.10.: Band-ranges and Band-indices for Mn3Sn.

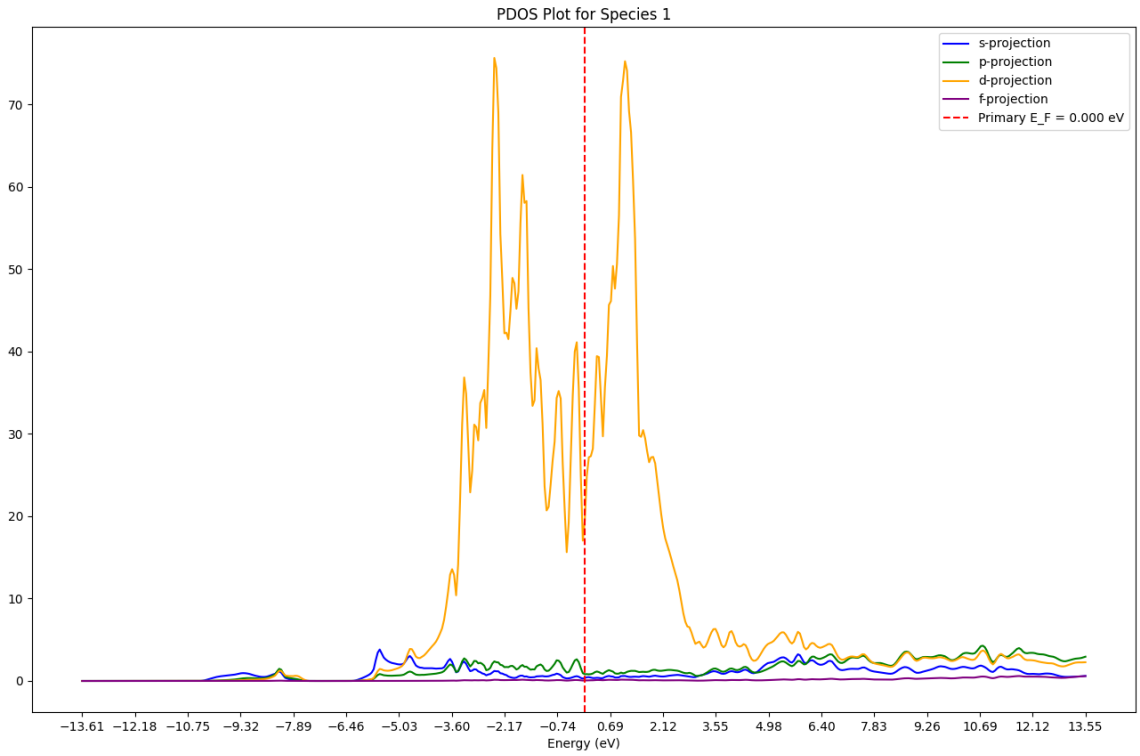

Figure 12.11.: Projected Density of States for Mn.

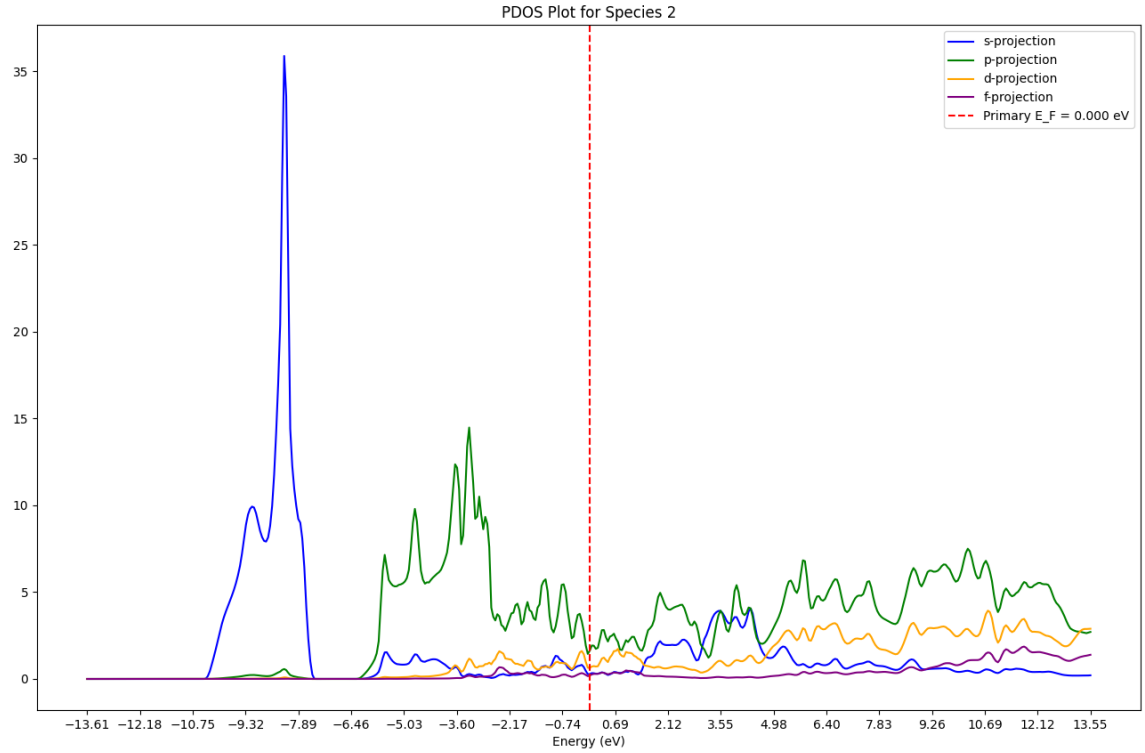

Figure 12.12.: Projected Density of States for Sn.

Now we wannierize the system with the following setting in the elk.in-file

```
! calculate the projectors for each spin, l and atom
projw90
 .true.

! list of species and angular momenta
lprojw90
  1   2    : species, l values
  2   1

! select the band indices for wannierisation to avoid the semicore states
idxw90
  85-156
```

Subsequently we add / replace the following lines in the wannier-input-file

```
num_wann  =
num_bands =

write_hr     = .true.

num_iter     =  750
dis_num_iter =  1000
dis_win_min   = TBD
```

```
dis_win_max   = TBD
dis_froz_min  = TBD
dis_froz_max  = TBD
```

12.4. Rb Crystal with Spacegroup Im3m Rubidium

TBImplemented.

13. Appendix for Proofs

Proof 13.1 of Theorem 2.7

We prove the theorem by explicitly constructing the (canonical) isomorphism. The isomorphism is given by the map

$$\Phi : (U \oplus V) \otimes W \xrightarrow{\simeq} (U \otimes W) \oplus (V \otimes W),$$

defined on simple tensors by

$$\Phi\big((u,0) \otimes w\big) = (\, u \otimes w,\ 0\,), \qquad \Phi\big((0,v) \otimes w\big) = (\, 0,\ v \otimes w\,),$$

and extended linearly.

We now prove that Φ is a well-defined linear isomorphism.

- linearity: By construction the map is defind on simple tensors extended linearly to all finite sums of simple tensors.
- invertibility: We define the map $\Psi : (U \otimes W) \oplus (V \otimes W) \to (U \oplus V) \otimes W$ on simple elements by

  $$\Psi(u \otimes w, 0) := (u,0) \otimes w, \qquad \Psi(0, v \otimes w) := (0,v) \otimes w,$$

  and extend linearly to the whole direct sum. Then it follows

  $$(\Psi \circ \Phi)\big((u,0) \otimes w\big) = \Psi\big(u \otimes w, 0\big) = (u,0) \otimes w,$$

  and similarly $(\Psi \circ \Phi)\big((0,v) \otimes w\big) = (0,v) \otimes w$. By linearity this equality extends to all elements of $(U \oplus V) \otimes W$, so $\Psi \circ \Phi = \mathrm{id}_{(U \oplus V) \otimes W}$. Conversely,

  $$(\Phi \circ \Psi)(u \otimes w, 0) = \Phi\big((u,0) \otimes w\big) = (u \otimes w, 0),$$

  and similarly on the other summand. Thus $\Phi \circ \Psi = \mathrm{id}_{(U \otimes W) \oplus (V \otimes W)}$. Hence Φ is bijective with inverse Ψ, so Φ is a linear isomorphism.

Proof 13.2 of Theorem 2.12

We construct the isomorphism by fixing an enumeration of the allowed wavevectors in the first Brillouin-zone as $\{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \dots\} = \mathrm{Q}$. Then we define the map

$$\varphi : \ L^2(\text{Crystal}) \to \bigoplus_{\mathbf{k} \in \mathrm{Q}} \mathcal{H}_{\mathbf{k}} \tag{13.1}$$

as

$$\varphi(\psi) = (c_{\mathbf{k}_1} u_{\mathbf{k}_1}, c_{\mathbf{k}_2} u_{\mathbf{k}_2}, c_{\mathbf{k}_3} u_{\mathbf{k}_3}, \dots) \tag{13.2}$$

with ψ, $c_{\mathbf{k}_\bullet}$ and $u_{\mathbf{k}_\bullet}$ as in Equation (2.7).

Proof 13.3 of Lemma 3.10

The proof is carried out straightforwardly by

$$c_l^{(k)}(m_1, m_2) = \left(\frac{4\pi}{2k+1}\right)^{1/2} \int \mathrm{d}\Omega\, Y^*_{lm_1}(\Omega) Y_{k,m_1-m_2}(\Omega) Y_{lm_2}(\Omega) \tag{13.3}$$

$$\overset{\text{Eq. (3.9)}}{=} \left(\frac{4\pi}{2k+1}\right)^{1/2} (-1)^{m_1} \int \mathrm{d}\Omega\, Y_{l,-m_1}(\Omega) Y_{k,m_1-m_2}(\Omega) Y_{lm_2}(\Omega) \tag{13.4}$$

$$\overset{\text{Eq. (3.11)}}{=} (-1)^{m_1}(2l+1) \begin{pmatrix} l & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_1 & m_1 - m_2 & m_2 \end{pmatrix} \tag{13.5}$$

Proof 13.4 of Lemma 3.11

We prove this, by using the fact, that the Gaunt-coefficients are real-valued, which can be easily seen by looking at Equation (3.19) and Definition 3.2 (i.e. using the fact, that the 3-j-symbols are real-valued).

Then we obtain

$$\begin{aligned}
(-1)^{m_1-m_3} c_l^{(k)}(m_3, m_1) &= (-1)^{m_1-m_3} \left(c_l^{(k)}(m_3, m_1)\right)^* \\
&= (-1)^{m_1-m_3} \left[\left(\frac{4\pi}{2k+1}\right)^{1/2} \int \mathrm{d}\Omega\, Y^*_{lm_3}(\Omega) Y_{k,m_3-m_1}(\Omega) Y_{lm_1}(\Omega)\right]^* \\
&= (-1)^{m_1-m_3} \left(\frac{4\pi}{2k+1}\right)^{1/2} \int \mathrm{d}\Omega\, Y_{lm_3}(\Omega) Y^*_{k,m_3-m_1}(\Omega) Y^*_{lm_1}(\Omega) \\
&\overset{\text{Eq. (3.9)}}{=} \left(\frac{4\pi}{2k+1}\right)^{1/2} \int \mathrm{d}\Omega\, Y^*_{lm_1}(\Omega) Y^*_{k,m_1-m_3}(\Omega) Y_{lm_3}(\Omega) \\
&= c_l^{(k)}(m_1, m_3)
\end{aligned}$$

Proof 13.5 of Theorem 3.23

To derive the matching-conditions for the expansion coefficients $A^\alpha_{jlm}(\mathbf{G}+\mathbf{p})$, one needs to ensure the continuity of the basis functions and their derivatives up to the appropriate order across the muffin-tin boundary.

We start by considering the case $M^\alpha_l = 1$ for all l and α. We consider a position-vector within the unit-cell $\mathbf{r} \in \mathrm{UC}$ and write

$$\mathbf{r} = \mathbf{r}_\alpha + \mathbf{R}_\alpha \tag{13.6}$$

whereas $\mathbf{r}_\alpha$ denotes the position of the atom α and $\mathbf{R}_\alpha$ denotes any vector, that connects the position of the atom to any point on the boundary of the corresponding muffin-tin MT_α. Having continuous APW-functions thus requires

$$\phi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}_\alpha + \mathbf{R}_\alpha) = \sum_{\substack{l=0,1,\ldots,\mathrm{lmaxo} \\ m=-l,\ldots,+l \\ j=1,\ldots,M_i^\alpha}} A^\alpha_{jlm}(\mathbf{G}+\mathbf{k}) u^\alpha_{jl}(R_\alpha) Y_{lm}(\hat{\mathbf{R}}_\alpha) \overset{!!!}{=} \frac{1}{\sqrt{\Omega}} e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{r}_\alpha} e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{R}_\alpha} \tag{13.7}$$

Expanding now $e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{R}_\alpha}$ w.r.t. Equation (3.24) gives

$$e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{R}_\alpha} = 4\pi \sum_{\substack{l=0,1,\ldots,\mathrm{lmaxo} \\ m=-l,\ldots,+l}} \mathrm{i}^l j_l(|\mathbf{G}+\mathbf{k}|R_\alpha) Y^*_{lm}(\widehat{\mathbf{G}+\mathbf{k}}) Y_{lm}(\hat{\mathbf{R}}_\alpha) \tag{13.8}$$

Plugging this into Equation (13.7) gives

$$\sum_{\substack{l=0,1,\ldots,\mathrm{lmaxo} \\ m=-l,\ldots,+l \\ j=1,\ldots,M_i^\alpha}} A^\alpha_{jlm}(\mathbf{G}+\mathbf{k}) u^\alpha_{jl}(R_\alpha) Y_{lm}(\hat{\mathbf{R}}_\alpha) = 4\pi \sum_{\substack{l=0,1,\ldots,\mathrm{lmaxo} \\ m=-l,\ldots,+l}} \mathrm{i}^l j_l(|\mathbf{G}+\mathbf{k}|R_\alpha) Y^*_{lm}(\widehat{\mathbf{G}+\mathbf{k}}) Y_{lm}(\hat{\mathbf{R}}_\alpha) \frac{1}{\sqrt{\Omega}} e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{r}_\alpha} \tag{13.9}$$

Comparing now the coefficients of $Y_{lm}(\hat{\mathbf{R}}_\alpha)$ for each l and m gives the equation

$$\sum_{j=1}^{M_i^\alpha} A^\alpha_{jlm}(\mathbf{G}+\mathbf{k}) u^\alpha_{jl}(R_\alpha) = 4\pi \mathrm{i}^l j_l(|\mathbf{G}+\mathbf{k}|R_\alpha) Y^*_{lm}(\widehat{\mathbf{G}+\mathbf{k}}) \frac{1}{\sqrt{\Omega}} e^{\mathrm{i}(\mathbf{G}+\mathbf{k})\cdot\mathbf{r}_\alpha} \tag{13.10}$$

This equation coincides exactly with Equations (3.32), (3.33) for $i = 1$ and with the literature[1]. For higher orders, one simply takes the radial derivatives on both sides and repeats the derivation.

Proof 13.6 of Theorem 4.6

We begin, by expanding a Bloch-wavefunction w.r.t. the given basis as

$$\psi^{\mathrm{fv}}_{\mathbf{k},\bullet}(\mathbf{r}) = \sum_{j=1}^{n_{\mathrm{fv}}(\mathbf{k})} C^{\mathrm{fv}\to\mathrm{apw+lo}}_{\bullet,j}(\mathbf{k}) \psi^{\mathrm{apw+lo}}_{j\mathbf{k}}(\mathbf{r}) \tag{13.11}$$

Putting this into Equation (4.1) gives

$$\begin{aligned}
E_{\mathrm{fv},\bullet}(\mathbf{k}) \sum_{i=1}^{n_{\mathrm{fv}}(\mathbf{k})} C^{\mathrm{fv}\to\mathrm{apw+lo}}_{\bullet,i}(\mathbf{k}) \psi^{\mathrm{apw+lo}}_{i\mathbf{k}}(\mathbf{r}) &= E_{\mathrm{fv},\bullet}(\mathbf{k}) \sum_{i,j=1}^{n_{\mathrm{fv}}(\mathbf{k})} C^{\mathrm{fv}\to\mathrm{apw+lo}}_{\bullet,i}(\mathbf{k}) \langle \mathbf{r}|\psi^{\mathrm{apw+lo}}_{j\mathbf{k}}\rangle \langle \psi^{\mathrm{apw+lo}}_{j\mathbf{k}}|\psi^{\mathrm{apw+lo}}_{i\mathbf{k}}\rangle \\
&= \sum_{i=1}^{n_{\mathrm{fv}}(\mathbf{k})} C^{\mathrm{fv}\to\mathrm{apw+lo}}_{\bullet,i}(\mathbf{k}) \left(H_0 \psi^{\mathrm{apw+lo}}_{i\mathbf{k}} \right)(\mathbf{r}) \\
&= \sum_{i=1}^{n_{\mathrm{fv}}(\mathbf{k})} C^{\mathrm{fv}\to\mathrm{apw+lo}}_{\bullet,i}(\mathbf{k}) \langle \mathbf{r}|H_0|\psi^{\mathrm{apw+lo}}_{i\mathbf{k}}\rangle \\
&= \sum_{i,j=1}^{n_{\mathrm{fv}}(\mathbf{k})} C^{\mathrm{fv}\to\mathrm{apw+lo}}_{\bullet,i}(\mathbf{k}) \langle \mathbf{r}|\psi^{\mathrm{apw+lo}}_{j\mathbf{k}}\rangle \langle \psi^{\mathrm{apw+lo}}_{j\mathbf{k}}|H_0|\psi^{\mathrm{apw+lo}}_{i\mathbf{k}}\rangle
\end{aligned} \tag{13.12}$$

[1] ELK-manual, match.f90 description

now, comparing coefficients w.r.t. $\psi_{\substack{\text{apw+lo}\\ j\mathbf{k}}}$ yields for each $j \in \{1, \dots, n_{\text{fv}}(\mathbf{k})\}$ the equation

$$\begin{aligned} \sum_{i=1}^{n_{\text{fv}}(\mathbf{k})} C_{\bullet,i}^{\text{fv}\to\text{apw+lo}}(\mathbf{k}) \langle \psi_{\substack{\text{apw+lo}\\ j\mathbf{k}}} | H_0 | \psi_{\substack{\text{apw+lo}\\ i\mathbf{k}}} \rangle &= E_{\text{fv},\bullet}(\mathbf{k}) \sum_{i=1}^{n_{\text{fv}}(\mathbf{k})} C_{\bullet,i}^{\text{fv}\to\text{apw+lo}}(\mathbf{k}) \langle \psi_{\substack{\text{apw+lo}\\ j\mathbf{k}}} | \psi_{\substack{\text{apw+lo}\\ i\mathbf{k}}} \rangle \\ \iff \qquad H_0(\mathbf{k}) \mathbf{C}_{\bullet}^{\text{fv}\to\text{apw+lo}}(\mathbf{k}) &= E_{\text{fv},\bullet}(\mathbf{k}) S(\mathbf{k}) \mathbf{C}_{\bullet}^{\text{fv}\to\text{apw+lo}}(\mathbf{k}) \end{aligned} \tag{13.13}$$

Proof 13.7 of Theorem 4.17

By expanding wavefunctions w.r.t. local orbitals, combining Equation (4.38) and Remark 4.42 we obtain by a straight-forward computation

$$\langle ab|g|cd\rangle = \sum\!\!\!\!\!\!\int_1 \sum\!\!\!\!\!\!\int_2 \psi_a^*(1)\psi_b^*(2) g(r_{12}) \psi_c(1)\psi_d(2) \tag{13.14}$$

$$= \sum_{s_1} \chi_a^*(s_1)\chi_c(s_1) \sum_{s_2} \chi_b^*(s_2)\chi_d(s_2) \int_{\mathbf{r}_1} \int_{\mathbf{r}_2} \varphi_a^*(\mathbf{r}_1)\varphi_b^*(\mathbf{r}_2) g(r_{12}) \varphi_c(\mathbf{r}_1)\varphi_d(\mathbf{r}_2) \tag{13.15}$$

$$= \dots \tag{13.16}$$

We now decompose the terms further …

- For the spin-part we have

$$\sum_{s_1} \chi_a^*(s_1)\chi_c(s_1) \sum_{s_2} \chi_b^*(s_2)\chi_d(s_2) = \delta_{s_a,s_c}\delta_{s_b,s_d} \tag{13.17}$$

- For the spatial part we obtain, using Equation (4.38)

$$\int_{\mathbf{r}_1} \int_{\mathbf{r}_2} \varphi_a^*(\mathbf{r}_1)\varphi_b^*(\mathbf{r}_2) g(r_{12}) \varphi_c(\mathbf{r}_1)\varphi_d(\mathbf{r}_2) = \sum_{k=0}^{\infty} F_l^{(k)} \, \mathrm{AI}_l^{(k)}(m_a, m_b, m_c, m_d) \tag{13.18}$$

with the Slater-integrals $F_l^{(k)}$ and the angular integrals $\mathrm{AI}_l^{(k)}(m_a, m_b, m_c, m_d)$ as defined in the main text.

Proof 13.8 of Corollary 4.18

By straightforward computation, starting with Equation (4.48), we obtain

$$\mathrm{AI}_l^k(m_a, m_b, m_c, m_d) \tag{13.19}$$

$$= \frac{4\pi}{2k+1} \sum_{q=-k}^{+k} \int \mathrm{d}\Omega_1 Y^*_{lm_a}(\Omega_1) Y^*_{kq}(\Omega_1) Y_{lm_c}(\Omega_1) \int \mathrm{d}\Omega_2 Y^*_{lm_b}(\Omega_2) Y_{kq}(\Omega_2) Y_{lm_d}(\Omega_2) \tag{13.20}$$

$$\overset{\text{Eq. (3.9)}}{=} \frac{4\pi}{2k+1} \sum_{q=-k}^{+k} (-1)^q \int \mathrm{d}\Omega_1 Y^*_{lm_a}(\Omega_1) Y_{k,-q}(\Omega_1) Y_{lm_c}(\Omega_1) \int \mathrm{d}\Omega_2 Y^*_{lm_b}(\Omega_2) Y_{kq}(\Omega_2) Y_{lm_d}(\Omega_2) \tag{13.21}$$

$$\overset{\text{Eq. (3.14)}}{=} \frac{4\pi}{2k+1} \sum_{q=-k}^{+k} \delta_{q,m_c-m_a} \delta_{q,m_b-m_d} (-1)^q \int \mathrm{d}\Omega_1 Y^*_{lm_a}(\Omega_1) Y_{k,-q}(\Omega_1) Y_{lm_c}(\Omega_1) \int \mathrm{d}\Omega_2 Y^*_{lm_b}(\Omega_2) Y_{kq}(\Omega_2) Y_{lm_d}(\Omega_2) \tag{13.22}$$

$$= \frac{4\pi}{2k+1} \delta_{m_b-m_d, m_c-m_a} (-1)^{m_b-m_d} \int \mathrm{d}\Omega_1 Y^*_{lm_a}(\Omega_1) Y_{k,m_a-m_c}(\Omega_1) Y_{lm_c}(\Omega_1) \int \mathrm{d}\Omega_2 Y^*_{lm_b}(\Omega_2) Y_{k,m_b-m_d}(\Omega_2) Y_{lm_d}(\Omega_2) \tag{13.23}$$

$$= \frac{4\pi}{2k+1} \delta_{m_a+m_b, m_c+m_d} (-1)^{m_b-m_d} \int \mathrm{d}\Omega_1 Y^*_{lm_a}(\Omega_1) Y_{k,m_a-m_c}(\Omega_1) Y_{lm_c}(\Omega_1) \int \mathrm{d}\Omega_2 Y^*_{lm_b}(\Omega_2) Y_{k,m_b-m_d}(\Omega_2) Y_{lm_d}(\Omega_2) \tag{13.24}$$

$$\overset{\text{Eq. (3.18)}}{=} \delta_{m_a+m_b, m_c+m_d} (-1)^{m_b-m_d} c_l^{(k)}(m_a, m_c) c_l^{(k)}(m_b, m_d) \tag{13.25}$$

$$\overset{\text{Eq. (3.21)}}{=} \delta_{m_a+m_b, m_c+m_d} c_l^{(k)}(m_a, m_c) c_l^{(k)}(m_d, m_b) \tag{13.26}$$

Proof 13.9 of Theorem 7.4

By inserting one equation and using Equation (7.8) into the other one gets

$$\begin{aligned}
\rho_{ab} &= (-1)^{m_b-l} \sum_{k=0}^{2l} \sum_{x=-k}^{+k} [k] \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} n_{lk} w_x^k \\
&= \sum_{k=0}^{2l} \sum_{x=-k}^{+k} \sum_{m_\bullet, \widetilde{m}_\bullet \in \{-l,\dots,+l\}} [k] (-1)^{m_b-m_\bullet} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} \rho_{\substack{\widetilde{m}_\bullet, m_\bullet \\ s_\bullet, s_\bullet}} \\
&= \dots
\end{aligned} \tag{13.27}$$

Now we use the identity TBImplemented: Sum-rule in fundamentals

$$\sum_{k=0}^{2l} \sum_{x=-k}^{+k} [k] \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} = \delta_{m_b, m_\bullet} \delta_{m_a, \widetilde{m}_\bullet} \tag{13.28}$$

such that we obtain

$$\dots = \sum_{m_\bullet, \widetilde{m}_\bullet \in \{-l,\dots,+l\}} \delta_{m_b, m_\bullet} \delta_{m_a, \widetilde{m}_\bullet} \rho_{\substack{\widetilde{m}_\bullet, m_\bullet \\ s_\bullet, s_\bullet}} = \rho_{\substack{m_a, m_b \\ s_\bullet, s_\bullet}} = \rho_{ab} \tag{13.29}$$

Proof 13.10 of Theorem 7.6

By inserting one equation into the other one gets

$$\begin{aligned}\rho_{ac} &= \sum_{k,x}[k]n_{lk}(-1)^{m_c-l}\begin{pmatrix} l & k & l \\ -m_c & x & m_a\end{pmatrix}\sum_{py}[p]n_{sp}(-1)^{s_c-s}\begin{pmatrix} s & p & s \\ -s_c & y & s_a\end{pmatrix} w_{xy}^{kp} \\ &= \sum_{k,x}[k]n_{lk}(-1)^{m_c-l}\begin{pmatrix} l & k & l \\ -m_c & x & m_a\end{pmatrix}\sum_{py}[p]n_{sp}(-1)^{s_c-s}\begin{pmatrix} s & p & s \\ -s_c & y & s_a\end{pmatrix}\sum_{\substack{m_\bullet,\widetilde{m}_\bullet\in\{-l,\dots,+l\}\\ s_\bullet,\widetilde{s}_\bullet\in\{-1/2,+1/2\}}}\left(t_y^p\otimes v_x^k\right)_{\substack{m_\bullet,\widetilde{m}_\bullet\\ s_\bullet,\widetilde{s}_\bullet}}\rho_{\substack{\widetilde{m}_\bullet,m_\bullet\\ \widetilde{s}_\bullet,s_\bullet}}\end{aligned} \tag{13.30}$$

Notice now, that the matrix-representation of the tensor-product is given as

$$\begin{pmatrix}\left(t_y^p\right)_{+1/2,+1/2} v_x^k & \left(t_y^p\right)_{+1/2,-1/2} v_x^k \\ \left(t_y^p\right)_{-1/2,+1/2} v_x^k & \left(t_y^p\right)_{-1/2,-1/2} v_x^k\end{pmatrix} \tag{13.31}$$

Thus, we obtain for the matrix-elements of the tensor-product simply the respective matrix element of the individual matrices, and we obtain

$$\rho_{ac} = \sum_{\substack{k,x\\ py\\ m_\bullet,\widetilde{m}_\bullet\in\{-l,\dots,+l\}\\ s_\bullet,\widetilde{s}_\bullet\in\{-1/2,+1/2\}}}[k][p](-1)^{m_c-m_\bullet+s_c-s_\bullet}\begin{pmatrix} l & k & l \\ -m_c & x & m_a\end{pmatrix}\begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet\end{pmatrix}\begin{pmatrix} s & p & s \\ -s_c & y & -s_a\end{pmatrix}\begin{pmatrix} s & p & s \\ -s_\bullet & y & \widetilde{s}_\bullet\end{pmatrix}\rho_{\substack{\widetilde{m}_\bullet,m_\bullet\\ \widetilde{s}_\bullet,s_\bullet}} \tag{13.32}$$

Now we want to use the following orthogonality-relation TBImplemented in Fundamentals

$$\sum_{j_3m_3}(2j_3+1)\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3\end{pmatrix}\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1' & m_2' & m_3\end{pmatrix} = \delta_{m_1,m_1'}\delta_{m_2,m_2'} \tag{13.33}$$

and we use

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3\end{pmatrix} = (-1)^{j_1+j_2+j_3}\begin{pmatrix} j_1 & j_3 & j_2 \\ m_1 & m_3 & m_2\end{pmatrix} \tag{13.34}$$

in order to write down the *adapted orthogonality-relation* as

$$\sum_{j_3m_3}(2j_3+1)\begin{pmatrix} j_1 & j_3 & j_2 \\ m_1 & m_3 & m_2\end{pmatrix}\begin{pmatrix} j_1 & j_3 & j_2 \\ m_1' & m_3 & m_2'\end{pmatrix} = \delta_{m_1,m_1'}\delta_{m_2,m_2'} \tag{13.35}$$

This yields

$$\sum_{kx}[k]\begin{pmatrix} l & k & l \\ -m_c & x & m_a\end{pmatrix}\begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet\end{pmatrix} = \delta_{m_c,m_\bullet}\delta_{m_a,\widetilde{m}_\bullet} \tag{13.36}$$

and

$$\sum_{py}[p]\begin{pmatrix} s & p & s \\ -s_c & y & -s_a\end{pmatrix}\begin{pmatrix} s & p & s \\ -s_\bullet & y & \widetilde{s}_\bullet\end{pmatrix} = \delta_{s_c,s_\bullet}\delta_{s_a,\widetilde{s}_\bullet} \tag{13.37}$$

Combining these two results with the equation above yields the desired result.

Proof 13.11 of Theorem 7.10

Inserting one equation into the other gives

$$\begin{aligned} w_t^{kpr} &= \underline{n}_{kpr}^{-1} \sum_{xy} (-1)^{k-x+p-y} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} w_{xy}^{kp} \\ &= \underline{n}_{kpr}^{-1} \sum_{xyr't'} (2r'+1) \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \begin{pmatrix} k & r' & p \\ -x & t' & -y \end{pmatrix} \underline{n}_{kpr'} w_{t'}^{kpr'} \\ &= \underline{n}_{kpr}^{-1} \sum_{r't'} \underline{n}_{kpr'} (2r'+1) w_{t'}^{kpr'} \sum_{xy} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \begin{pmatrix} k & r' & p \\ -x & t' & -y \end{pmatrix} \\ &= \dots \end{aligned} \tag{13.38}$$

Now we will / want to use the orthogonality-relation

$$(2j_3+1) \sum_{m_1 m_2} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3' \\ m_1 & m_2 & m_3' \end{pmatrix} = \delta_{j_3,j_3'} \delta_{m_3,m_3'} \{j_1\, j_2\, j_3\} \tag{13.39}$$

whereas $\{j_1\, j_2\, j_3\}$ is equal to one, if the triangle-condition is fulfilled

$$|j_1 - j_2| \leq j_3 \leq j_1 + j_2 \tag{13.40}$$

This way we obtain

$$(2r'+1) \sum_{xy} \begin{pmatrix} p & k & r \\ -y & -x & t \end{pmatrix} \begin{pmatrix} p & k & r' \\ -y & -x & t' \end{pmatrix} = \delta_{r,r'} \delta_{t,t'} \cdot 1 \tag{13.41}$$

whereas the triangle-condition is fulfilled, since $|k-p| \leq r \leq |k+p|$. This yields then

$$\dots = \underline{n}_{kpr}^{-1} \sum_{r't'} \underline{n}_{kpr'} w_{t'}^{kpr'} \delta_{r,r'} \delta_{t,t'} = w_t^{kpr} \tag{13.42}$$

Proof 13.12 of Theorem 7.12

<u>The Left Formula:</u> We obtain the first formula by straight forward expansion

$$\begin{aligned} w_t^{kpr} &= \underline{n}_{kpr}^{-1} \sum_{xy} (-1)^{k-x+p-y} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} w_{xy}^{kp} \\ &= \underline{n}_{kpr}^{-1} \sum_{xy} (-1)^{k-x+p-y} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \mathrm{Tr}\left[\left(t_y^p \otimes v_x^k \right) \cdot \rho \right] \\ &= \mathrm{Tr}\left(\Gamma_t^{kpr} \rho \right) \end{aligned} \tag{13.43}$$

with

$$
\begin{aligned}
\left(\Gamma_t^{kpr}\rho\right)_{\substack{m_\bullet,\widetilde{m}_\bullet\\ s_\bullet,\widetilde{s}_\bullet}} &= \underline{n}_{kpr}^{-1}\sum_{xy}(-1)^{k-x+p-y}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}\mathrm{Tr}\left(t_y^p\otimes v_x^k\right)_{\substack{m_\bullet,\widetilde{m}_\bullet\\ s_\bullet,\widetilde{s}_\bullet}}\\
&= \underline{n}_{kpr}^{-1}\sum_{xy}(-1)^{k-x+p-y}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}(-1)^{s-s_\bullet}\begin{pmatrix} s & p & s\\ -s_\bullet & y & \widetilde{s}_\bullet\end{pmatrix}n_{sp}^{-1}(-1)^{l-m_\bullet}\begin{pmatrix} l & k & l\\ -m_\bullet & x & \widetilde{m}_\bullet\end{pmatrix}n_{lk}^{-1}
\end{aligned}
\tag{13.44}
$$

The Right Formula:

We insert the left formula into the right one and obtain

$$
\begin{aligned}
\rho_{ab} &= \sum_{kprt}\left(\Lambda_t^{kpr}\right)_{ab}w_t^{kpr} = \sum_{kprt}\left(\Lambda_t^{kpr}\right)_{ab}\sum_{\widetilde{a},\widetilde{b}}\left(\Gamma_t^{kpr}\right)_{\widetilde{b},\widetilde{a}}\rho_{\widetilde{a},\widetilde{b}}\\
&= \sum_{kprt}N_{kpr}[kpr](-1)^{m_b-l+s_b-s}\sum_{xy}(-1)^{x-k+y-p}\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}\times\\
&\quad\times N_{kpr}^{-1}(-1)^{l-m_{\widetilde{b}}+s-s_{\widetilde{b}}}\sum_{\widetilde{x},\widetilde{y}}(-1)^{k-\widetilde{x}+p-\widetilde{y}}\begin{pmatrix} l & k & l\\ -m_{\widetilde{b}} & \widetilde{x} & m_{\widetilde{a}}\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_{\widetilde{b}} & \widetilde{y} & s_{\widetilde{a}}\end{pmatrix}\begin{pmatrix} k & r & p\\ -\widetilde{x} & t & -\widetilde{y}\end{pmatrix}\rho_{\widetilde{a},\widetilde{b}}\\
&= \sum_{\substack{kp\\ xy\\ \widetilde{x},\widetilde{y}\\ \widetilde{a},\widetilde{b}}}[kp](-1)^{m_b-m_{\widetilde{b}}+s_b-s_{\widetilde{b}}}(-1)^{x+y-\widetilde{x}-\widetilde{y}}\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\begin{pmatrix} l & k & l\\ -m_{\widetilde{b}} & \widetilde{x} & m_{\widetilde{a}}\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_{\widetilde{b}} & \widetilde{y} & s_{\widetilde{a}}\end{pmatrix}\times\\
&\quad\times\sum_{rt}[r]\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}\begin{pmatrix} k & r & p\\ -\widetilde{x} & t & -\widetilde{y}\end{pmatrix}\rho_{\widetilde{a},\widetilde{b}}\\
&= \dots
\end{aligned}
\tag{13.45}
$$

now we use

$$
\sum_{rt}[r]\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}\begin{pmatrix} k & r & p\\ -\widetilde{x} & t & -\widetilde{y}\end{pmatrix} = \delta_{x,\widetilde{x}}\delta_{y,\widetilde{y}}
\tag{13.46}
$$

as a special case of the orthogonality-relations TBReferenced and obtain

$$
\begin{aligned}
\dots &= \sum_{\widetilde{a},\widetilde{b}}(-1)^{m_b-m_{\widetilde{b}}+s_b-s_{\widetilde{b}}}\rho_{\widetilde{a},\widetilde{b}}\sum_{kx}[k]\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} l & k & l\\ -m_{\widetilde{b}} & x & m_{\widetilde{a}}\end{pmatrix}\sum_{py}[p]\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_{\widetilde{b}} & y & s_{\widetilde{a}}\end{pmatrix}\\
&= \dots
\end{aligned}
\tag{13.47}
$$

using now the orthogonality-relations TBReferenced

$$
\sum_{kx}[k]\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} l & k & l\\ -m_{\widetilde{b}} & x & m_{\widetilde{a}}\end{pmatrix} = \delta_{m_b,m_{\widetilde{b}}}\delta_{m_a,m_{\widetilde{a}}}
\tag{13.48}
$$

and

$$\sum_{py}[p]\begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_{\widetilde{b}} & y & s_{\widetilde{a}} \end{pmatrix} = \delta_{s_b,s_{\widetilde{b}}}\delta_{s_a,s_{\widetilde{a}}} \tag{13.49}$$

we obtain

$$\ldots = \rho_{ab} \tag{13.50}$$

Proof 13.13 of Theorem 7.13

Since

$$w_t^{kpr} = \mathrm{Tr}\left(\Gamma_t^{kpr}\rho\right) \tag{13.51}$$

and

$$(-1)^t\left(w_{-t}^{kpr}\right)^* = (-1)^t\,\mathrm{Tr}\left(\Gamma_{-t}^{kpr}\rho\right)^* = \mathrm{Tr}\left[(-1)^t\left(\Gamma_{-t}^{kpr}\rho\right)^*\right] = \mathrm{Tr}\left[(-1)^t\left(\Gamma_{-t}^{kpr}\right)^\dagger\rho\right] \tag{13.52}$$

it suffices (although this is not a neccessary condition / implication) to show that

$$\Gamma_t^{kpr} = (-1)^t\left(\Gamma_{-t}^{kpr}\right)^\dagger \tag{13.53}$$

Using now Equation (TBReferenced), we obtain

$$\left(\Gamma_t^{kpr}\right)_{\substack{m_\bullet,\widetilde{m}_\bullet \\ s_\bullet,\widetilde{s}_\bullet}} = \underline{n}_{kpr}^{-1}\sum_{xy}(-1)^{k-x+p-y}\begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}(-1)^{s-s_\bullet}\begin{pmatrix} s & p & s \\ -s_\bullet & y & \widetilde{s}_\bullet \end{pmatrix} n_{sp}^{-1}(-1)^{l-m_\bullet}\begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} n_{lk}^{-1} \tag{13.54}$$

and

$$\begin{aligned}
&(-1)^t\left(\Gamma_{-t}^{kpr}\right)^\dagger_{\substack{m_\bullet,\widetilde{m}_\bullet \\ s_\bullet,\widetilde{s}_\bullet}} \\
&= (-1)^t\left(\Gamma_{-t}^{kpr}\right)^*_{\substack{\widetilde{m}_\bullet,m_\bullet \\ \widetilde{s}_\bullet,s_\bullet}} \\
&= (-1)^t\left(\underline{n}_{kpr}^{-1}\right)^*\sum_{xy}(-1)^{k-x+p-y}\begin{pmatrix} k & r & p \\ -x & -t & -y \end{pmatrix}(-1)^{s-\widetilde{s}_\bullet}\begin{pmatrix} s & p & s \\ -\widetilde{s}_\bullet & y & s_\bullet \end{pmatrix} n_{sp}^{-1}(-1)^{l-\widetilde{m}_\bullet}\begin{pmatrix} l & k & l \\ -\widetilde{m}_\bullet & x & m_\bullet \end{pmatrix} n_{lk}^{-1} \\
&\overset{\substack{x\to -x \\ y\to -y}}{=} (-1)^{-t}\left(\underline{n}_{kpr}^{-1}\right)^*\sum_{xy}(-1)^{k-x+p-y}\begin{pmatrix} k & r & p \\ x & -t & y \end{pmatrix}(-1)^{s-\widetilde{s}_\bullet}\begin{pmatrix} s & p & s \\ -\widetilde{s}_\bullet & -y & s_\bullet \end{pmatrix} n_{sp}^{-1}(-1)^{l-\widetilde{m}_\bullet}\begin{pmatrix} l & k & l \\ -\widetilde{m}_\bullet & -x & m_\bullet \end{pmatrix} n_{lk}^{-1} \\
&= \ldots
\end{aligned} \tag{13.55}$$

now we note the following simplifications / modifications

- $$\begin{pmatrix} k & r & p \\ x & -t & y \end{pmatrix} = (-1)^{k+r+p}\begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \tag{13.56}$$

- $$\begin{pmatrix} s & p & s \\ -\widetilde{s}_\bullet & -y & s_\bullet \end{pmatrix} = (-1)^{2s+p} \begin{pmatrix} s & p & s \\ \widetilde{s}_\bullet & y & -s_\bullet \end{pmatrix} = \begin{pmatrix} s & p & s \\ -s_\bullet & y & \widetilde{s}_\bullet \end{pmatrix} \tag{13.57}$$
- $$\begin{pmatrix} l & k & l \\ -\widetilde{m}_\bullet & -x & m_\bullet \end{pmatrix} = (-1)^{2l+k} \begin{pmatrix} l & k & l \\ -\widetilde{m}_\bullet & -x & m_\bullet \end{pmatrix} = \begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} \tag{13.58}$$

this yields

$$\begin{aligned} \ldots &= (-1)^{t+k+r+p} \left(\underline{n}^{-1}_{kpr}\right)^* \sum_{xy} (-1)^{k-x+p-y} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} (-1)^{s-\widetilde{s}_\bullet} \begin{pmatrix} s & p & s \\ -s_\bullet & y & \widetilde{s}_\bullet \end{pmatrix} n^{-1}_{sp} (-1)^{l-\widetilde{m}_\bullet} \begin{pmatrix} l & k & l \\ -m_\bullet & x & \widetilde{m}_\bullet \end{pmatrix} n^{-1}_{lk} \\ &= \ldots \end{aligned} \tag{13.59}$$

Now we use

- $-s_\bullet + y + \widetilde{s}_\bullet = 0 \quad \Rightarrow \quad \widetilde{s}_\bullet = s_\bullet - y$
- $-m_\bullet + x + \widetilde{m}_\bullet = 0 \quad \Rightarrow \quad \widetilde{m}_\bullet = m_\bullet - x$
- $-x + t - y = 0 \quad \Rightarrow \quad t = x + y$

This makes us look at the prefactor

$$(-1)^{t+k+r+p} \cdot (-1)^y \cdot (-1)^x \left(\underline{n}^{-1}_{kpr}\right)^* \overset{t=x+y}{=} (-1)^{k+r+p} \left(\underline{n}^{-1}_{kpr}\right)^* \tag{13.60}$$

Now since

$$\left(\underline{n}^{-1}_{kpr}\right)^* = (-1)^{k+p+r} \underline{n}^{-1}_{kpr} \tag{13.61}$$

the Lemma is proven.

Proof 13.14 of Lemma 7.15

<u>Case 1: $t > 0$:</u>

Then we have

$$\widetilde{w}^{kpr}_t = \frac{w^{kpr}_t + (-1)^t w^{kpr}_{-t}}{\sqrt{2}} = \frac{w^{kpr}_t + \left(w^{kpr}_t\right)^*}{\sqrt{2}} = \sqrt{2}\,\mathrm{Re}\, w^{kpr}_t \tag{13.62}$$

<u>Case 1: $t = -|t| < 0$:</u>

Then we have

$$\widetilde{w}^{kpr}_t = \widetilde{w}^{kpr}_{-|t|} = \frac{w^{kpr}_{|t|} + (-1)^{|t|+1} w^{kpr}_{-|t|}}{i\sqrt{2}} = \frac{w^{kpr}_{|t|} - \left(w^{kpr}_{|t|}\right)^*}{i\sqrt{2}} = \sqrt{2}\,\mathrm{Im}\, w^{kpr}_{|t|} \tag{13.63}$$

Proof 13.15 of Lemma 7.15

It suffices to consider the case $t \neq 0$. We construct the inverse mapping by considering for $c = i\widetilde{c} \in \mathbb{C}$ the linear combination

$$\widetilde{w}^{kpr}_{|t|} + c\widetilde{w}^{kpr}_{-|t|} = \frac{\widetilde{w}^{kpr}_{|t|} + (-1)^{|t|} w^{kpr}_{-|t|}}{\sqrt{2}} + c\frac{\widetilde{w}^{kpr}_{|t|} + (-1)^{|t|+1} w^{kpr}_{-|t|}}{i\sqrt{2}} = \frac{1}{\sqrt{2}}\left(\widetilde{w}^{kpr}_{|t|}(1+\widetilde{c}) + (-1)^{|t|}(1-\widetilde{c})\widetilde{w}^{kpr}_{-|t|}\right) \tag{13.64}$$

This reveals: If we want to obtain …

- $w^{kpr}_{|t|}$ we choose $\widetilde{c} = 1$ and scale the whole expression with a global prefactor of $1/\sqrt{2}$,
- $w^{kpr}_{-|t|}$ we choose $\widetilde{c} = -1$ and scale the whole expression with a global prefactor of $(-1)^{|t|}/\sqrt{2}$.

Summarizing this into a single expression yields

$$w^{kpr}_{t} = \frac{(\mathrm{sign}(t))^{t}}{\sqrt{2}}\left(\widetilde{w}^{kpr}_{|t|} + \mathrm{i}\,\mathrm{sign}(t)\widetilde{w}^{kpr}_{-|t|}\right) \tag{13.65}$$

Proof 13.16 of Lemma 7.20 Using Equations (7.44) and (7.45) we obtain

$$\begin{aligned}\langle s_b|S^{(1)}_{-1}|s_a\rangle &= \frac{1}{\sqrt{2}^3}\delta_{s_b,-s_a}\left(1 - \mathrm{i}^2(-1)^{s+s_b}\right)\\ &= \frac{1}{\sqrt{2}^3}\delta_{s_b,-s_a}\left(1 + (-1)^{s+s_b}\right)\end{aligned} \tag{13.66}$$

and

$$\langle s_b|S^{(1)}_{0}|s_a\rangle = \frac{1}{2}\delta_{s_b,s_a}(-1)^{s-s_a} \tag{13.67}$$

and

$$\begin{aligned}\langle s_b|S^{(1)}_{1}|s_a\rangle &= -\frac{1}{\sqrt{2}^3}\delta_{s_b,-s_a}\left(1 + \mathrm{i}^2(-1)^{s+s_b}\right)\\ &= \frac{1}{\sqrt{2}^3}\delta_{s_b,-s_a}\left((-1)^{s+s_b} - 1\right)\end{aligned} \tag{13.68}$$

Proof 13.17 of Lemma 7.21

We use Equation (7.44) and Corollary 6.19 to obtain

$$\begin{aligned}\langle lm|L^{(1)}_{-1}|l\widetilde{m}\rangle &= \frac{1}{\sqrt{2}}\left(\langle lm|L_x|l\widetilde{m}\rangle - \mathrm{i}\langle lm|L_y|l\widetilde{m}\rangle\right)\\ &= \frac{1}{\sqrt{2}}\delta_{m,\widetilde{m}-1}\sqrt{l(l+1) - \widetilde{m}(\widetilde{m}-1)}\end{aligned} \tag{13.69}$$

and

$$\begin{aligned}\langle lm|L_0^{(1)}|l\widetilde{m}\rangle &= \langle lm|L_z|l\widetilde{m}\rangle \\ &= \delta_{m,\widetilde{m}} m\end{aligned} \tag{13.70}$$

and

$$\begin{aligned}\langle lm|L_1^{(1)}|l\widetilde{m}\rangle &= -\frac{1}{\sqrt{2}}\left(\langle lm|L_x|l\widetilde{m}\rangle + \mathrm{i}\langle lm|L_y|l\widetilde{m}\rangle\right) \\ &= \frac{1}{\sqrt{2}}\delta_{m,\widetilde{m}+1}\sqrt{l(l+1)-\widetilde{m}(\widetilde{m}+1)}\end{aligned} \tag{13.71}$$

Proof 13.18 of Theorem 6.9

We seek to compute the electronic particle-number of the system. For this, we choose the one-body operator in Equation (13.74) to be the identity-operator

$$\mathrm{End}(\mathcal{H}) \ni X = \mathbb{1}_{\mathcal{H}} \tag{13.72}$$

This yields for its second quantized version w.r.t. the basis given by the solutions of the second variational problem

$$\mathbb{1}_{\mathcal{H}}^{2\mathrm{Q}} = \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k}\in\mathrm{Q}} c_{N\mathbf{k}}^{\dagger} c_{N\mathbf{k}} \equiv N^{2\mathrm{Q}} \tag{13.73}$$

Using now Fact 6.5, Equation (6.5), the linearity of the trace and the standard-basis in Notation 5.1 we obtain

$$\begin{aligned}\mathrm{Tr}_{\mathscr{F}}\left(\rho_{\mathscr{F}} N^{2\mathrm{Q}}\right) = \langle N^{2\mathrm{Q}}\rangle &= \mathrm{Tr}_{\mathcal{H}}\left(\rho_{\mathcal{H}}\right) \\ &= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k}\in\mathrm{Q}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)\,\mathrm{Tr}_{\mathcal{H}}\left(|\Psi_{N\mathbf{k}}^{\mathrm{sv}}\rangle\langle\Psi_{N\mathbf{k}}^{\mathrm{sv}}|\right) \\ &= \sum_{N=1}^{\mathrm{nstsv}} \sum_{\mathbf{k}\in\mathrm{Q}} \mathrm{F}_{\beta}(\epsilon_{N\mathbf{k}}-\mu)\int_{\mathrm{Crystal}} \mathrm{d}^3\mathbf{r}\,\mathrm{Tr}_{2\times 2}\begin{pmatrix} |\alpha_{N\mathbf{k}}(\mathbf{r})|^2 & \alpha_{N\mathbf{k}}(\mathbf{r})\beta_{N\mathbf{k}}^*(\mathbf{r}) \\ \alpha_{N\mathbf{k}}^*(\mathbf{r})\beta_{N\mathbf{k}}(\mathbf{r}) & |\beta_{N\mathbf{k}}(\mathbf{r})|^2 \end{pmatrix}\end{aligned} \tag{13.74}$$

Proof 13.19 of Theorem 6.12

We compute the matrix-elements of the composition as

$$\begin{aligned}\left\langle \mathbf{r}, s|\rho_{N\mathbf{k}} S_z|\mathbf{r}', s'\right\rangle &= \left(\mathrm{mat}_{\mathbb{B}_{\mathrm{standard}}} \rho_{N\mathbf{k}} S_z\right)(\mathbf{r}, s; \mathbf{r}', s') \\ &= \frac{\hbar}{2}\sum\!\!\!\!\!\!\int_{\mathbf{r}'',s''} \begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha_{N\mathbf{k}}^*(\mathbf{r}'') & \alpha_{N\mathbf{k}}(\mathbf{r})\beta_{N\mathbf{k}}^*(\mathbf{r}'') \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha_{N\mathbf{k}}^*(\mathbf{r}'') & \beta_{N\mathbf{k}}(\mathbf{r})\beta_{N\mathbf{k}}^*(\mathbf{r}'') \end{pmatrix}_{s,s''} \delta(\mathbf{r}''-\mathbf{r}')\left(\sigma_z\right)_{s'',s'} \\ &= \frac{\hbar}{2}\left(\begin{pmatrix} \alpha_{N\mathbf{k}}(\mathbf{r})\alpha_{N\mathbf{k}}^*(\mathbf{r}') & \alpha_{N\mathbf{k}}(\mathbf{r})\beta_{N\mathbf{k}}^*(\mathbf{r}') \\ \beta_{N\mathbf{k}}(\mathbf{r})\alpha_{N\mathbf{k}}^*(\mathbf{r}') & \beta_{N\mathbf{k}}(\mathbf{r})\beta_{N\mathbf{k}}^*(\mathbf{r}') \end{pmatrix}\cdot\sigma_z\right)_{s,s'}\end{aligned}$$

and now taking the diagonal elements proves the theorem.

Proof 13.20 of Theorem 6.15

We will first prove

$$\langle \Psi_{N\mathbf{k}}^{\text{sv}} | S_z | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle = \left(\text{mat}_{\mathbb{B}_{\text{sv}}} S_z\right)(N\mathbf{k}; N'\mathbf{k}') = \delta_{\mathbf{k},\mathbf{k}'} \left(\left(\mathbf{C}^{\text{sv}\to\text{fv}}(\mathbf{k})\right)^* \cdot \left(\sigma_z \otimes \mathbb{1}_{\text{nstfv}}\right) \cdot \left(\mathbf{C}^{\text{sv}\to\text{fv}}(\mathbf{k}')\right)^{\text{T}} \right)_{N,N'} \tag{13.75}$$

The spin-operator, e.g. the component S_z, must be computed from it's matrix-elements of the standard-basis in Equation (6.25), e.g.

$$\langle \Psi_{N\mathbf{k}}^{\text{sv}} | S_z | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle = \left(\text{mat}_{\mathbb{B}_{\text{sv}}} S_z\right)(N\mathbf{k}; N'\mathbf{k}') = \dots \tag{13.76}$$

now we have to insert resolutions of identity from Equation (5.2) and expand $\Psi_{N\mathbf{k}}^{\text{sv}}$ w.r.t. the first-variational eigenvectors as in Equation (4.27) to compute the matrix-elements.

$$\begin{aligned}
\langle \Psi_{N\mathbf{k}}^{\text{sv}} | S_z | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle &= \sum\!\!\!\!\!\!\int_{x,x'} \langle \Psi_{N\mathbf{k}}^{\text{sv}} | x \rangle \langle x | S_z | x' \rangle \langle x' | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle \\
&= \sum_{s,s'} \int \mathrm{d}^3\mathbf{r}\mathrm{d}^3\mathbf{r}' \langle \Psi_{N\mathbf{k}}^{\text{sv}} | \mathbf{r}, s \rangle \delta(\mathbf{r} - \mathbf{r}') \left(\sigma_z\right)_{s,s'} \langle \mathbf{r}', s' | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle \\
&= \sum_{s,s'} \int \mathrm{d}^3\mathbf{r} \langle \Psi_{N\mathbf{k}}^{\text{sv}} | \mathbf{r}, s \rangle \left(\sigma_z\right)_{s,s'} \langle \mathbf{r}, s' | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle \\
&= \star \star \star
\end{aligned}$$

Now we expand according to Equation (4.27)

$$\langle \Psi_{N\mathbf{k}}^{\text{sv}} | = \sum_{m=1}^{\text{nstfv}} \sum_{\pi \in \{\uparrow,\downarrow\}} \left(C_{N,m\pi}^{\text{sv}\to\text{fv}}(\mathbf{k}) \right)^* \langle \Psi_{m\mathbf{k}\pi}^{\text{fv}} | = \sum_{m=1}^{\text{nstfv}} \sum_{\pi \in \{\uparrow,\downarrow\}} \left(C_{N,m\pi}^{\text{sv}\to\text{fv}}(\mathbf{k}) \right)^* \langle \Psi_{m\mathbf{k}}^{\text{fv}} | \otimes \langle \pi | \tag{13.77}$$

Thus

$$\begin{aligned}
\langle \Psi_{N\mathbf{k}}^{\text{sv}} | \mathbf{r}, s \rangle &= \sum_{m=1}^{\text{nstfv}} \sum_{\pi \in \{\uparrow,\downarrow\}} \left(C_{N,m\pi}^{\text{sv}\to\text{fv}}(\mathbf{k}) \right)^* \langle \Psi_{m\mathbf{k}\pi}^{\text{fv}} | \mathbf{r}, s \rangle = \sum_{m=1}^{\text{nstfv}} \sum_{\pi \in \{\uparrow,\downarrow\}} \left(C_{N,m\pi}^{\text{sv}\to\text{fv}}(\mathbf{k}) \right)^* \langle \Psi_{m\mathbf{k}}^{\text{fv}} | \mathbf{r} \rangle \langle \pi | s \rangle = \dots \\
\dots &= \sum_{m=1}^{\text{nstfv}} \sum_{\pi \in \{\uparrow,\downarrow\}} \left(C_{N,m\pi}^{\text{sv}\to\text{fv}}(\mathbf{k}) \right)^* \psi_{m\mathbf{k}}^{*\,\text{fv}}(\mathbf{r}) \delta_{\pi,s} = \sum_{m=1}^{\text{nstfv}} \left(C_{N,ms}^{\text{sv}\to\text{fv}}(\mathbf{k}) \right)^* \psi_{m\mathbf{k}}^{*\,\text{fv}}(\mathbf{r})
\end{aligned}$$

Analogously we have

$$\langle \mathbf{r}, s' | \Psi_{N'\mathbf{k}'}^{\text{sv}} \rangle = \sum_{m'=1}^{\text{nstfv}} C_{N',m's'}(\mathbf{k}') \psi_{m'\mathbf{k}'}^{\text{fv}}(\mathbf{r})$$

And therefore we obtain

$$
\begin{aligned}
\star\star\star &= \sum_{s,s'} \sum_{m,m'=1}^{\text{nstfv}} C^*_{N,ms}(\mathbf{k}) \left(\sigma_z\right)_{s,s'} C_{N',m's'}(\mathbf{k}') \int \mathrm{d}^3\mathbf{r}\psi^*_{\substack{\text{fv}\\ m\mathbf{k}}}(\mathbf{r})\psi_{\substack{\text{fv}\\ m'\mathbf{k}'}}(\mathbf{r}) \\
&= \sum_{s,s'} \sum_{m,m'=1}^{\text{nstfv}} C^*_{N,ms}(\mathbf{k}) \left(\sigma_z\right)_{s,s'} C_{N',m's'}(\mathbf{k}')\delta_{\mathbf{k},\mathbf{k}'}\delta_{m,m'} \\
&= \sum_{s,s'} \sum_{m=1}^{\text{nstfv}} C^*_{N,ms}(\mathbf{k}) \left(\sigma_z\right)_{s,s'} C_{N,ms'}(\mathbf{k})\delta_{\mathbf{k},\mathbf{k}'} \\
&= \left(C^*(\mathbf{k}) \cdot \left(\sigma_z \otimes \mathbb{1}_{\text{nstfv}}\right) \cdot C^{\mathrm{T}}(\mathbf{k})\right)_{N,N'} \delta_{\mathbf{k},\mathbf{k}'}
\end{aligned}
$$

Proof 13.21 of Theorem 8.2

We continue with Equation (8.3) and compute

$$
\begin{aligned}
\mathrm{Tr}\left[\rho\mathbb{1}\right] &= \sum_{\substack{kprt\\ x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \sum_{a,b} (-1)^{m_b-l+s_b-s} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} (\mathbb{1})_{b,a} \\
&= \dots
\end{aligned}
\tag{13.78}
$$

now we use

$$
(\mathbb{1})_{b,a} = \delta_{m_b,m_a}\delta_{s_b,s_a} = (-1)^{m_b-l+s_b-s} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} \begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} n_{s0}^{-1} n_{l0}^{-1} \tag{13.79}
$$

which leads us to consider the sums and use Equation (3.4) in the first step, relabel $m_b \mapsto -m_b$ and use Equation (3.4) in the second step, use Equation (3.5) in the third step and Equation (3.6) in the last step

$$
\begin{aligned}
[k] \sum_{m_b,m_a} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} &= [k](-1)^{2l+k+2l} \sum_{m_b,m_a} \begin{pmatrix} l & l & k \\ -m_b & m_a & x \end{pmatrix} \begin{pmatrix} l & l & 0 \\ -m_a & m_b & 0 \end{pmatrix} \\
&= (-1)^k[k] \sum_{m_b,m_a} \begin{pmatrix} l & l & k \\ m_b & m_a & x \end{pmatrix} \begin{pmatrix} l & l & 0 \\ -m_b & -m_a & 0 \end{pmatrix} \\
&= (-1)^k[k] \sum_{m_b,m_a} \begin{pmatrix} l & l & k \\ m_b & m_a & x \end{pmatrix} \begin{pmatrix} l & l & 0 \\ +m_b & +m_a & 0 \end{pmatrix} \\
&= (-1)^k\delta_{k,0}\delta_{x,0}\{l,l,k\} \\
&= (-1)^k\delta_{k,0}\delta_{x,0}
\end{aligned}
\tag{13.80}
$$

we use Equation (3.4) twice in the first step, and another time in the second step, Equation (3.5) in the third step and Equation (3.6) in the last step

$$
\begin{aligned}
[p]\sum_{s_b,s_a}\begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix}\begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} &= (-1)^{2s+p}(-1)^{2s}[p]\sum_{s_b,s_a}\begin{pmatrix} s & s & p \\ -s_b & s_a & y \end{pmatrix}\begin{pmatrix} s & s & 0 \\ -s_a & s_b & 0 \end{pmatrix} \\
&= (-1)^{2s+p}[p]\sum_{s_b,s_a}\begin{pmatrix} s & s & p \\ -s_b & s_a & y \end{pmatrix}\begin{pmatrix} s & s & 0 \\ s_b & -s_a & 0 \end{pmatrix} \\
&= (-1)^{p}[p]\sum_{s_b,s_a}\begin{pmatrix} s & s & p \\ -s_b & s_a & y \end{pmatrix}\begin{pmatrix} s & s & 0 \\ -s_b & s_a & 0 \end{pmatrix} \\
&= (-1)^{p}\delta_{p,0}\delta_{y,0}\{s,s,p\}
\end{aligned}
\tag{13.81}
$$

inserting this above yields

$$
\begin{aligned}
\ldots &= \sum_{\substack{kprt \\ xy}} w_t^{kpr} N_{kprls}[r](-1)^{x-k+y-p}\begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}(-1)^{k+p}\delta_{k,0}\delta_{x,0}\delta_{p,0}\delta_{y,0}n_{s0}^{-1}n_{l0}^{-1} \\
&= \sum_{rt} w_t^{00r} N_{00rls}[r]\begin{pmatrix} 0 & r & 0 \\ 0 & t & 0 \end{pmatrix} n_{s0}^{-1}n_{l0}^{-1} \\
&= w_0^{000} N_{000ls} n_{s0}^{-1} n_{l0}^{-1}
\end{aligned}
\tag{13.82}
$$

Finally we simplify the expression with Equation (7.33)

$$
N_{000ls}n_{l0}^{-1}n_{s0}^{-1} = \underline{n}_{000}n_{l0}n_{s0}n_{l0}^{-1}n_{s0}^{-1} = \tag{13.83}
$$

noting now that Equation (7.23) gives

$$
\underline{n}_{000} = 1 \tag{13.84}
$$

we obtain the final result

$$
\operatorname{Tr}\rho = w_0^{000} \tag{13.85}
$$

Proof 13.22 of Theorem 8.3

We continue with Equation (8.3), use Equation (7.47) and compute

$$
\begin{aligned}
\operatorname{Tr}\left[\rho S_q^{(1)}\right] &= \sum_{\substack{kprt \\ x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p}\begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}\sum_{a,b}(-1)^{m_b-l+s_b-s}\begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix}\left(S_q^{(1)}\right)_{b,a} \\
&= \sum_{\substack{kprt \\ x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p}\begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}\sum_{a,b}(-1)^{m_b-l+s_b-s}\begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix}\begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix}\times \\
&\qquad \times\delta_{m_b,m_a}(-1)^{s-s_b}\sqrt{s(s+1)(2s+1)}\begin{pmatrix} s & 1 & s \\ -s_b & q & s_a \end{pmatrix} \\
&= \ldots
\end{aligned}
\tag{13.86}
$$

whereas we used Theorem 7.22 in the second step. Now we use

$$\delta_{m_b,m_a} = (-1)^{m_b-l} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} n_{l0}^{-1} \tag{13.87}$$

now we consider the sums and use Equation (3.4) twice in the first step, relabel $s_b \mapsto -s_b$ and use Equation (3.6) in third step

$$\begin{aligned} [p] \sum_{s_b,s_a} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \begin{pmatrix} s & 1 & s \\ -s_b & q & s_a \end{pmatrix} &= (-1)^{4s+p+1} [p] \sum_{s_b,s_a} \begin{pmatrix} s & s & p \\ -s_b & s_a & y \end{pmatrix} \begin{pmatrix} s & s & 1 \\ -s_b & s_a & q \end{pmatrix} \\ &= (-1)^{p+1} \delta_{p,1} \delta_{y,q} \{s, s, p\} \\ &= (-1)^{p+1} \delta_{p,1} \delta_{y,q} \end{aligned} \tag{13.88}$$

and we obtain, using various identities from Lemma 3.3 and Theorem 3.4

$$\begin{aligned} [k] \sum_{m_b,m_a} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} l & 0 & l \\ -m_a & 0 & m_b \end{pmatrix} &= (-1)^k [k] \sum_{m_b,m_a} \begin{pmatrix} l & l & k \\ -m_b & m_a & x \end{pmatrix} \begin{pmatrix} l & l & 0 \\ -m_b & m_a & 0 \end{pmatrix} \\ &= (-1)^k [k] \sum_{m_b,m_a} \begin{pmatrix} l & l & k \\ m_b & m_a & x \end{pmatrix} \begin{pmatrix} l & l & 0 \\ m_b & m_a & 0 \end{pmatrix} \\ &= (-1)^k \delta_{k,0} \delta_{x,0} \{l, l, k\} \\ &= (-1)^k \delta_{k,0} \delta_{x,0} \end{aligned} \tag{13.89}$$

so we obtain

$$\begin{aligned} \ldots &= \sum_{\substack{kprt \\ x,y}} w_t^{kpr} N_{kprls}[r] (-1)^{x-k+y-p} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} (-1)^{k+p+1} \delta_{k,0} \delta_{x,0} \delta_{p,1} \delta_{y,q} \sqrt{s(s+1)(2s+1)} n_{l0}^{-1} \\ &= \sum_{rt} w_t^{01r} N_{01rls}[r] (-1)^{q-1} \begin{pmatrix} 0 & r & 1 \\ 0 & t & -q \end{pmatrix} \sqrt{s(s+1)(2s+1)} n_{l0}^{-1} \\ &= \ldots \end{aligned} \tag{13.90}$$

now $r = 1$ and therefore $t \in \{-1, 0, +1\}$ and we obtain

$$\begin{aligned} \ldots &= \sum_{t} w_t^{011} N_{011ls}[r = 1] (-1)^{q-1} \begin{pmatrix} 0 & 1 & 1 \\ 0 & t & -q \end{pmatrix} \sqrt{s(s+1)(2s+1)} n_{l0}^{-1} \\ &= w_q^{011} N_{011ls}[r = 1] (-1)^{q-1} \begin{pmatrix} 0 & 1 & 1 \\ 0 & q & -q \end{pmatrix} \sqrt{s(s+1)(2s+1)} n_{l0}^{-1} \\ &= \ldots \end{aligned} \tag{13.91}$$

Now we simplify the factors further... We obtain ...

- With Equation (7.3) we obtain $[r = 1] = 3$
- With Equation (7.33) we obtain $N_{011ls} = \underline{n}_{011} n_{l0} n_{s1}$
- With Lemma 3.3 we obtain

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & q & -q \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ -q & 0 & +q \end{pmatrix} \tag{13.92}$$

plugging this into the equation above yields

$$\ldots = \sqrt{s(s+1)(2s+1)}\underline{n}_{011} n_{s1} \cdot 3 \cdot (-1)^{1-q} \begin{pmatrix} 1 & 0 & 1 \\ -q & 0 & +q \end{pmatrix} n_{10}^{-1} n_{10} w_q^{011} = \ldots \tag{13.93}$$

Now we note that ...

- With Equation (7.7) we have $n_{s1} = 1/\sqrt{3}$
- With Lemma 7.18 we have
$$(-1)^{1-q} \begin{pmatrix} 1 & 0 & 1 \\ -q & 0 & +q \end{pmatrix} n_{10}^{-1} = 1 \tag{13.94}$$
- With Equation (7.6) we have $n_{10} = \frac{2}{\sqrt{2!\cdot 3!}}$
- With Equation (7.23) we have
$$\underline{n}_{011} = -\sqrt{\frac{1\cdot 1\cdot 2}{3!}} \cdot \frac{2}{1\cdot 1\cdot 2} = -\frac{1}{\sqrt{3}} \tag{13.95}$$

putting this into the result above yields

$$\begin{aligned} \ldots &= \sqrt{\frac{1}{2}\cdot\frac{3}{2}\cdot 2} \cdot \left(-\frac{1}{\sqrt{3}}\right) \cdot \frac{1}{2\cdot 3} \cdot 3 \cdot 1 \cdot \frac{2}{\sqrt{2\cdot 2\cdot 3}} w_q^{011} \\ &= \sqrt{\frac{3}{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{3}} w_q^{011} = -\frac{1}{2} w_q^{011} = -s w_q^{011} \end{aligned} \tag{13.96}$$

which proves the theorem.

Proof 13.23 of Theorem 8.4

We continue with Equation (8.3), use Equation (7.47) and compute

$$\begin{aligned} \mathrm{Tr}\left[\rho L_q^{(1)}\right] &= \sum_{\substack{kprt \\ x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \sum_{a,b} (-1)^{m_b-l+s_b-s} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \left(L_q^{(1)}\right)_{b,a} \\ &= \sum_{\substack{kprt \\ x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p} \begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix} \sum_{a,b} (-1)^{m_b-l+s_b-s} \begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix} \begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix} \times \\ &\qquad \times \delta_{s_b,s_a} (-1)^{l-m_b} \sqrt{l(l+1)(2l+1)} \begin{pmatrix} l & 1 & l \\ -m_b & q & m_a \end{pmatrix} \\ &= \ldots \end{aligned} \tag{13.97}$$

whereas we used Theorem 7.22 in the second step. Now we use Lemma 7.18

$$\delta_{s_b,s_a} = (-1)^{s_b-s} \begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} n_{s0}^{-1} \tag{13.98}$$

and we compute the sums with Equation (3.4) in the first step and Equation (3.5) in the second step and Equation (3.6) in the last step

$$\begin{aligned}
[p]\sum_{s_b,s_a}\begin{pmatrix} s & p & s \\ -s_b & y & s_a \end{pmatrix}\begin{pmatrix} s & 0 & s \\ -s_a & 0 & s_b \end{pmatrix} &= (-1)^{2s+p}(+1)^{2s}[p]\sum_{s_b,s_a}\begin{pmatrix} s & s & p \\ -s_b & s_a & y \end{pmatrix}\begin{pmatrix} s & s & 0 \\ s_b & -s_a & 0 \end{pmatrix} \\
&= (-1)^{p+1}(-1)^{2s}[p]\sum_{s_b,s_a}\begin{pmatrix} s & s & p \\ -s_b & s_a & y \end{pmatrix}\begin{pmatrix} s & s & 0 \\ -s_b & s_a & 0 \end{pmatrix} \\
&= (-1)^{p}\delta_{p,0}\delta_{y,0}\{s,s,p\} \\
&= (-1)^{p}\delta_{p,0}\delta_{y,0}
\end{aligned} \tag{13.99}$$

and likewise (after relabelling $m_b \mapsto -m_b$)

$$\begin{aligned}
[k]\sum_{m_b,m_a}\begin{pmatrix} l & k & l \\ -m_b & x & m_a \end{pmatrix}\begin{pmatrix} l & 1 & l \\ -m_b & q & m_a \end{pmatrix} &= (-1)^{2l+k+2l+1}[k]\sum_{m_b,m_a}\begin{pmatrix} l & l & k \\ m_b & m_a & x \end{pmatrix}\begin{pmatrix} l & l & 1 \\ m_b & m_a & q \end{pmatrix} \\
&= (-1)^{k+1}\delta_{k,1}\delta_{x,q}\{l,l,k\} \\
&= (-1)^{k+1}\delta_{k,1}\delta_{x,q}
\end{aligned} \tag{13.100}$$

inserting this above yields

$$\begin{aligned}
\ldots &= \sum_{kprtxy} w_t^{kpr} N_{kprls}[r](-1)^{x-k+y-p}\begin{pmatrix} k & r & p \\ -x & t & -y \end{pmatrix}(-1)^{k+p+1}\delta_{k,1}\delta_{x,q}\delta_{p,0}\delta_{y,0}\sqrt{l(l+1)(2l+1)}n_{s0}^{-1} \\
&= \sum_{rt} w_t^{10r} N_{10rls}[r](-1)^{q-1}\begin{pmatrix} 1 & r & 0 \\ -q & t & 0 \end{pmatrix}\sqrt{l(l+1)(2l+1)}n_{s0}^{-1} \\
&= \ldots
\end{aligned} \tag{13.101}$$

now the selection rule in Equation (3.2) gives that only $t = q$ contributes and we obtain

$$\begin{aligned}
\ldots &= \sum_{r} w_q^{10r} N_{10rls}[r](-1)^{q-1}\begin{pmatrix} 1 & r & 0 \\ -q & q & 0 \end{pmatrix}\sqrt{l(l+1)(2l+1)}n_{s0}^{-1} \\
&= \ldots
\end{aligned} \tag{13.102}$$

now we have $r \in \{1\}$ from Definition 7.9 and the triangle-condition in Equation (3.3) gives $|1-r| \leq 0 \leq |1+r|$ s.t. $r = 1$ can contribute such that we obtain the final result

$$\begin{aligned}
\ldots &= w_q^{101} N_{101ls}[r=1](-1)^{q-1}\begin{pmatrix} 1 & 1 & 0 \\ -q & q & 0 \end{pmatrix}\sqrt{l(l+1)(2l+1)}n_{s0}^{-1} \\
&= \ldots
\end{aligned} \tag{13.103}$$

now we simplify the final result by using ...

- Equation (7.3) as $[r = 1] = 3$
- Equation (7.33) as $N_{101ls} = \underline{n}_{101} n_{l1} n_{s0}$

- Lemma 7.18 as

$$(-1)^{1-q}\begin{pmatrix}1 & 0 & 1\\ -q & 0 & q\end{pmatrix} n_{10}^{-1} = 1 \tag{13.104}$$

- Equation (7.6) as $n_{10} = \frac{2}{\sqrt{2!\cdot 3!}}$
- Equation (7.23) as

$$\underline{n}_{101} = \mathrm{i}^2\sqrt{\frac{0!\cdot 2!\cdot 0!}{3!}}\cdot\frac{3!!}{0!!\cdot 2!!\cdot 0!!} = -1\cdot\frac{1}{\sqrt{3}}\cdot\frac{3}{2} = -\frac{\sqrt{3}}{2} \tag{13.105}$$

Plugging this into the equation above yields

$$\begin{aligned}\ldots &= \sqrt{l(l+1)(2l+1)}\underline{n}_{101}n_{l1}\cdot 3\cdot(-1)^{1-q}\begin{pmatrix}1 & 0 & 1\\ -q & 0 & q\end{pmatrix} n_{10}^{-1}n_{10}^{-1}w_q^{101}\\ &= \sqrt{l(l+1)(2l+1)}\cdot\left(-\frac{\sqrt{3}}{2}\right)n_{l1}\cdot 3\cdot 1\cdot\frac{2}{\sqrt{2\cdot 2\cdot 3}}w_q^{101}\\ &= \sqrt{l(l+1)(2l+1)}\cdot(-1)\cdot n_{l1}\cdot 3\cdot\frac{1}{2}w_q^{101}\\ &= \ldots\end{aligned} \tag{13.106}$$

Now we use Equation (7.6) to obtain

$$n_{l1}^2 = \frac{(2l)!\cdot(2l)!}{(2l-1)!\cdot(2l+2)!} = \frac{2l}{(2l+2)(2l+1)} \quad\Longleftrightarrow\quad n_{l1} = \sqrt{\frac{2l}{(2l+2)(2l+1)}} = \sqrt{\frac{l}{(l+1)(2l+1)}} \tag{13.107}$$

plugging this into the Equation above yields

$$\begin{aligned}\ldots &= \sqrt{l(l+1)(2l+1)}\cdot\sqrt{\frac{2l}{(2l+2)(2l+1)}}\cdot\left(-\frac{3}{2}\right)w_q^{101}\\ &= -\frac{3}{2}l\,w_q^{101}\end{aligned} \tag{13.108}$$

Proof 13.24 of Theorem 8.5

We start from Equation (8.3)

$$\begin{aligned}\mathrm{Tr}\left[\rho\,(\mathbf{L}\cdot\boldsymbol{S})\right] &= \ldots\\ &= \sum_{\substack{kprt\\ x,y}} w_t^{kpr}N_{kprls}[kpr](-1)^{x-k+y-p}\begin{pmatrix}k & r & p\\ -x & t & -y\end{pmatrix}\sum_{a,b}(-1)^{m_b-l+s_b-s}\begin{pmatrix}l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix}s & p & s\\ -s_b & y & s_a\end{pmatrix}(\mathbf{L}\cdot\boldsymbol{S})_{b,a}\\ &= \ldots\end{aligned} \tag{13.109}$$

Now as

$$\mathbf{L}\cdot\boldsymbol{S} = L_x S_x + L_y S_y + L_z S_z = \sum_{q=-1}^{+1} L^{(1)}_{-q} S^{(1)}_q \tag{13.110}$$

we can use Equation (7.49) and obtain

$$\begin{aligned}
\ldots &= \sum_{\substack{kprt\\a,b}} w_t^{kpr} N_{kprls}[kpr](-1)^{m_b-l+s_b-s} \sum_{x,y}(-1)^{x-k+y-p} \begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}(\mathbf{L}\cdot\boldsymbol{S})_{b,a}\\
&= \sum_{\substack{kprt\\x,y}} w_t^{kpr} N_{kprls}[kpr](-1)^{x-k+y-p}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}\sum_{a,b}(-1)^{m_b-l+s_b-s}\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\\
&\quad\times\sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}(-1)^{l-m_b}(-1)^{s-s_b}\sum_{q=-1}^{+1}(-1)^q\begin{pmatrix} l & 1 & l\\ -m_b & -q & m_a\end{pmatrix}\begin{pmatrix} s & 1 & s\\ -s_b & q & s_a\end{pmatrix}\\
&= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}\sum_{\substack{kprt\\x,y,q}} w_t^{kpr} N_{kprls}[r](-1)^{x-k+y-p}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}(-1)^q\\
&\quad\times[k]\sum_{m_a,m_b}\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} l & 1 & l\\ -m_b & -q & m_a\end{pmatrix}[p]\sum_{s_a,s_b}\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\begin{pmatrix} s & 1 & s\\ -s_b & q & s_a\end{pmatrix}\\
&= \ldots
\end{aligned} \tag{13.111}$$

Now we use the odd-permutation symmetry in Equation (3.4) to rewrite

$$\begin{aligned}
[k]\sum_{m_a,m_b}\begin{pmatrix} l & k & l\\ -m_b & x & m_a\end{pmatrix}\begin{pmatrix} l & 1 & l\\ -m_b & -q & m_a\end{pmatrix} &= (-1)^{k+1}[k]\sum_{m_a,m_b}\begin{pmatrix} l & l & k\\ -m_b & m_a & x\end{pmatrix}\begin{pmatrix} l & l & 1\\ -m_b & m_a & -q\end{pmatrix}\\
&= (-1)^{k+1}\delta_{k,1}\delta_{x,-q}\{l,l,k\}\\
&= (-1)^{k+1}\delta_{k,1}\delta_{x,-q}
\end{aligned} \tag{13.112}$$

and we obtain also

$$\begin{aligned}
[p]\sum_{s_a,s_b}\begin{pmatrix} s & p & s\\ -s_b & y & s_a\end{pmatrix}\begin{pmatrix} s & 1 & s\\ -s_b & q & s_a\end{pmatrix} &= (-1)^{4s+p+1}[p]\sum_{s_a,s_b}\begin{pmatrix} s & s & p\\ -s_b & s_a & y\end{pmatrix}\begin{pmatrix} s & s & 1\\ -s_b & s_a & q\end{pmatrix}\\
&= \delta_{p,1}\delta_{y,q}(-1)^{p+1}\{s,s,p\}\\
&= \delta_{p,1}\delta_{y,q}(-1)^{p+1}
\end{aligned} \tag{13.113}$$

inserting these equations above yields

$$\begin{aligned}
\ldots &= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}\sum_{\substack{kprt\\x,y,q}} w_t^{kpr} N_{kprls}[r](-1)^{x-k+y-p}\begin{pmatrix} k & r & p\\ -x & t & -y\end{pmatrix}(-1)^q(-1)^{k+1}\delta_{k,1}\delta_{x,-q}\delta_{p,1}\delta_{y,q}(-1)^{p+1}\\
&= \ldots
\end{aligned} \tag{13.114}$$

now we use the Kronecker-deltas to terminate the summations over k, p, x, y and obtain

$$
\begin{aligned}
\ldots &= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}\sum_{rtq} w_t^{11r} N_{11rls}[r](-1)^{-q-1+q-1}\begin{pmatrix}1 & r & 1\\ q & t & -q\end{pmatrix}(-1)^q(-1)^{1+1}(-1)^{1+1}\\
&= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}\sum_{rtq} w_t^{11r} N_{11rls}[r]\begin{pmatrix}1 & r & 1\\ q & t & -q\end{pmatrix}(-1)^q\\
&= \ldots
\end{aligned}
\tag{13.115}
$$

Now due to the sum-rule we have that only those terms with $q+t-q=0 \Leftrightarrow t=0$ contribute, i.e. we obtain

$$
\begin{aligned}
\ldots &= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}\sum_{rq} w_0^{11r} N_{11rls}[r]\begin{pmatrix}1 & r & 1\\ q & 0 & -q\end{pmatrix}(-1)^q\\
&= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}\sum_{r\in\{0,1,2\}} w_0^{11r} N_{11rls}[r]\sum_{q=-1}^{+1}\begin{pmatrix}1 & r & 1\\ q & 0 & -q\end{pmatrix}(-1)^q\\
&= \ldots
\end{aligned}
\tag{13.116}
$$

now since we can use the odd-permutation-symmetry we obtain

$$
\sum_{q=-1}^{+1}\begin{pmatrix}1 & r & 1\\ q & 0 & -q\end{pmatrix}(-1)^q = (-1)^{2+r}\sum_{q=-1}^{+1}\begin{pmatrix}1 & r & 1\\ -q & 0 & q\end{pmatrix} \overset{q\to -q}{=} (-1)^r\sum_{q=-1}^{+1}\begin{pmatrix}1 & r & 1\\ q & 0 & -q\end{pmatrix}(-1)^q \tag{13.117}
$$

thus only $r=0$ and $r=2$ contribute. Computing the sum, we see[2] that only the $r=0$ term contributes, which yields

$$
\begin{aligned}
\ldots &= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}w_0^{110} N_{110ls}[r=0]\sum_{q=-1}^{+1}\begin{pmatrix}1 & 0 & 1\\ q & 0 & -q\end{pmatrix}(-1)^q\\
&= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}N_{110ls}[r=0]\sum_{q=-1}^{+1}\begin{pmatrix}1 & 0 & 1\\ q & 0 & -q\end{pmatrix}(-1)^q w_0^{110}\\
&= \ldots
\end{aligned}
\tag{13.118}
$$

Now we simplify the prefactor

$$
\begin{aligned}
&\sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}N_{110ls}[r=0]\sum_{q=-1}^{+1}\begin{pmatrix}1 & 0 & 1\\ q & 0 & -q\end{pmatrix}(-1)^q\\
&= \sqrt{l(l+1)(2l+1)}\sqrt{s(s+1)(2s+1)}N_{110ls}[r=0]\sum_{q=-1}^{+1}\begin{pmatrix}1 & 0 & 1\\ q & 0 & -q\end{pmatrix}(-1)^q\cdot(-1)^{-1}\cdot n_{10}^{-1}\cdot(-1)^{+1}\cdot n_{10}^{+1}\\
&= \star
\end{aligned}
\tag{13.119}
$$

Now we will use ...

- Lemma 7.18 + Equation (3.4) to find

[2] See Python-script, Code-Folder, ELK-related, Topic 12.

$$\sum_{q=-1}^{+1} \begin{pmatrix} 1 & 0 & 1 \\ q & 0 & -q \end{pmatrix} (-1)^q \cdot (-1)^{-1} \cdot n_{10}^{-1} = 3 \tag{13.120}$$

- Equation (7.33) to find

$$N_{110ls} = \underline{n}_{110} n_{l1} n_{s0} \tag{13.121}$$

- Equation (7.3) to find $[r = 0] = 1$,
- Equation (13.107) to find $n_{10} = \sqrt{\frac{1}{2\cdot 3}}$
- Equation (7.7) to find $n_{s1} = 1/\sqrt{2\cdot 3}$,
- Equation (13.107) to find $n_{l1} = \sqrt{\frac{l}{(l+1)(2l+1)}}$
- Equation (7.23) to find $\underline{n}_{110} = -1\sqrt{\frac{0!\cdot 0!\cdot 2!}{3!}} \cdot \frac{2!!}{0!!\cdot 0!!\cdot 2!!} = -\frac{1}{\sqrt{3}}$

this yields

$$\begin{aligned} \star &= l \cdot \sqrt{\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2}} \cdot \left(-\frac{1}{\sqrt{3}}\right) \cdot 3 \cdot \frac{1}{\sqrt{2\cdot 3}} \cdot (-1) \cdot \frac{1}{\sqrt{2\cdot 3}} \\ &= l \cdot \frac{1}{2}\sqrt{5} \cdot \frac{3}{2\cdot 3} = l \cdot s \cdot \frac{\sqrt{5}}{2} \end{aligned} \tag{13.122}$$

inserting this above yields

$$\ldots = l \cdot s \cdot \frac{\sqrt{5}}{2} w_0^{110} \tag{13.123}$$

and incorporating the dependencies again as

$$w_0^{110} \equiv w_{\substack{0 \\ N\mathbf{k}\alpha l}}^{110} \tag{13.124}$$

and summarizing the results we obtain the desired result

$$\begin{aligned} \langle (\mathbf{L}\cdot\mathbf{S})^{2\mathrm{Q}} \rangle &= N_p \sum_{\mathbf{k}\in\mathrm{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} \mathrm{Tr}_{\mathcal{H}_{\mathrm{LO}_\alpha(l)}} \left(\rho_{\substack{N\mathbf{k} \\ \mathrm{LO}_\alpha(l)}} \mathbf{L}\cdot\mathbf{S} \right) \\ &= N_p \sum_{\mathbf{k}\in\mathrm{Q}} \sum_{N=1}^{\mathrm{nstsv}} \mathrm{F}_\beta(\epsilon_{N\mathbf{k}} - \mu) \sum_{\alpha=1}^{r} \sum_{l=0}^{\infty} l \cdot s \cdot \frac{\sqrt{5}}{2} w_{\substack{0 \\ N\mathbf{k}\alpha l}}^{110} \end{aligned} \tag{13.125}$$